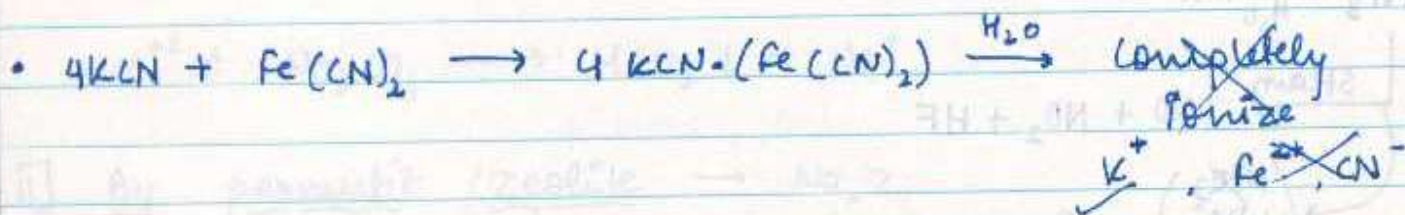
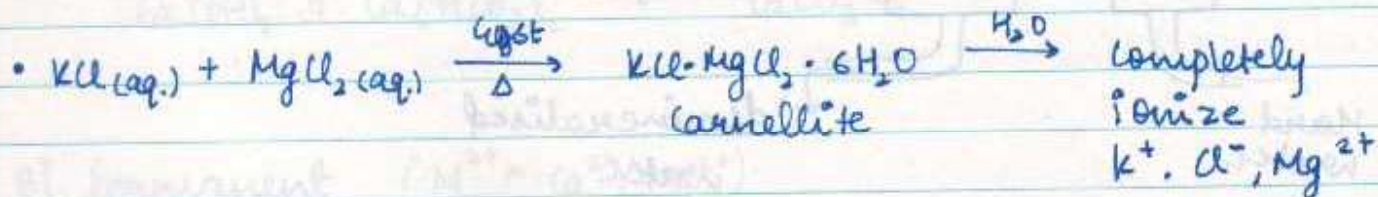
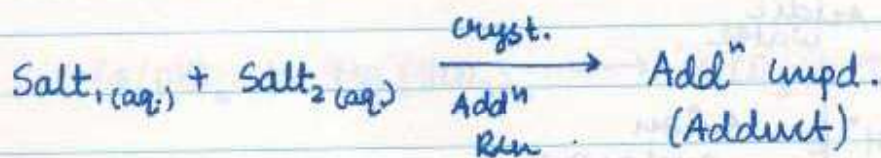


5/5/2018

Co-ordination Compounds

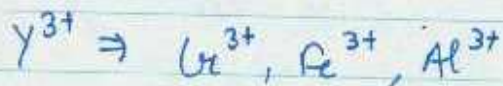
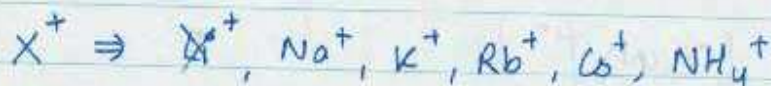
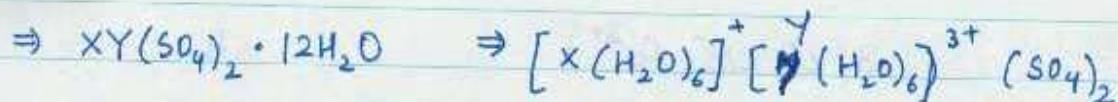


* Types of Addition compound :

1] Double salt :- which ionise completely in aq. medium. All ions show their presence in aq. medium

	Type of ions in aq.	No. of ions
A) Carnallite $KCl \cdot MgCl_2 \cdot 6H_2O$	$3 (K^+, Mg^{2+}, Cl^-)$	5
B) Mohr's salt $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$	$3 (Fe^{2+}, SO_4^{2-}, NH_4^+)$	5
C) Alums ↓	$3 (X^+, Y^{3+}, SO_4^{2-})$	8

General formula $\Rightarrow X_2SO_4 \cdot Y_2(SO_4)_3 \cdot 24H_2O$

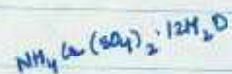


Q. Find the formula of an Alum which gives following observation:

i) gives ppt with $BaCl_2$
BaSO₄

ii) gives a volatile product & coloured ppt when reacted with NaOH
 $Y^{3+} = Cr^{3+}, Fe^{3+}$

iii) gives a volatile product & coloured solⁿ when reacted with Na_2O_2
 $X = NH_4^+$



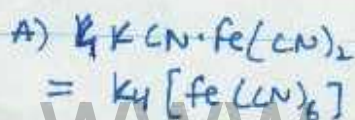
Also find the colour of solⁿ when it reacts with excess Na_2O_2 .



2) Complex Salt :- does not ionise completely in aq. medium. some of the ions lose their identity in aq. medium.

Type of ions
in aq. medium

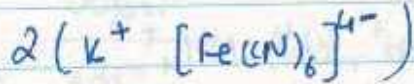
No. of ions
in aq. med.



Theo.



Pract.



5

- Charge on complex is sum of O.S. of C.M.I & ligands
- Space in 3-D where co-ordinate bond form b/w ligand & C.M.I is called co-ordination sphere.
- Simple ions which ionic bond with complex ion are called counter ions.
- Space in 3-D where complex ion forms ionic bond with opp. charge ions is called ionization sphere.
- Species in co-ordinⁿ sphere is non-ionisable while species in ionisation sphere is ionisable.
- Geometry around C.M.I is called co-ordination polyhedra.
eg, tetrahedral (4), Sq. planar (4).
- Species in sq. bracket represent co-ordination entity.
- Co-ordination no. is property of CMI, no. of co-ordinate bond formed by CMI is called its C.N.O.

+3 O.S. of 3d series = 6 always

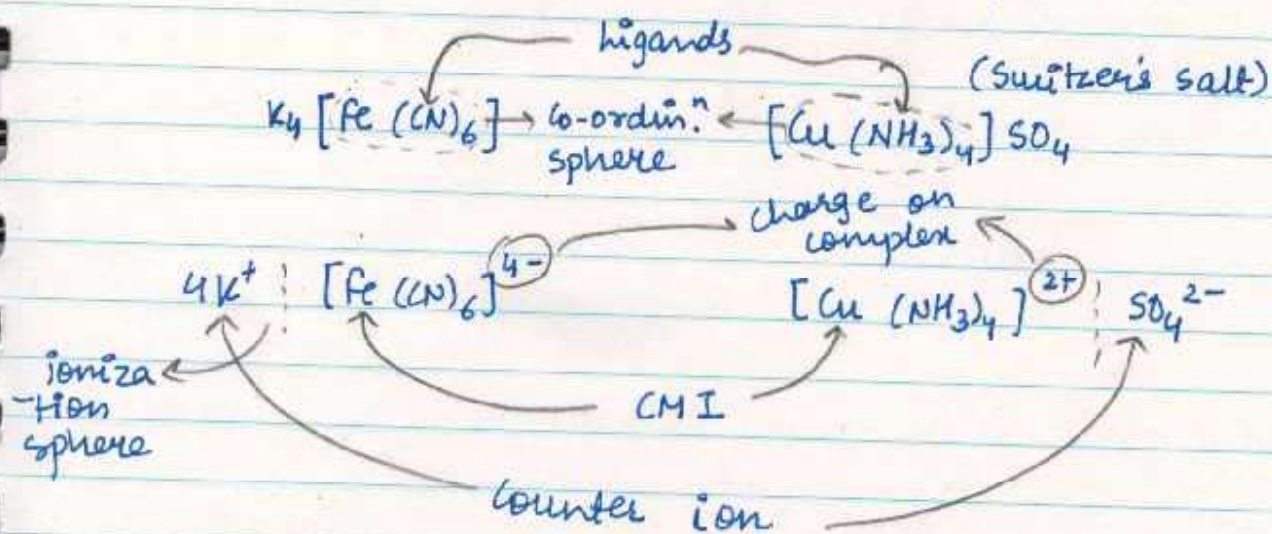
+2 O.S. of 3d series = 4/6

$\text{Ni}^{2+} (\text{OH}_2 / \text{NH}_3) = 6$

$(\text{Cl}^- / \text{CN}^-) = 4$

* Basic Terminology :-

- Ions which lose their identity in aq. solⁿ are written in square bracket are called complex ions & compound containing such ions are called complex compound.
- In complex ion at least one metal ion is present which is centre of complex ion, acts as Lewis acid is called central metal ion [CMI].
- CMI is surrounded by neutral molecule and ions which act as Lewis base and are called ligands.
- Complex formation is a Lewis acid-base rxn in which co-ordinate bond is present b/w CMI & ligands, that's why, complex compounds are also known as co-ordination compounds.



* classification of ligands :

1) On the basis of charge :

<u>Cationic</u>	<u>Anionic</u>	<u>Neutral</u>
-ium suffix is used	ide $\rightarrow$ ido ite $\rightarrow$ ito ate $\rightarrow$ ato	- Common name <u>except</u>
$\text{NO}^+ \rightarrow$ Nitrosylium/ Nitrosonium	$\text{SO}_4^{2-} \rightarrow$ Sulphato	$\text{H}_2\text{O} \rightarrow$ aqua
$\text{N}_2\text{H}_5^+ \rightarrow$ Hydrazinium	$\text{O}^{2-} \rightarrow$ Oxido	$\text{NH}_3 \rightarrow$ ammine
	$\text{SO}_3^{2-} \rightarrow$ Sulphito	$\text{CO} \rightarrow$ Carbonyl
	$\text{C}_6\text{H}_5^- \rightarrow$ phenyl	$\text{NO} \rightarrow$ Nitrosyl
	$\text{CH}_3^- \rightarrow$ methyl	$\text{C}_6\text{H}_6 \rightarrow$ benzene
		$\text{PH}_3 \rightarrow$ Phosphene

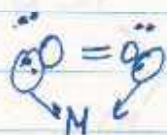
2) On the basis of denticity :

- Having one donor site = monodentate

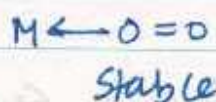
eg : H^- , F^- , Cl^- , Br^- , I^- , H_2O , OH^- , NH_3 , NH_2^- , NH^{2-}

- Having two donor site

A) form three membered ring with $\text{CMI} = \text{monodentate}$



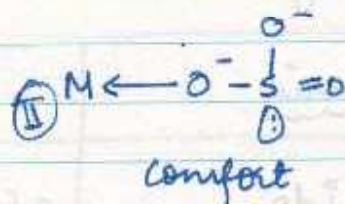
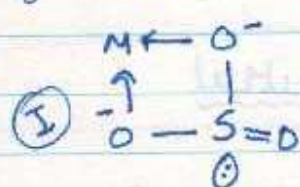
3-membered ring
Angle strain & (unstable)



eg : CN^- , SCN^-

B) Form four membered ring with CHI

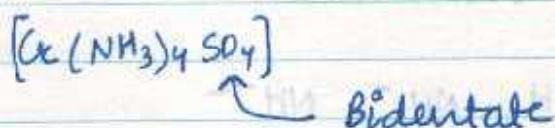
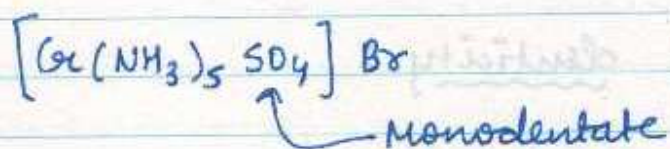
eg: SO_3^{2-}



- Stable in comp. of 3-m ring but unstable due to angle strain.

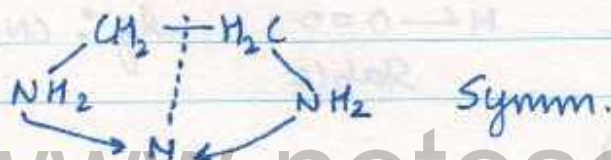
- More stable in comp. of 1st structure.

Such ligands are called flexident coz they show variable denticity, as per requirement of C. No. of central metal ion they can act as bidentate ligand.

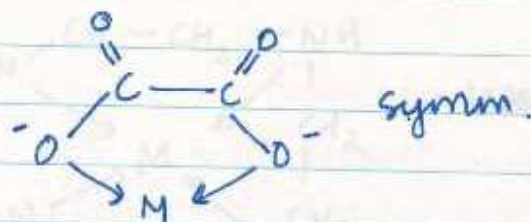


c) Form 5/6/7 membered ring are called bidentate/didentate

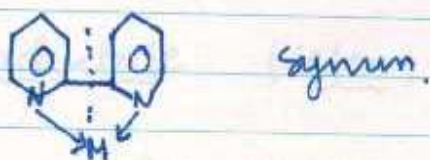
- (en) ethylenediamine (ethane 1-2 diamine)



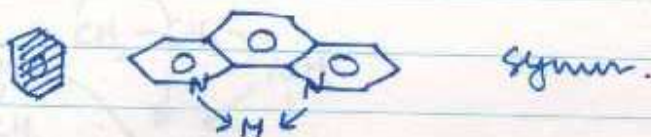
- (OX^{-2}) Oxalato $C_2O_4^{2-}$



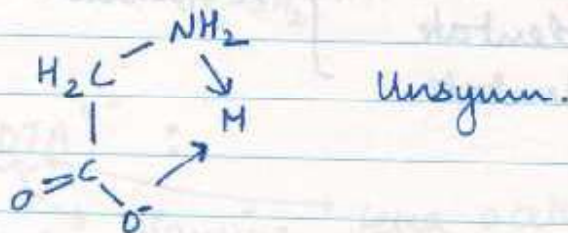
- dipyrizidine (dipy)



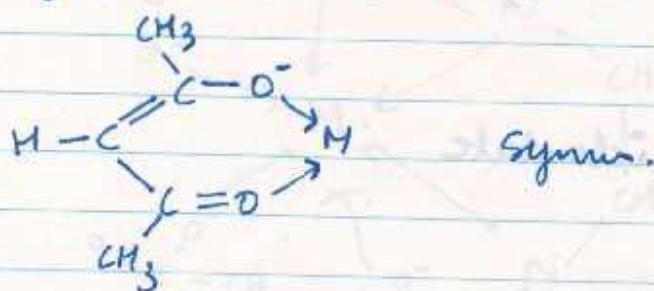
- Phenanthroline (Phen)



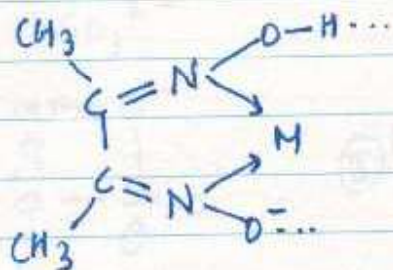
- glycinate (gly⁻)



- Acetylacetonato (acac⁻)

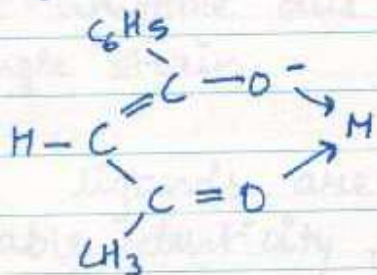


- dimethylglyoximate (dmg^-)



Unsym.

- benzylacetato (bcac^-)



Unsym.

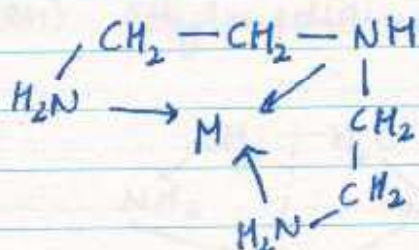
- having,

3 donor site $\rightarrow$ tridentate
 4 donor site $\rightarrow$ tetradentate
 5 donor site $\rightarrow$ pentadentate
 6 donor site $\rightarrow$ Hexadentate

} Polydentate

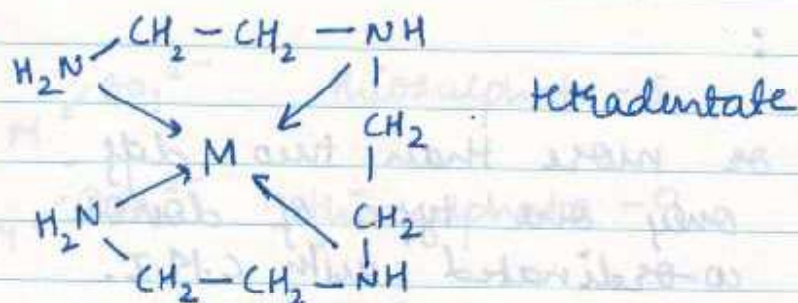
eg,

dien (diethylene tri amine)



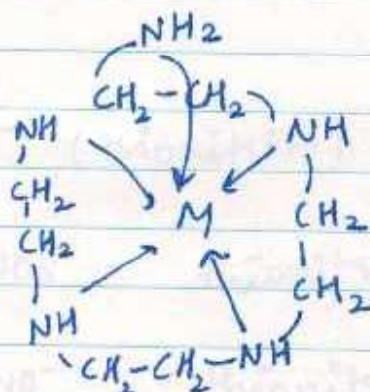
tridentate

trien (triethylene tetra amine)



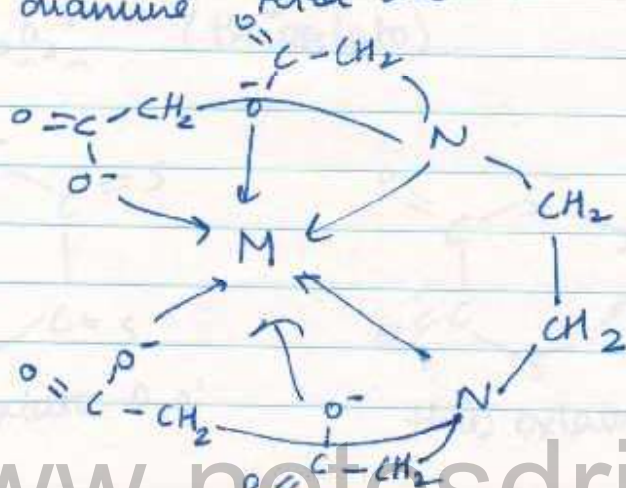
• Pentadentate (trien) :

tetra ethylene penta amine



• EDTA⁴⁻ :

ethylene diamine tetra acetato



(MNO = 8)

• special case :

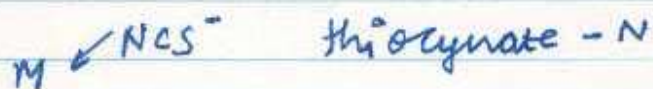
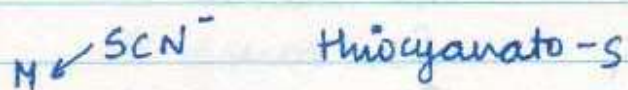
1) Ambidentate ligands :

- ligands having two or more than two diff. donor site but uses only one type of donor site at a time when co-ordinated with C.M.I.
- In naming of such ligands, capital letter of donor site is written as suffix.

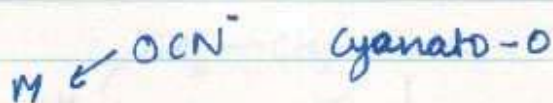
a) CN⁻ [:C≡N:]⁻

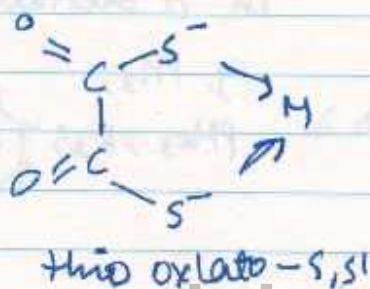
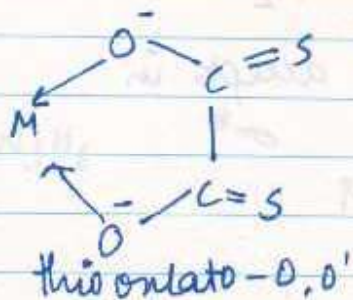
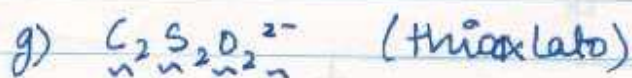
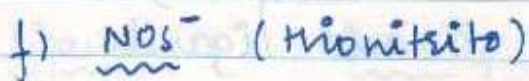
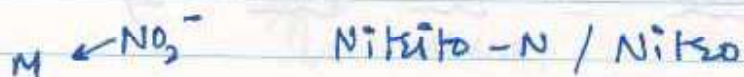
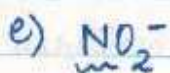
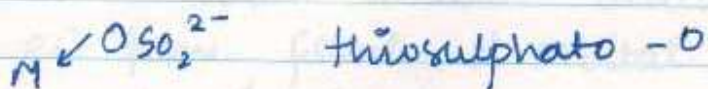
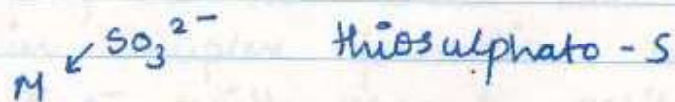


b) SCN⁻ [:S=C=N:]⁻



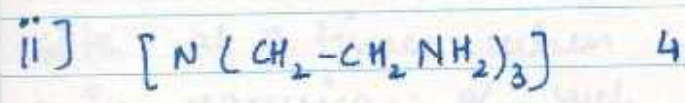
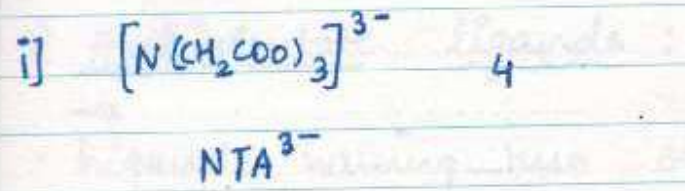
c) OCN⁻



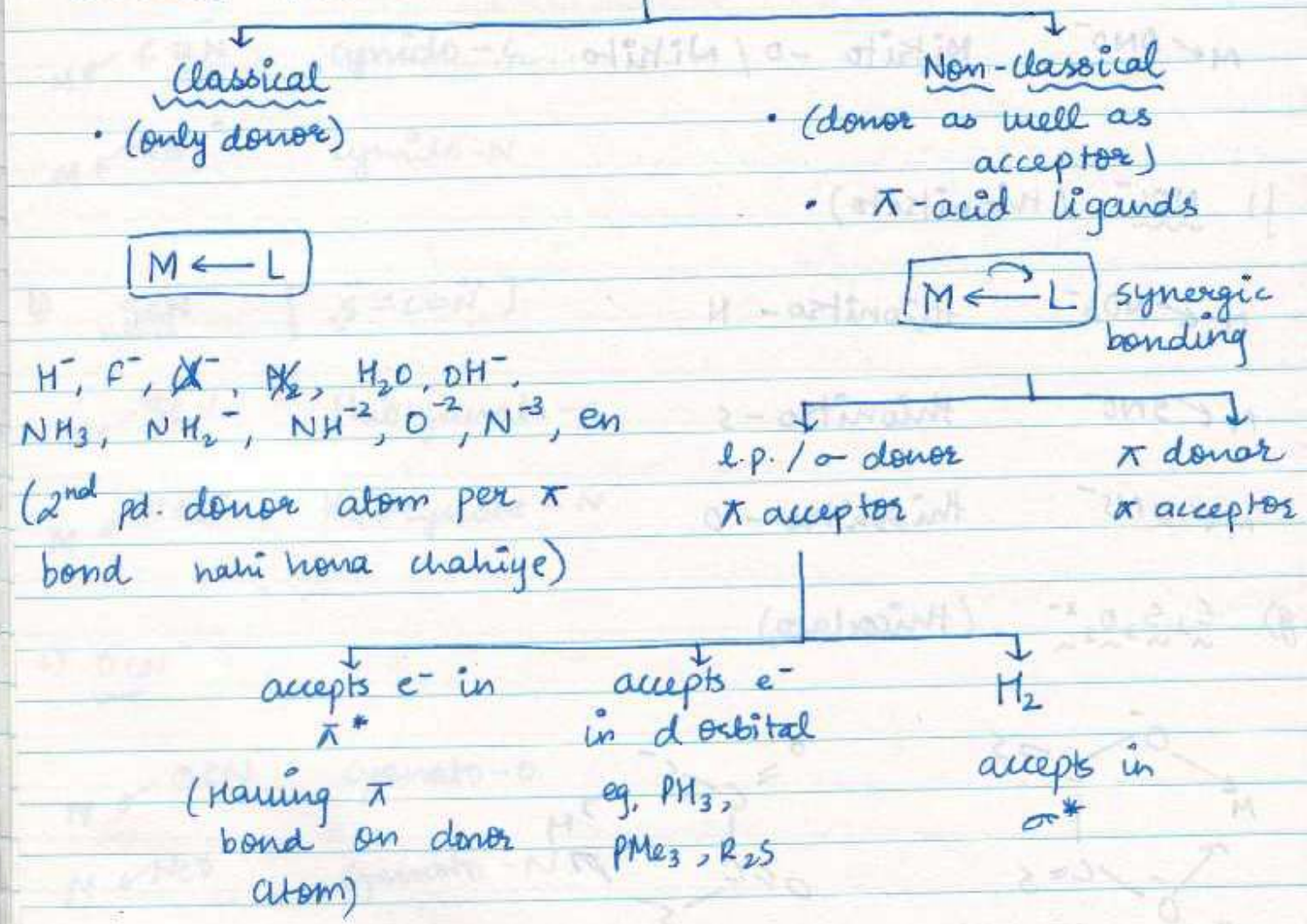




g. calculate denticity of :



* Classification of ligands on the basis of bonding pattern :

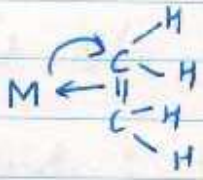
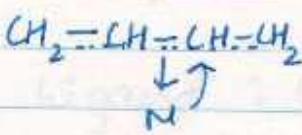
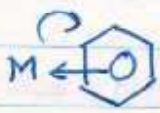
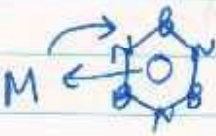


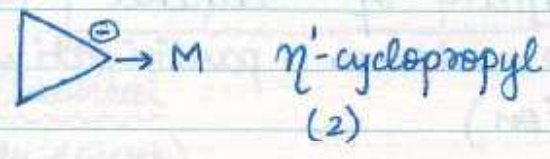
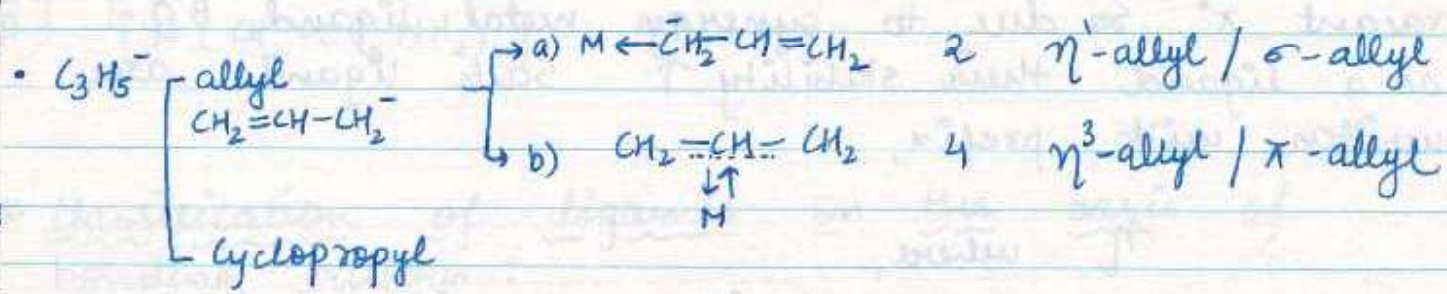
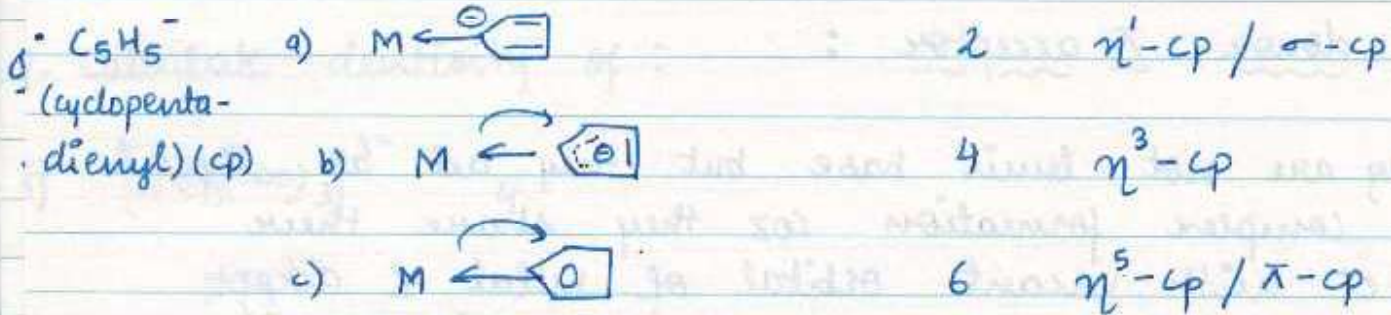
* π -donor, π -acceptor :

- They are not Lewis base but they act as ligand in complex formation coz they share their π -e⁻ with vacant orbital of metal & accept e⁻ from filled d-orbital of metal into their vacant π^* . So due to synergy metal ligand B.O. ↑ as a ligand their stability ↑. Such ligands are written with prefix,

η^x where,

$x \rightarrow$ hapticity of ligand (no. of donor atoms participating in donation)

<u>ligand</u>	<u>Bonding Pattern</u>	<u>no. of e⁻ in do.</u>	<u>name</u>
• C_2H_4		2	η^2 -ethylene / π -ethylene
• $CH_2=CH-CH=CH_2$		4	η^4 -butadiene / π -butadiene
• C_6H_6		6	η^6 -benzene / π -benzene
• $B_3N_3H_6$		6	η^6 -borazine / π -borazine



Q. Match

- | | | |
|---------------------|----------------------------|--|
| A) gly ⁻ | i) same donor atoms | $\text{CH}_2-\text{CH}_2-\text{NH}$

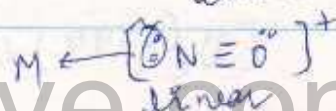
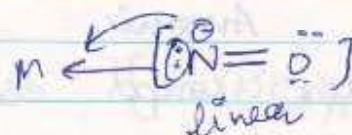
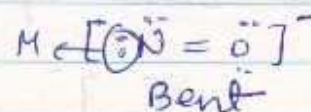
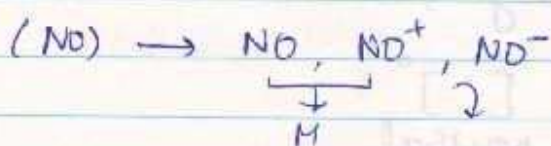
CH_2

CH_2 |
| B) dipy | ii) diff. donor atoms | $\text{H}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}$
$\text{H}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{NH}$ |
| C) bn | iii) optically active | |
| D) pn | iv) forms 5-membered ring |  |
| | v) unsymmetrical bidentate | |

→ C → i; ii; iv
 A → ii; iv; v
 B → i; iv;
 D → i; ii; iv; v

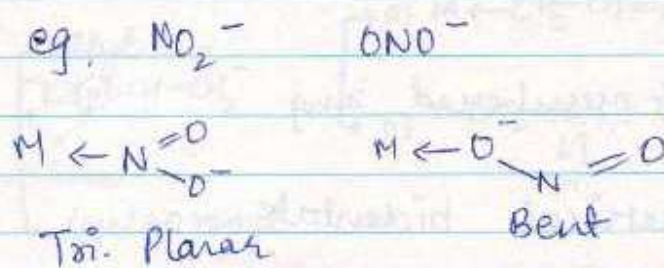
* On the basis of fixed & variable charge:

- 1] Innocent ligand : Having fixed O.S.
 eg, Cl^- , OH^- , H_2O , NH_3 , etc.
- 2] Non-innocent ligand : Can show diff. diff. O.S.
 eg. $(\text{O}_2) \rightarrow \text{O}_2, \text{O}_2^-, \text{O}_2^{2-}$



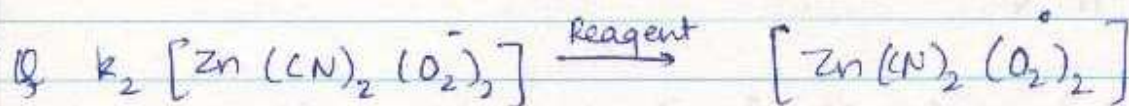
They can be distinguished by,

- i) B.O.
- ii) B.L / B.E.
- iii) No. of unpaired e^-
- iv) Geometry around donor site



$\{ \text{NO}_2^- \text{ is an innocent ligand} \}$

- v) Conf. of central metal ion
- vi) V.P. e^- of CMI
- vii) O.S. of CMI

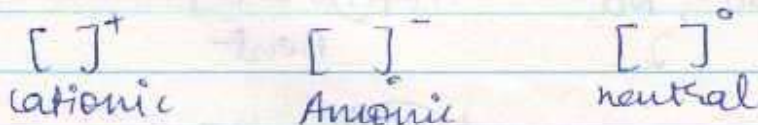


Find diff in no. of π b/w I & II

$\rightarrow 1 \times 2 = 2$

* Classification of Complexes:

i] on the basis of charge,



So complex compd can be

i] simple cation + complex anion

ii] complex cation + simple anion

iii] complex anion + complex anion

iv] Neutral complex

eg of i)

a) Fischer's salt $K_3 [Co(NO_2)_6]$
(Yellow ppt)

b) Sodium nitro prusside $Na_2 [Fe^{+2}(CN)_5 NO^{+1}]$
 $\searrow$
 $[Fe(CN)_5X]^{-n}$

eg of ii)

a) Brownian complex $[Re(H_2O)_5NO]SO_4$

eg of iii)

a) $[Cr(NH_3)_6][Co(NO_2)_6]$

eg of iv)

a) cis-platin (anticancer) $cis - [Pt(NH_3)_2Cl_2]$

b) Wilkinson catalyst $[Rh(PPh_3)_3Cl]$

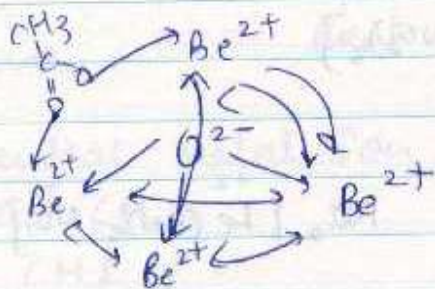
ii] On the basis of no. of CMI

a) Mononuclear : Having one CMI

eg. cis-platin

b) bi/polynuclear : Having more than one CMI

eg. $[\text{Be}_4\text{O}(\text{CH}_3\text{COO})_6]$



iii] On the basis of types of ligand

a) Homoleptic : only one type of ligand

b) Heteroleptic : more than one type of ligand

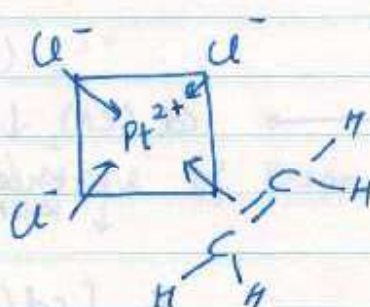
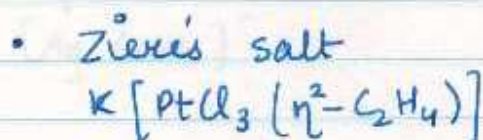
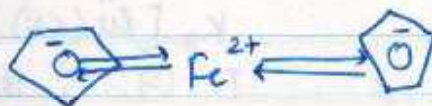
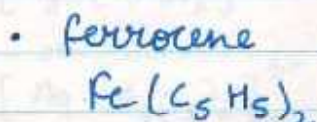
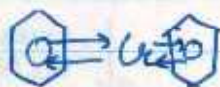
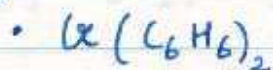
iv] On the basis of donation of ligands

A) σ complexes : Having only lone pair donor

cis-platin

B] π -complexes : Having atleast one π donor

eg.



C_2H_4 & PtCl_3
plane is $\perp$ or
to each
other

v] On the basis of stability :



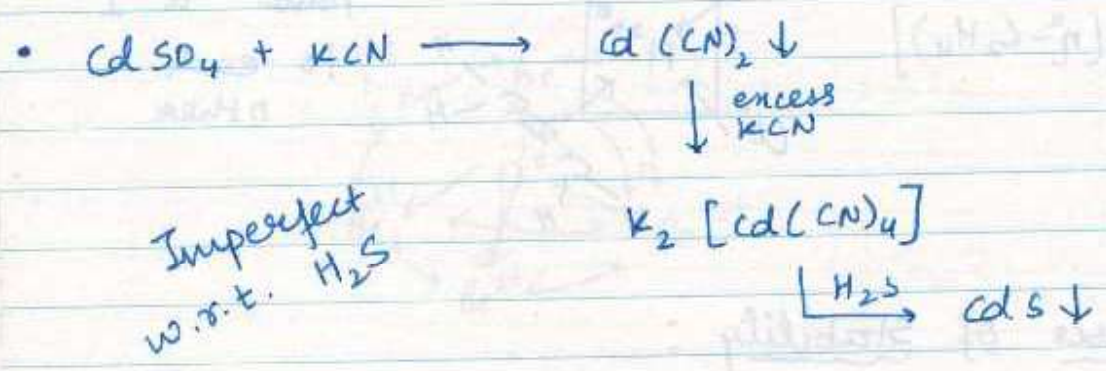
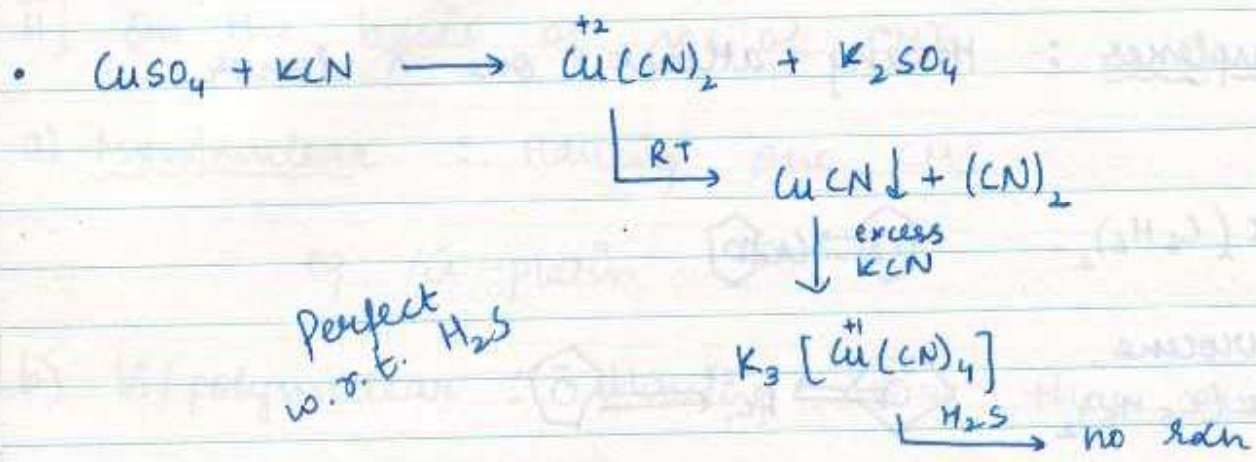
$$\frac{K_s}{K_f} = \frac{[\text{ML}_n^{+x}]}{[\text{M}^{+x}][\text{L}]^n}$$

$\uparrow$
stability
constant

$K_f \uparrow$ forward rxn $\uparrow$
stability of complex $\uparrow$
conc. of simple ions $\downarrow$

a) Perfect complex : Stable in presence of particular reagent.
Having high value of K_f .

b) Imperfect complex : Unstable in presence of particular reagent.
Having low value of K_f .

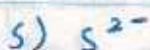
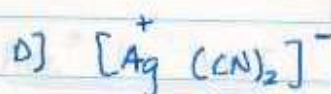
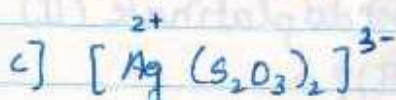
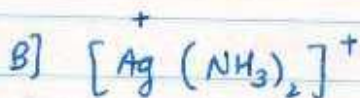
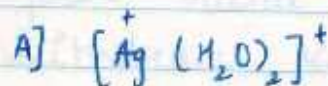


	AgCl	AgBr	AgI	Ag ₂ S
in H ₂ O	X	X	X	X
cold NH ₃	✓	X	X	X
iv Hot NH ₃	✓	✓	X	X
A S ₂ O ₃ ²⁻	✓	✓	✓	X
CN ⁻	✓	✓	✓	✓

✓ = soluble
 X = insoluble

Matein,

Imperfect in presence of



T) None of these

→ A - P, Q, R, S

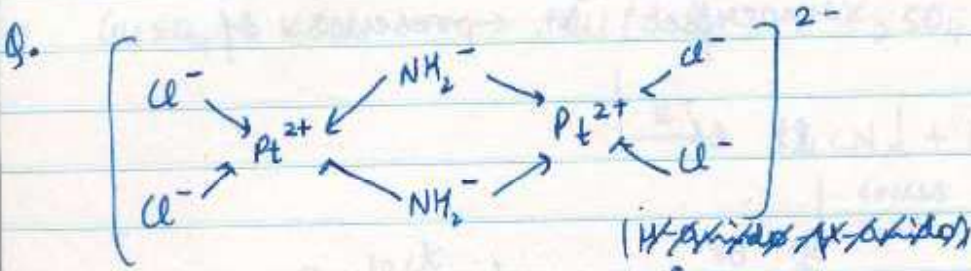
B - P, Q, R, S or Q, R, S

C - P, Q, R, S

D - P, Q, R, S, T

• Chelating ligands :- those who form ring like structure with CMI (bidentate, polydentate) ring formation is called chelation.

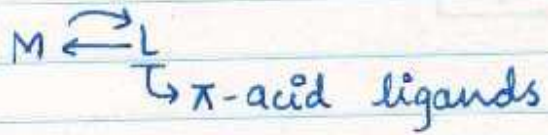
• Bridging ligands :- In polynuclear complexes, ligands joins two or more than two CMI. Such ligands are written with μ_n prefix where n = no. of CMI which are directly bonded with a ligand.



-
- dichlorido platinate (II) μ -amido-dichlorido platinate (II)
 - di- μ -amido bis(dichlorido platinate (II))
 - di- μ -amido tetrachlorido platinate (II)
 - bis- μ -amido dichlorido platinate (II)

* Synergic Bonding :

↖ give & take



i] in metal carbonyls ($M(CO)_x$) :

Accⁿ to VBT,

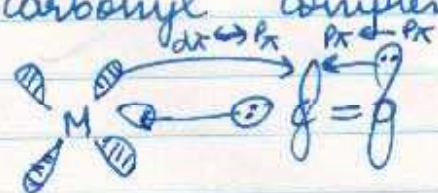


BO = 3 EN C < O

donor site 'C'

more EN 'O' donates e.p. to less EN 'C'

in carbonyl complexes



EN, $M < C < O$

'C' ko e^- देने mai majha metal se jyada aayega, jaise metal se donation $\uparrow$ hoga O' apna l.p. withdraw kar lega, metal-C BO $\uparrow$

CO BO $\downarrow$

CO BE $\downarrow$ CO BL $\uparrow$

$1 < BO_{MC} < 2$

$2 < BO_{CO} < 3$

Accⁿ to MOT

NBMO of 'C' $\rightarrow$ $:\text{C} \equiv \text{O}:$ $\leftarrow$ NBMO of 'O'

BO = 3, 14 e^- system

energy of NBMO of 'C' > energy of NBMO of 'O'
So donor atom is 'C'.

HOMO of CO is σ_{NBMO} having slight AB ch^r
LUMO of CO is π^* having 100% AB ch^r

due to donation, BO of CO $\uparrow$ slightly
due to acceptance, BO of CO $\downarrow$ rapidly
So, due to synergic bonding
Net CO BO $\downarrow$ $2 < CO_{BO} < 3$



Q. If CO BD in $\text{Cr}(\text{CO})_6$ is x then comment on
CO BD in

- A) $\text{Co} > x$
 B) $\text{BH}_3 \leftarrow \text{Co} > x$
 C) $\text{Cr}(\text{CO})_6^+ > x$
 D) $\text{Cr}(\text{CO})_6^- < x$
 E) $\text{V}(\text{CO})_6^- < x$
 F) $\text{Mn}(\text{CO})_5^+ > x$
 G) $[\text{Cr}(\text{CO})_5(\text{NH}_3)] < x$
 H) $[\text{Cr}(\text{CO})_5\text{PF}_3] > x$
 I) $[\text{Cr}(\text{CO})_5\text{PtCl}_3] < x$
- Acceptance tendency of
 $\text{PF}_3 > \text{CO} > \text{PtCl}_3$
- classical one d.p. decol- b/w 5 CO

Practical evidence of CO BL / stretching freq. Method :

—|—|—|

As strength of spring $\uparrow$
 to stretch string required energy $\uparrow$

$$E = h\nu$$

Stretching frequency (ν) $\uparrow$

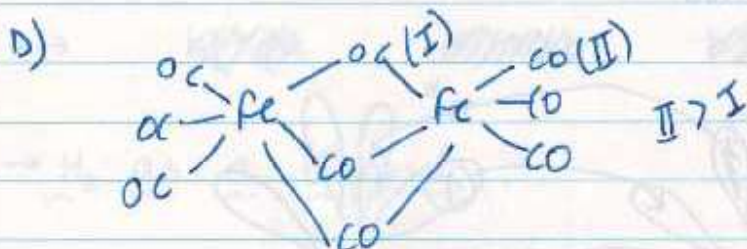
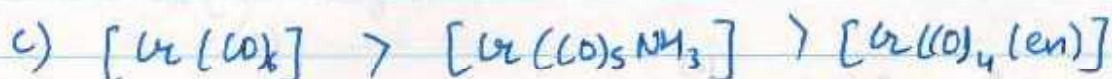
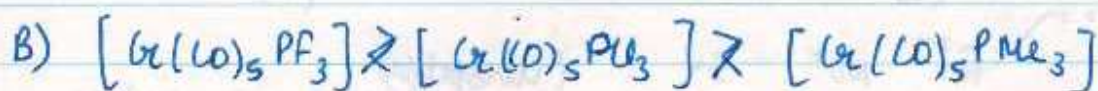
As like spring,

As CO BD $\uparrow$

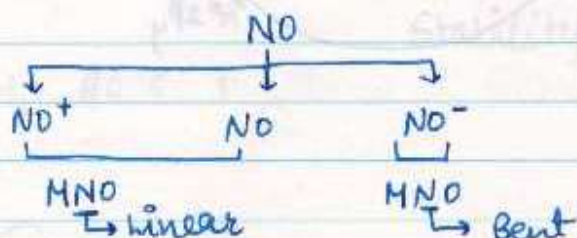
CO BF $\uparrow$

Stretching frequency of CO $\uparrow$

g. Compare stretching frequency of CO

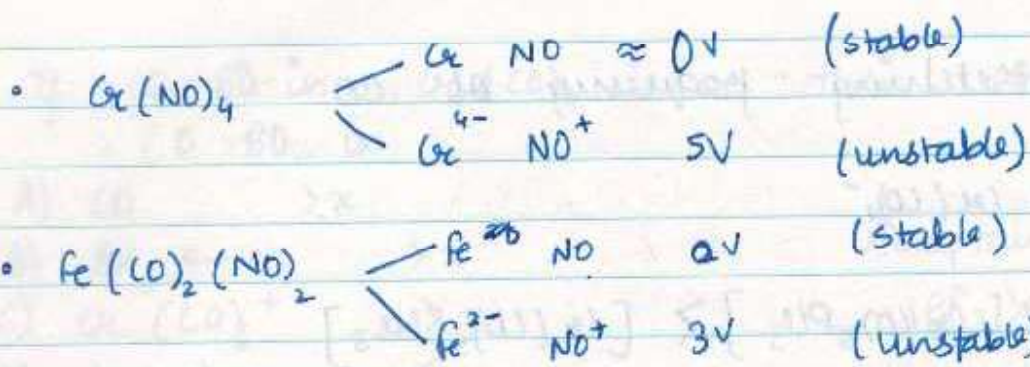


* NO as a ligand :



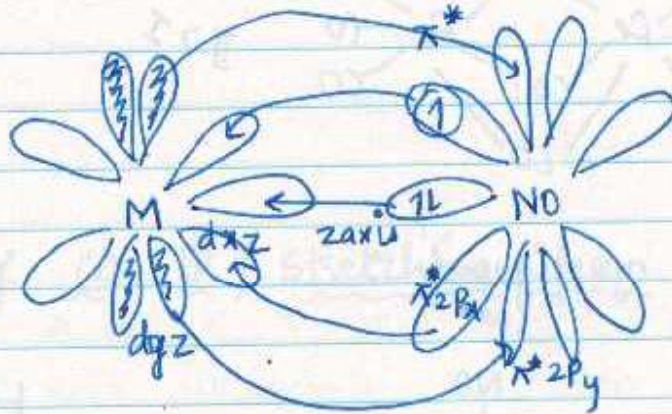
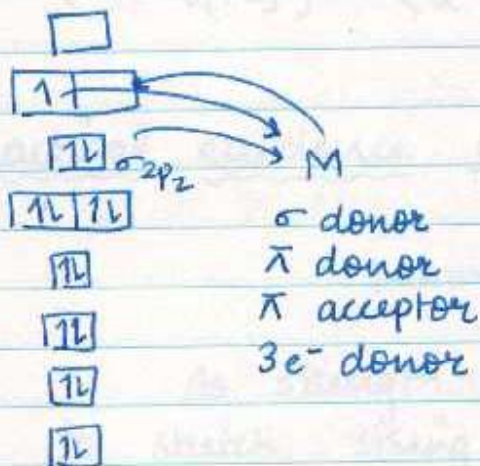
• How to distinguish b/w NO & NO⁺



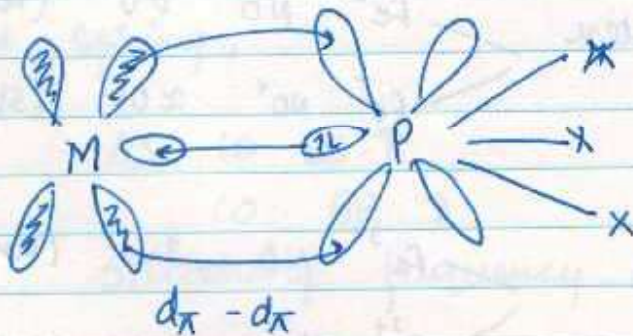


→ Synergic bonding in $\text{MNO}^{(0)}$

NO 1Se^-

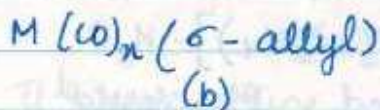
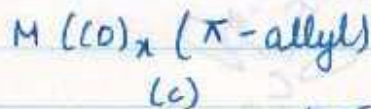
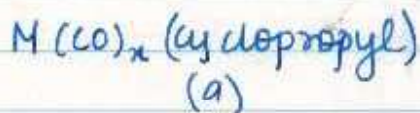


→ Synergic bonding in PX_3 , SX_2 , AsX_3



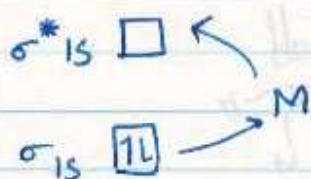
No change in P-X BO due to synergy because e^- from metal ion enters in vacant d-orbital of 'P' not in σ^* of P-X bond.

Q. compare stretching frequency of CO



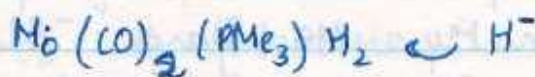
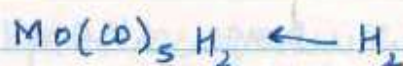
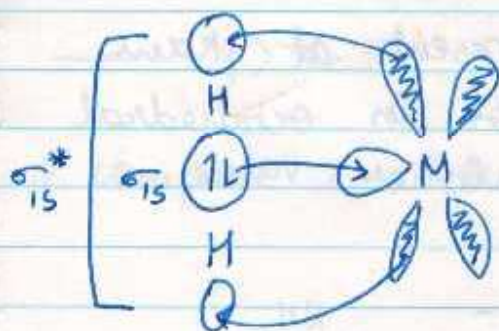
→ ~~very high~~ ~~very high~~ ~~very high~~ $c > b > a$

→ H₂ as a ligand.



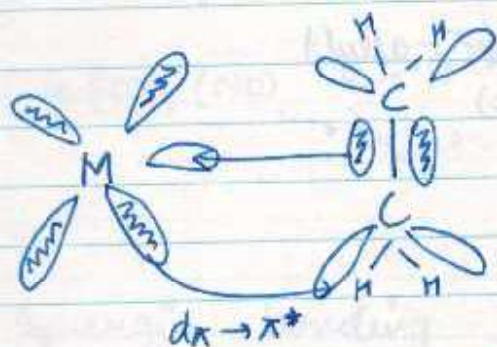
H₂ as a ligand stable when good acceptor present in complex, so e^- cloud in σ^* of H₂ ↓, Stability of H₂ ↑

$0 < H_2 \text{ BO} < 1$



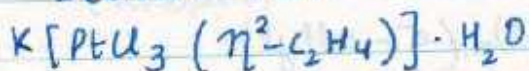
acceptance tendency of $CO > PMe_3$

* C₂H₄ as a ligand



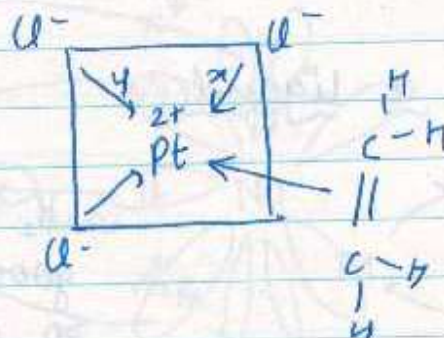
example :-

Zeise's salt



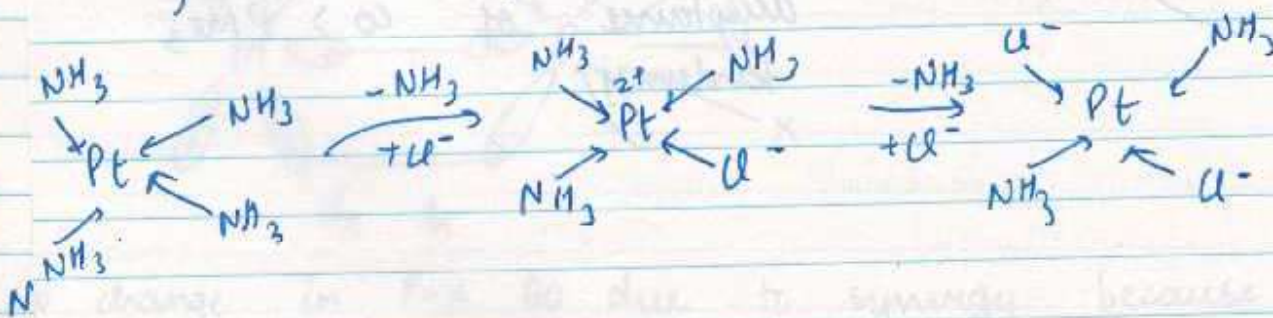
first compound observed which has π -donor ligand.

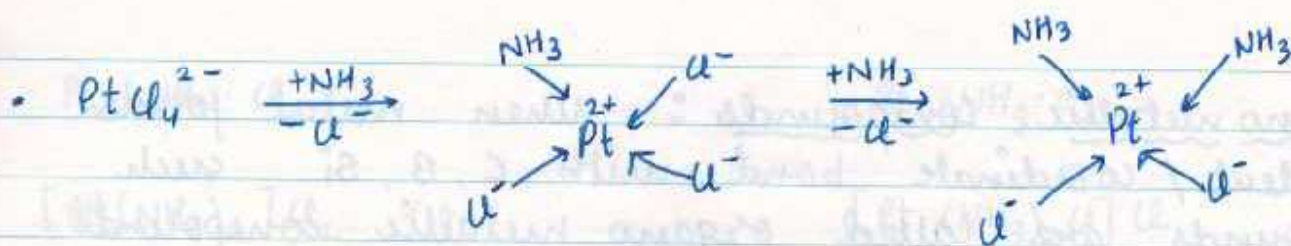
- $PtCl_3$ & C_2H_4 plane are $\perp$ to each other
- C_2H_4 is not planar
- All H atoms are in a plane but not with 'C' atoms due to σ repⁿ b/w $d\pi \rightarrow \pi^*$ & $C-H$ σ bond



BL $y > x$
due to trans effect

trans effect :- when strong a.c. present at trans position in sq. planar or in octahedral geometry M-ligand BL has higher value from its expected value.

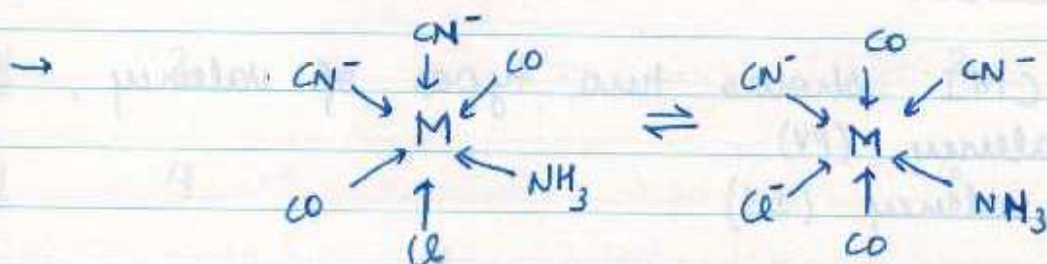




Q. Draw possible correct structure of



If two type of stretching frequency obtained for CN^- & one type of stretching frequency obtained for CO .



Q. If difference in C-C BL in C_2H_4 & $\text{K}[\text{PtCl}_3(\text{C}_2\text{H}_4)]$ is x & in C_2F_4 & $\text{K}[\text{PtCl}_3(\text{C}_2\text{F}_4)]$ is y . then compare x & y

$\rightarrow y > x$ C_2F_4 is better acceptor than C_2H_4

Q. Compare C-C BL in C_5H_5^- & Ferrocene
 C_6H_6 & $\text{Cr}(\text{C}_6\text{H}_6)_2$

$\rightarrow \text{C}_6\text{H}_6 < \text{Cr}(\text{C}_6\text{H}_6)_2$

$\text{C}_5\text{H}_5^- < \text{Ferrocene}$

* Organo metallic compounds : When metal forms covalent / coordinate bond with C, B, Si such compounds are called organo metallic compounds.

• cyanide salts & ionic carbide are not organo metallic compounds.

* WERNER'S THEORY :-

Accⁿ to him, CMI shows two types of valency,

- i) Primary Valency (PV)
- ii) Secondary valency (SV)

PV

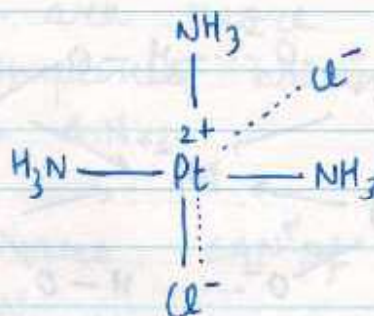
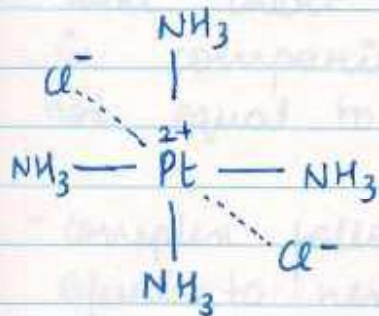
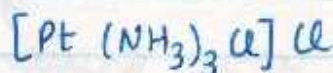
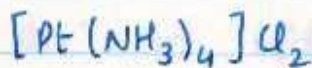
SV

- O.S. of CMI
- Satisfied by anions.
- Form ionic bond
- Generally ionisable
- Non-directional
- ~~Satisfied~~ represented by dotted (---) line

- C.N. of CMI
- Satisfied by anions / neutral molecules (ligands)
- Co-ordinate bond
- Non-ionisable
- Directional
- by solid (—) line

Note :

- 1) Anionic ligands are represented by solid with dotted line. If nothing is specified, always consider neutral ligands inside coordination sphere in conversion of werner's formula into modern formula



PV 2

2

SV 4

4

electrolyte 1:2 (cation: anion)

1:1

cond. >

ppt with 2 mol
excess AgNO3

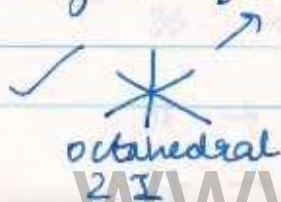
1 mol

• Application of Werner's theory:

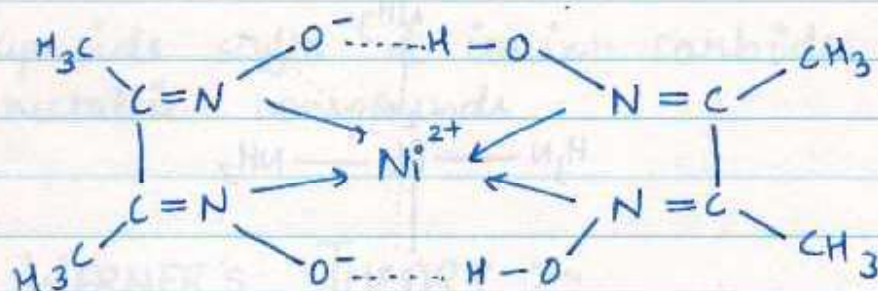
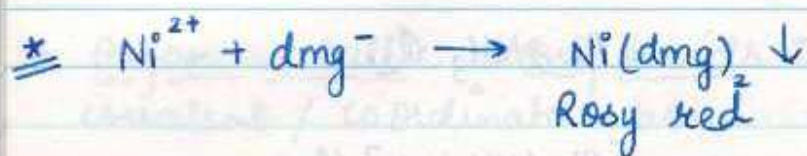
i] to predict exact formula of each complex.

ii] to predict the structure of diff. complexes with C.N. 4 & 6.

eg: $[Cr(NH_3)_4Cl_2]Cl \rightarrow 2 \text{ isomers}$



trigonal prismatic
3I



* E.A.N (Effective Atomic Number) :-

- Total no. of e^- in CMI after complex formation.

Q.

1. $[\text{Sc}(\text{H}_2\text{O})_6]^{3+} = 30$

8. $[\text{NiEDTA}]^- = 40 - 3$

2. $[\text{Ti}(\text{CO})_6]^{2-} = 36$

9. $[\text{Ag}(\text{CN})_2]^- = 50$

3. $[\text{VO}_4]^{3-} = 26$

10. $\text{CuCl}_4^{2-} = 35$

4. $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3 = 33$

11. $[\text{Zn}(\text{acac})_2] = 32 + 4$

5. $[\text{Mn}(\text{en})_3]\text{SO}_4 = 29 + 6$

12. $\text{HgI}_4^{2-} = \text{Hg} + 6 = 86$

6. $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3] = 35$

13. $[\text{Pt}(\text{en})(\text{dien})\text{NH}_3]^{4+} = 16 + 8$

7. $\text{Co}(\text{gly})_3 = 36$

Sidgwick EAN Rule (as like octet rule) :

Accⁿ to Sidgwick, complexes having EAN equal to next inert gas atomic no. are more stable in comparison of other complexes having EAN not equal to next inert gas A.N.

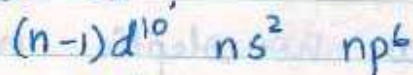
- complex follows EAN rule means, EAN of CMI is equal to next inert gas A.N.,

$$3d \rightarrow 36$$

$$4d \rightarrow 54$$

$$5d \rightarrow 86$$

- EAN rule is also called $18e^-$ rule due to 18 valence shell es^- ,

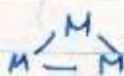


which is confⁿ of next inert gas atomic no.

- This rule is applicable for metal carbonyls & π -complexes.

* Metal Carbonyls :

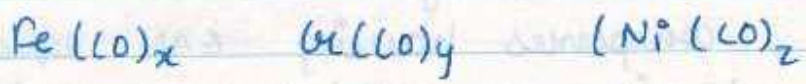
EAN	
Oxidising agent	33 $\rightarrow$ tetramerize
	34 $\rightarrow$ trimerize
	35 $\rightarrow$ dimerize
	36 $\rightarrow$ stable



36 $\rightarrow$ stable

37 $\rightarrow$ Reducing agent

Q. Find value of x, y, z for

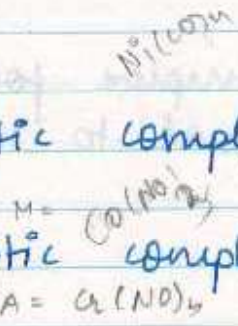


→ $x=5, y=6, z=4$

Q. Find the formula of :

a) lightest stable metal carbonyl homoleptic complex

b) lightest stable metal nitrosyl homoleptic complex.

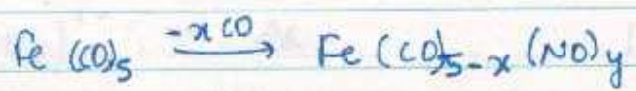


• Nitrosyl complexes are formed by replacement of CO from metal carbonyls.

• $\text{Cr}(\text{NO})_6$ is only mononuclear homoleptic nitrosyl complex of 3d series.

Q. Find the value of $x+y$ (x, y are integers)

$x=3, y=2$

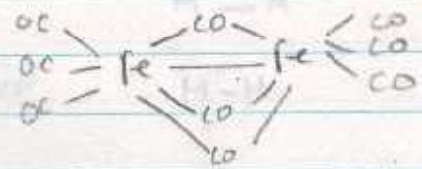


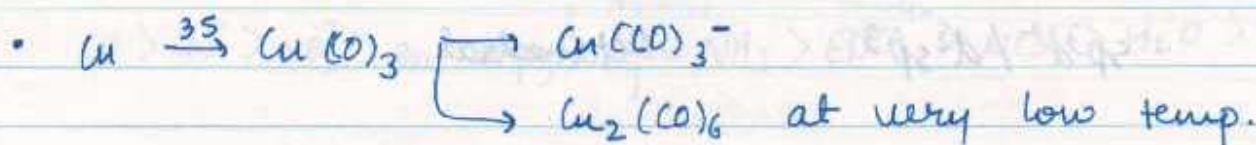
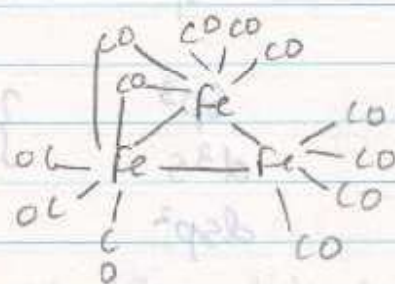
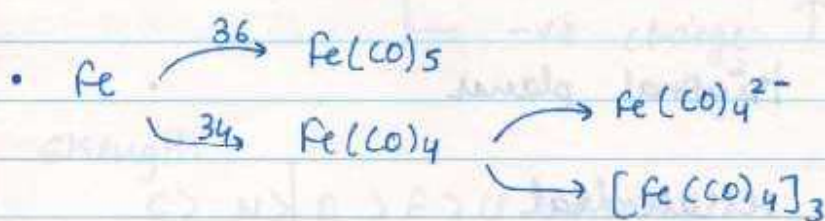
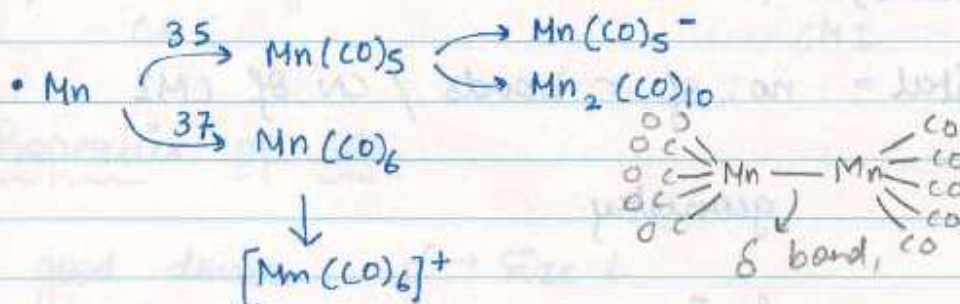
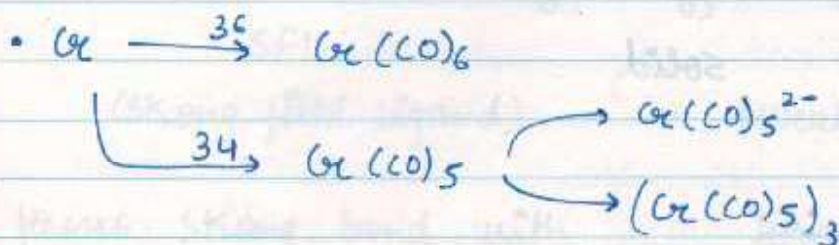
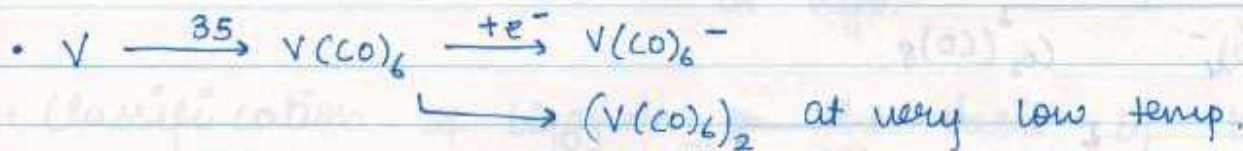
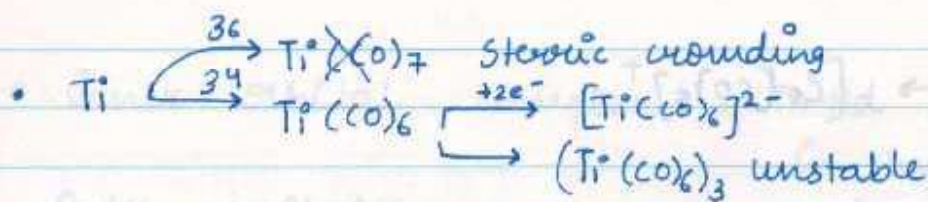
(5)

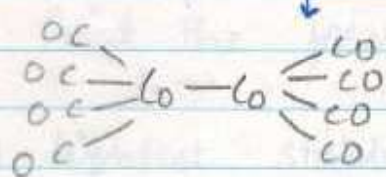
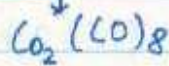
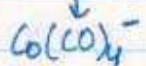
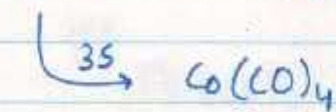
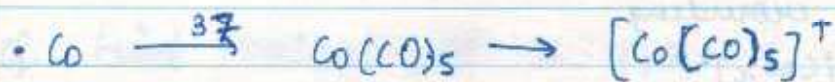
• 3 CO are replaced by 2 NO in formation of nitrosyl complex from metal carbonyls.

Q. Find the value of x for $\text{Fe}_2(\text{CO})_x$

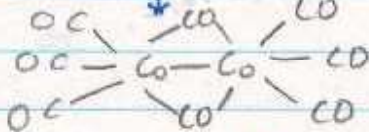
→ 9.







solution



solid

* Valence Bond Theory :

No. of hybrid orbital = no. of σ bonds / CN of CMI

CN	hyb.	geometry
2	sp	linear
3	sp ²	trigonal planar
4	sp ³	} tetrahedral
	d ³ s	
	dsp ²	
		Sq. planar
5	dsp ³ / sp ³ d (dz ²)	TBP
	dsp ³ / sp ³ d (dx ² -y ²)	Sq. pyramidal
6	sp ³ d ² / d ² sp ³	Octahedral

• inner orbital complex = $(n-1)d$ orbital participates in hyb.

Outer orbital complex = when nd orbital participate in hyb.

• Classification of ligands on the basis of bond strength $\therefore$

SFL

(Strong field ligand)

forms strong bond with CMI

WFL

(Weak field ligand)

forms weak bond with CMI

Properties of SFL

good donor

(Lewis B.N.T)

Size $\downarrow$
EN $\downarrow$
-ve charge $\uparrow$

Strength

$C > N > O > F > U > Br > I$

SFL

WFL

* spectrochemical series (Strength of ligands) :

$I^- < Br^- < SCN^- < U^- < S^{2-} < N_3^- / NO_3^- < F^- < OH^- < CH_3COO^-$

$CO > CN^- > NO_2^- > en \approx dipy > py \approx NH_3 > EDTA^{4-} > NCS^- > H_2O > C_2O_4^{2-}$

Trick

इतना बुरा था कालू श्यामू नाई कुले दाय ऐसे उसके पानी
भरा हो जैसे
नवीन चंद्र बसुसेना इन दिनों अमन पाने के लिए double double
नौ सेना कोर्ट के चक्कर लगा रहे हैं।

In presence of SFL, e^- are forced to pair as per requirement of d-orbitals for hybridisation. Due to pairing of e^- value of spin only Magnetic moment $\downarrow$ such complexes are called low spin complexes, in presence of WFL, no change in no. of unpaired e^- so they have high value of spin only magnetic moment in comparison of low spin complexes such complexes are called high spin complexes. (low spin w.r.t ground state)

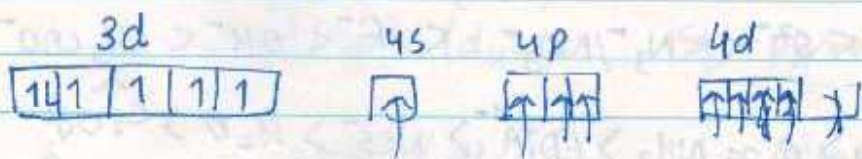
Note : High spin & low spin complexes are defined only when ligand's effect in that configuration occurs.

• $K_4[FeF_6]$

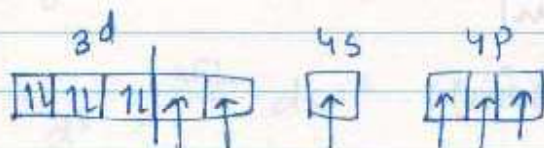
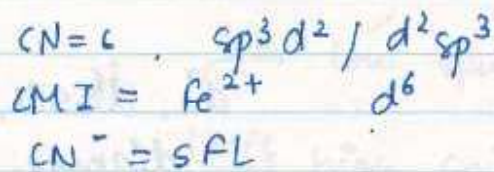
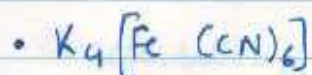
CN = 6, sp^3d^2/d^2sp^3

CMI = $Fe^{2+} d^6$

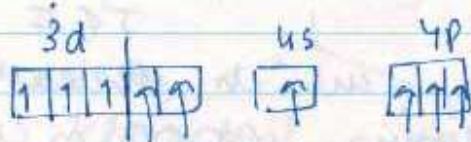
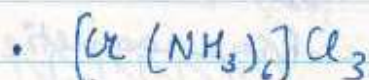
F^- = WFL



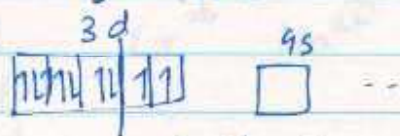
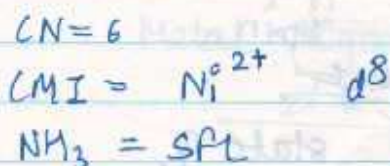
sp^3d^2 octahedral
outer orbital
 $\uparrow$ spin



Inner orbital
 d^2sp^3 octahedral
 diamagnetic
 low spin



d^2sp^3 (para.)



sp^3d^2 (para.)

For $CN = 6$.

 $d^{0,1,2,3}$

d 4, 5, 6, 7

 $d^{8,9,10}$ $d^2 sp^3$

SFL $\rightarrow d^2sp^3$ (low spin)

WFL $\rightarrow$ sp^3d^2 (high spin)

 $sp^3 d_2$

d^0, d^6-SF_6, d^{10}
diamag.

g.

$$d^5 \quad (1) \quad Fe^{3+}$$
 $d^3 \quad G^{3+}$
$$d^6 \quad 6d^3$$
$$d^8 \quad Ni^{2+}$$

(ii)

en SFL

CN⁻ SFL

F- WPL

SN - WFL

(iii)

low spin, Dia $\rightarrow$ paramagnetik
high spin, Para $\rightarrow$ diamagnetik
inner orbital, Dia
outer orbital, Para

(2)

outer orbital, Para

$$I \neq \Pi$$

Find total no. of combination in which ~~at least~~
~~three~~ ~~three~~ columns are used containing ~~(I & II)~~ in CN 6
species

[illegible]

16

$$g. d^8 NiCl_4^{2-}$$
 sp^3
$$d^8 \text{ Ni}^{2+} (\text{CN})_4^{2-}$$
$$sp^3 \quad dsp^2$$
$$d^{10} \text{ Ni}^+ (\text{CO})_4$$

Sp

рана,

high spin

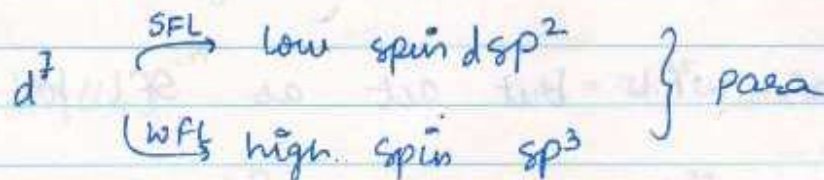
di ~~pot~~

low spin

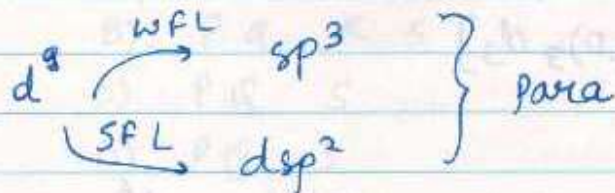
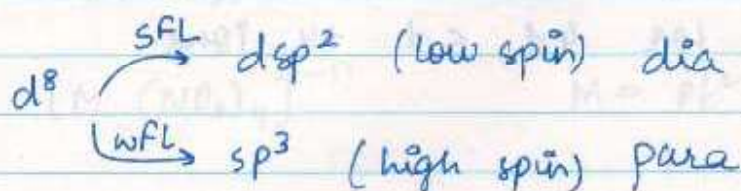
di

w.r.t. d^8s^2 { ground state }

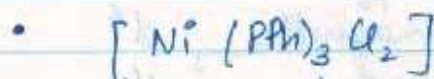
conclusion,
CN = 4



} para

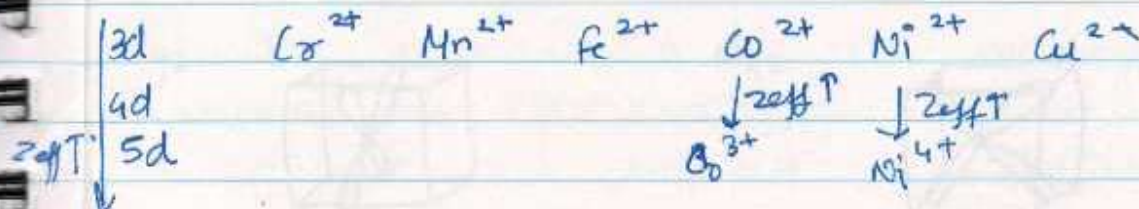


} para



tetrahedral (paramagnetic)
 sq. planar (diamagnetic)

As the charge on CMI $\uparrow$
 or Z_{eff} of CMI $\uparrow$
 Metal-ligand bond strength $\uparrow$
 $d e^- \uparrow$ $Z_{eff} \uparrow$



1) generally NH_3 acts as SFL but for Cr^{2+} , Mn^{2+} , Fe^{2+} , Co^{2+} it acts as WFL

2) gen. 'O' donors are WFL but act as SFL for Co^{3+} , Cu^{2+}

3) All ligands are SFL for 4d, 5d +4 ions

4) If $\text{WFL} \leq \text{SFL}$
always consider effect of SFL
except $[\text{Co}(\text{H}_2\text{O})_3\text{F}_3]$, $[\text{Co}(\text{H}_2\text{O})_3\text{Cl}_3]$

Q.		If $M =$	
A) $[\text{MCl}_6]^{-x}$	WFL	P) Fe^{2+}	Outer orbital d^6
B) $[\text{M}(\text{H}_2\text{O})_6]^{+x}$	WFL or (S,T)	Q) Cr^{3+}	Inner orbital d^3
C) $[\text{M}(\text{C}_2\text{O}_4)_3]^{-x}$	WFL (S,T)	R) Fe^{3+}	Paramagnetic d^5
D) $[\text{M}(\text{NH}_3)_6]^{+x}$	SFL or	S) Co^{3+}	Diamagnetic d^6
E) $[\text{M}(\text{en})_3]^{+x}$	SFL	T) Pt^{4+}	low spin d^6
F) $[\text{M}(\text{CN})_6]^{-x}$	SFL	U) Mn^{2+}	high spin d^5

→ A) P, Q, R, T, U

B) P, Q, R, U, T, S

C) P, Q, R, S, T, U

D) P, Q, R, T, U

E) x, Q, R, T, S

F) x, Q, R, T, S

8.

$[M(C_2O_4)_2]^{-n}$	$M = Cu^{2+}$, Para	d^9	Always
$[MCl_4]^{-n}$	$M = Zn^{2+}$, dia.	d^{10}	Always
$[M(en)_2]^{+n}$	$M = Ni^{0+}$, tetrahedral	d^8	sp^3 (WFL)
$[M(NO_2)_4]^{-n}$	$M = Pt^{2+}$, Sq. planar	d^8	dsp^2 (SFL)

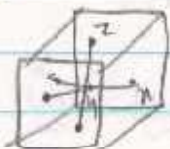
- A) P Q R S
 B) P Q R S
 C) P Q S
 D) P Q S

Types of d-orbitals

↓
 along the axis
 $d_{x^2-y^2}$, d_{z^2}
 at face centre

↓
 b/w the axis
 d_{xy} , d_{yz} , d_{zx}
 at edge centre

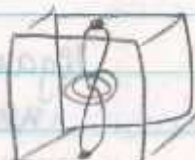
let



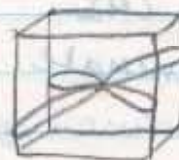
$d_{x^2-y^2}$



d_{z^2}



d_{xy}



d_{yz}



d_{xz}



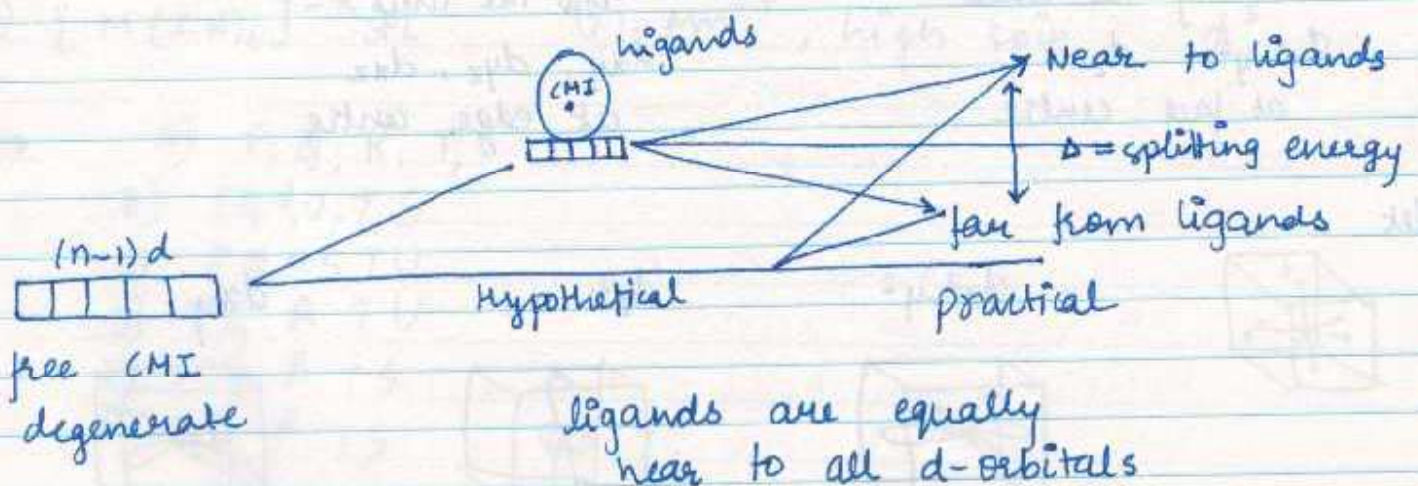
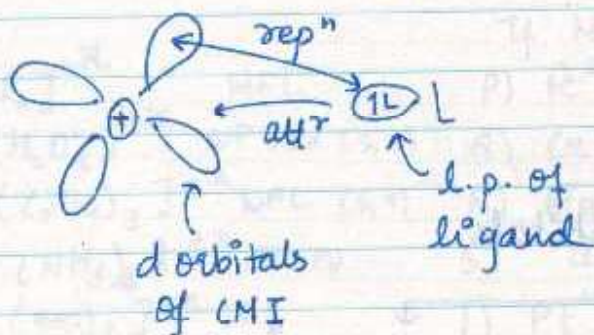
* Crystal Field Theory (CFT) :

Acc. to CFT,

CMI = point charge

ligands $\begin{cases} \rightarrow \text{Neutral} = \text{dipole} \\ \rightarrow \text{Anionic} = -ve \text{ point charge} \end{cases}$

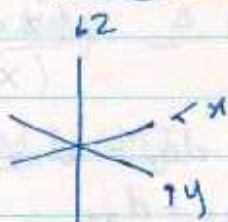
Interaction b/w ligand & CMI = electrostatic



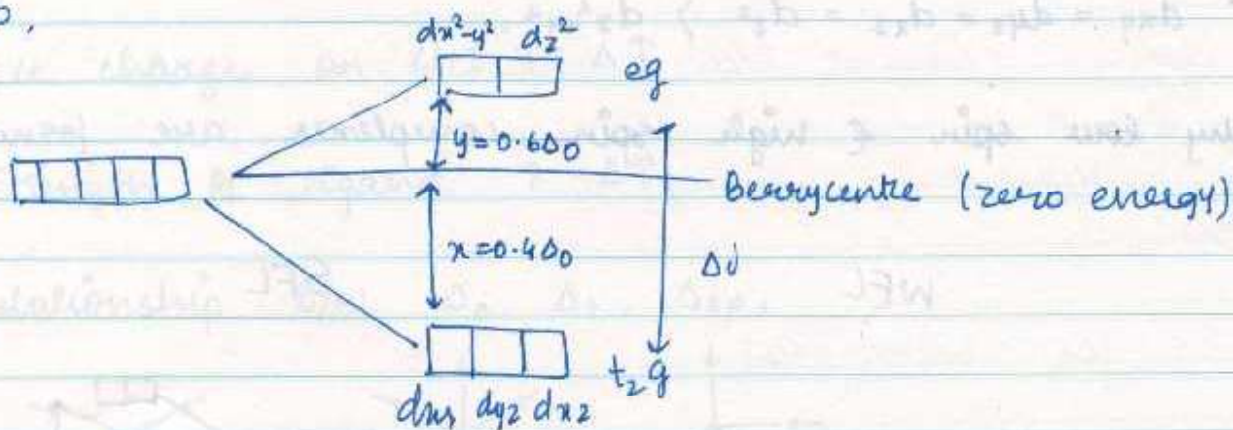
- When ligand approaches CMI in a particular geometry some of the d-orbital exp. more repⁿ as compared to other d-orbitals coz they are near to ligands

as compared to other d-orbitals. So, d-orbitals are split into diff. energy set, loss of degeneracy of d-orbitals is called crystal field splitting & energy diff. b/w them is called splitting energy (Δ).

- Splitting in Octahedral :

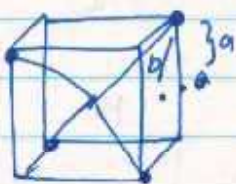


ligands are near to axial d-orbitals and far from non-axial d-orbitals



$$d_1^2 = d_{x^2-y^2} \rightarrow dx_1 = dy_2 = dz_2$$

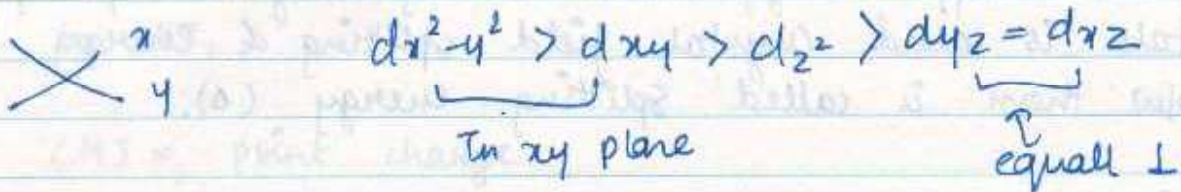
- in tetrahedral :



b) a ligands are near to non axial d-orbitals

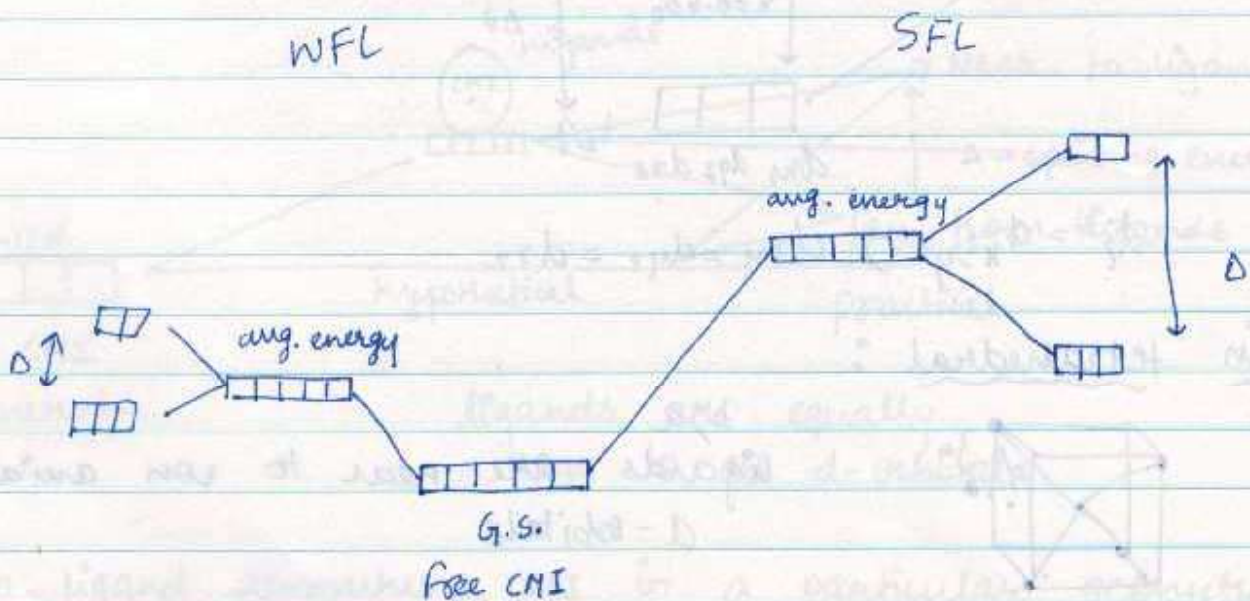
$$dm = dy_2 = dx_2 \Rightarrow dx_2^2 - dy_2^2 = dz^2$$

• in sq. planar:



- $sp \Rightarrow d_{z^2} > d_{yz} = d_{xz} > d_{x^2-y^2} = d_{xy}$ (z axis)
- $sp^2 \Rightarrow d_{x^2-y^2} = d_{xy} > d_{z^2} > d_{yz} = d_{xz}$ (xy plane)
- dsp^3 (TBP) $\Rightarrow d_{z^2} > d_{x^2-y^2} = d_{xy} > d_{yz} = d_{xz}$
- dsp^3 (S.P) $\Rightarrow d_{x^2-y^2} > d_{z^2} > d_{xy} > d_{yz} = d_{xz}$
- square antiprismatic $\Rightarrow d_{z^2} \neq d_{x^2-y^2} > d_{xy} \neq d_{yz} \neq d_{xz}$
- Dodecahedron $\Rightarrow d_{xy} \neq d_{yz} \neq d_{xz} \neq d_{x^2-y^2} > d_{z^2}$
- $d_{xy} = d_{yz} = d_{xz} = d_{x^2-y^2} > d_{z^2}$
- $d_{xy} = d_{yz} = d_{xz} = d_{z^2} > d_{x^2-y^2}$

Q. Why low spin & high spin complexes are formed?



In presence of SFL $\Delta >$ Pairing energy
 WFL $\Delta <$ Pairing energy.

from d^n conf. energy is required to fill e^- either in form of Δ or in form of pairing energy.

• Factors affecting Δ (splitting energy) :-

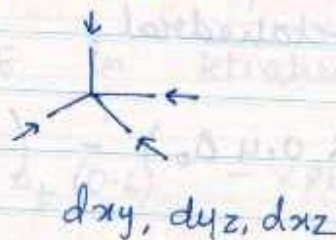
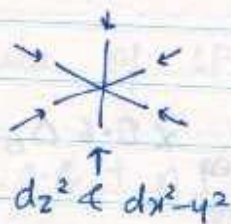
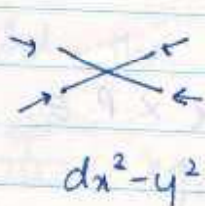
(A) As Z_{eff} of CMI $\uparrow$

{ distance b/w ligands $\propto$ CMI $\downarrow$
 keepⁿ b/w d-orbitals of CMI $\propto$ ligands $\uparrow$
 $\Delta \uparrow$ }

(B) +ve charge on CMI $\uparrow \Delta \uparrow$

(C) Strength of ligand $\uparrow \Delta \uparrow$

* Relationship b/w Δ_o , Δ_t , Δ_{sp} ,



Orbital 1

2

3

Ligands 4

6

4 (not in direcⁿ of orbitals)

$$\Delta_{sp} > \Delta_o > \Delta_t$$

$$\# \Delta_{sp} = \frac{4}{3} \Delta_0$$

$$\# \Delta_t \approx \frac{4}{9} \Delta_0$$

Application of CFT:

1] To find stability of complexes $\rightarrow$

$$\Delta G = \Delta H - T \Delta S$$

$\uparrow$ $\downarrow$
 Bond strength Chelation.
 (Δ)/CFSE

* C.F.S.E (Crystal field stabilization Energy) :

Decrement in energy due to crystal field splitting

CFSE for octahedral,

$$(\overset{\substack{\uparrow \\ e^- \text{ in } t_{2g}}}{n_{t_{2g}}} \times 0.4 \Delta_0) - (\overset{\substack{\uparrow \\ e^- \text{ in } e_g}}{n_{e_g}} \times 0.6 \Delta_0) - x \times P.E.$$

$\uparrow$
no. of extra pair

Q. Find CFSE value & confⁿ of (n-1)d orbitals in form of t_{2g} & e_g in coordination 6 in presence of SFL as well as WFL.

	t_{2g}	SFL e_g		t_{2g}	WFL e_g	
d^0	0	0		0	0	= 0
d^1	1	0		1	0	= $0.4\Delta_0$
d^2	2	0		2	0	= $0.8\Delta_0$
d^3	3	0		3	0	= $1.2\Delta_0$
d^4	4	0		3	1	= $1.6\Delta_0 - 1\text{PE}$
d^5	5	0		3	2	= $2.0\Delta_0 - 2\text{PE}$
d^6	6	0		4	2	= $2.4\Delta_0 - 3\text{PE}$
d^7	6	1		5	2	= $1.8\Delta_0 - 3\text{PE} + 0.4\Delta_0$
d^8	6	2		6	2	= $1.2\Delta_0 - 3\text{PE} + 0.8\Delta_0$
d^9	6	3		6	3	= $0.6\Delta_0 - 3\text{PE} + 1.2\Delta_0$
d^{10}	6	4		6	4	= $0\Delta_0 - 3\text{PE} + 0.6\Delta_0$

$\Delta_0 = 10 Dq$
↓
Dequantita.

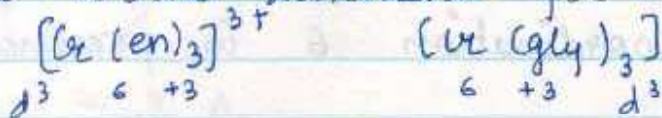
sp^3

Q. Find the formula of CFSE in tetrahedral (filled).

→ $\boxed{\uparrow\uparrow\uparrow\uparrow}^{sp^3} - \eta_{t_{2g}} \times \Delta_t \times 0.4 + \eta_{e_g} \times \Delta_t \times (0.6) - \times \text{PE}$

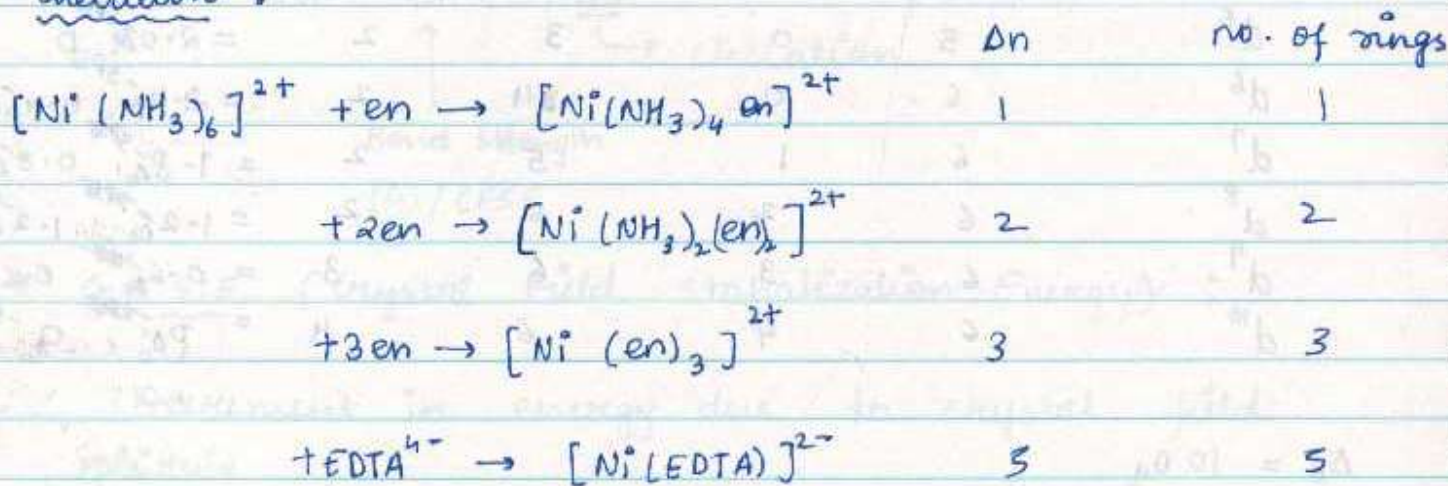
Q.

Q. Select correct statements for



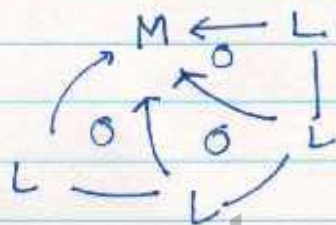
- A) Both are inner orbital complex
- B) Both are low spin complex
- ✓ C) Both are paramagnetic
- D) Both have same confⁿ of CMI
- E) Both have same CFSE

* Chelation :



$\Delta n \uparrow \quad \Delta s \uparrow \quad \text{TDs} \uparrow \quad \Delta O \downarrow$
Stability $\uparrow$

$\Delta n \propto \text{no. of rings}$
no. of rings $\uparrow$ chelation $\uparrow$ stability $\uparrow$



No. of rings formed by a ligand = Denticity - 1

* Priority order for comparison of stability of complexes for same CMI, same C.N. & diff. ligands :

⇒ C-donor > chelation
for same no. of ring(s)
Stability $\propto \Delta$ (CFSE).

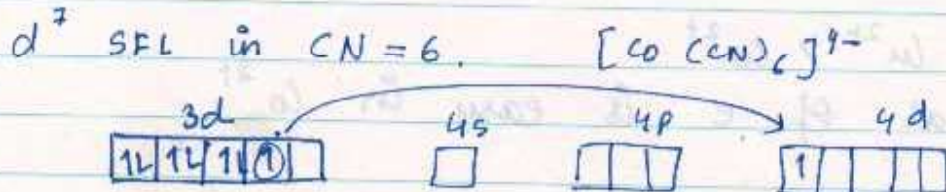
Q. Stability order :

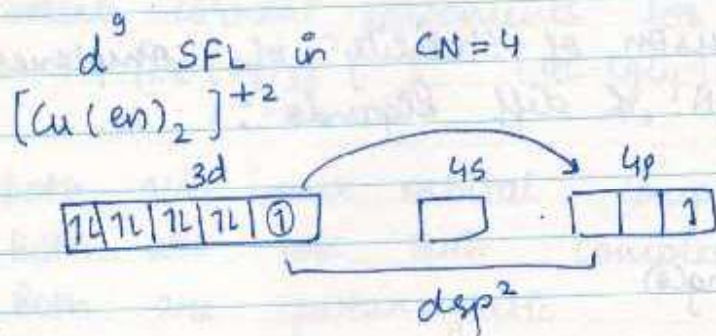
- 1) PtCl_4^{2-} PdCl_4^{2-} NiCl_4^{2-} $a > b > c$ $\{Z_{\text{eff}}\}$
- 2) $\text{K}_4\text{Fe}(\text{CN})_6$ $\text{K}_3\text{Fe}(\text{CN})_6$ $b > a$ $\{+ve \text{ charge on CMI}\}$
- 3) $\text{Ni}(\text{dmg})_2$ $[\text{Ni}(\text{en})_2]^{2+}$ $a > b$ $\{H\text{-bonding}\}$
- 4) $\text{Cr}(\text{CN})_6^{3-}$, $\text{Cr}(\text{C}_2\text{O}_4)_3^{3-}$ $a) > c) > b) > d) > e)$ $\{c > \text{chelate if no chelate} > N > O > F\}$
 $\text{Cr}(\text{en})_3^{3+}$ $[\text{Cr}(\text{NH}_3)_6]^{3+}$
 $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$
- 5) $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ $[\text{Fe}(\text{acac})_3]$ $b > a$ $\{\text{no. of atoms in a ring}\}$

* Interference of electrons :-

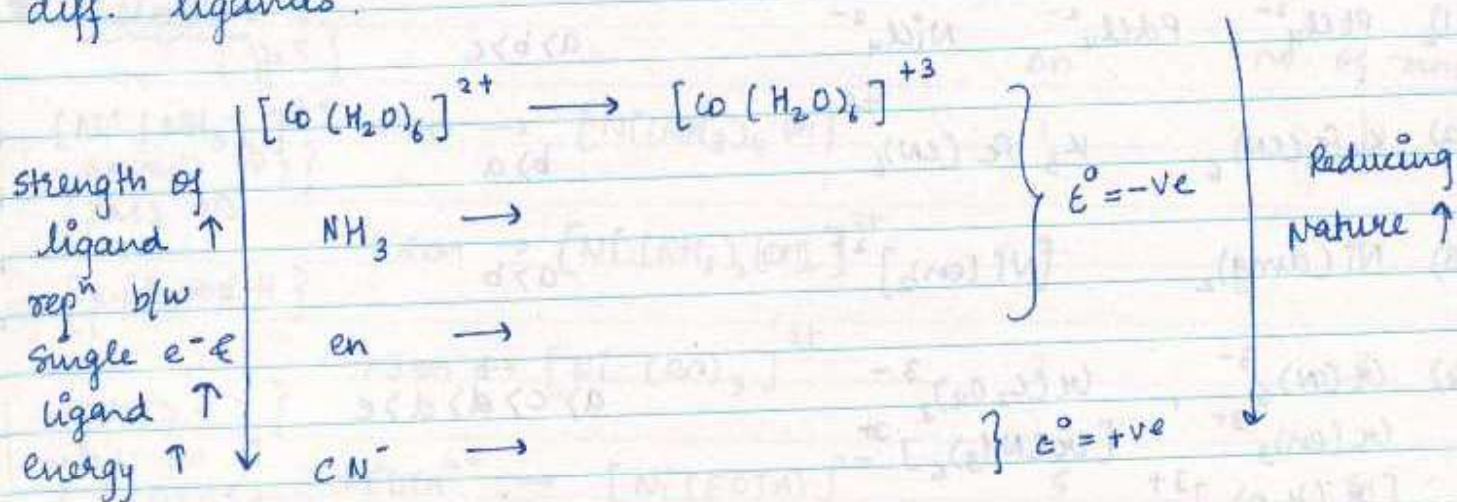
In presence of strong field ligand when single e^- is present in highest energy $(n-1)d$ orbital, excitation of e^- takes place.

eg.





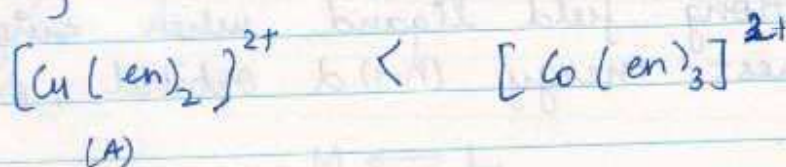
With the help of transference we can compare reducing nature of Co^{2+} in presence of different diff. ligands.



$$\Delta G = -nFE$$

$\uparrow$
electrode potential

Reducing nature of



1) Z_{eff} of $Cu^{2+} > Co^{2+}$
 removal of e^- is easy in Co^{2+}

2) in complex A outermost e^- present in $4p$ while in complex B outermost e^- present in $4d$
 distance from nucleus
 $4p < 4d$

* Jahn Teller distortion $\Rightarrow$

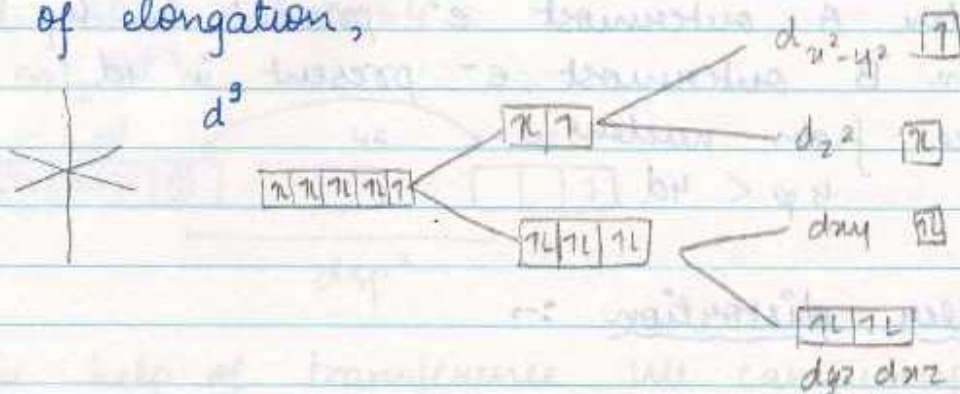
When t_{2g} & e_g subsets (set of orbitals) are unsymmetrically filled, octahedral complex is no more symmetrical. This is called Jahn Teller distortion.

When t_{2g} set of orbitals are unsymmetrically filled distortion is very weak but when e_g orbitals are unsymmetrically filled, strong distortion is observed.

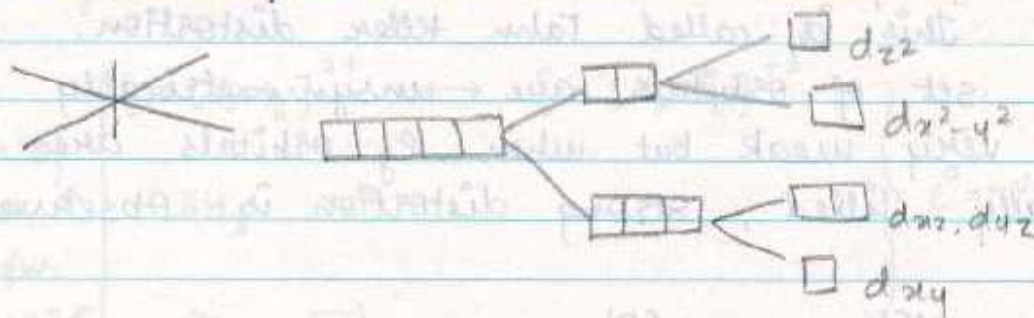
	WFL	SFL
d^0	X	X
d^1	W	W
d^2	W	W
d^3	X	X
d^4	S	W
d^5	X	W
d^6	W	X
d^7	W	S
d^8	X	X
d^9	S	S
d^{10}	X	X

$[Cu(H_2O)_6]^{2+}$
 all Cu-O bond
 length are not
 identical,

In case of elongation,



In case of compression,



In CN=5,

In case of SFL (except CO) $\rightarrow$ sq. pyramidal

In case of WFL & CO $\rightarrow$ TBP.

* Isomerism $\Rightarrow$ structural & stereo

Structural

- 1) ionization
- 2) hydrate
- 3) linkage
- 4) co-ordination
- 5) co-ordination position
- 6) ligand
- 7) polymerisation

stereo

- 1) geometrical
- 2) optical
- 3) Allotropy isomerism.

• Stereo :

$$CMI = M$$

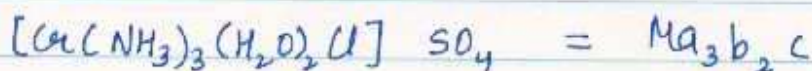
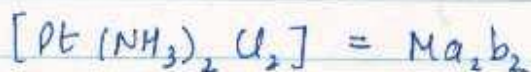
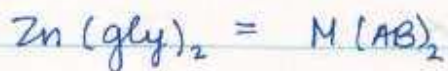
Monodentate ligands = a, b, c, d, e, ...

Symm. bidentate = AA, BB, CC, ...

($C_2O_4^{2-}$, en, dipy, acac⁻)

unsymm. bidentate = AB, AC, BC, ...

(gly⁻, bcac⁻)



1. geometrical : diff. diff. Bond angles

2. Optical : POS $\times$

CN=2
linear

Mab

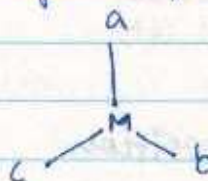


POS $\checkmark$ OI $\times$

Only one type
of BA GI $\times$

CN=3

Trigonal planar



GI $\times$

POS $\checkmark$, OI $\times$

CN=4

tetrahedral



Only one type
of BA GI $\times$
Can show OI

Sq. planar



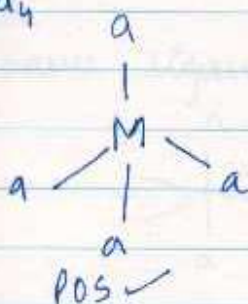
POS $\checkmark$ OI $\times$

two types of BA GI $\checkmark$

• OI in tetrahedral

Ma_4

*

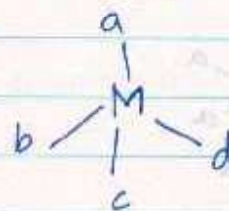


* Ma_3b (POS ✓)

* Ma_2b_2 (POS ✓)

* Ma_2bc (POS ✓)

* $Maabcd$



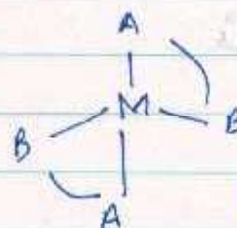
POS X OI ✓
OI = 2

* $M(AA)_2$, $M(AA)b_2$, $M(AA)bc$
 $M(AA)(AB)$



POS ✓ OI X

* $M(AB)_2$



POS X OI ✓
OI = 2

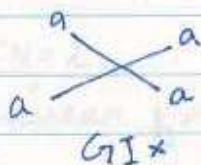
* $M(AB)(CD)$ OI ✓

* $M(AB)C_2$ OI X

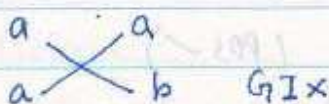
* $M(AB)cd$ OI ✓

• GI in Sq. planar

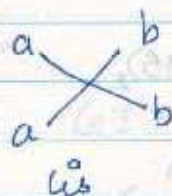
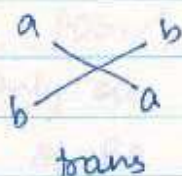
Ma_4



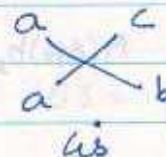
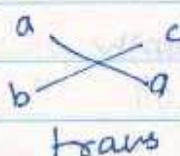
Ma_3b



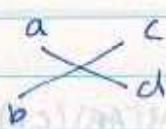
Ma_2b_2



Ma_2bc

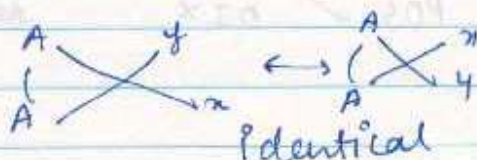


$Maabcd$

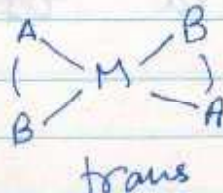
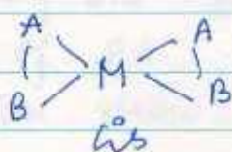


③ 2 cis, 1 trans

$M(AA)_2, M(AA)b_2, M(AA)(BC), M(AA)bc$



$M(AB)_2$ GI=2



$M(AB)(CD)$ GI=2

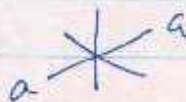
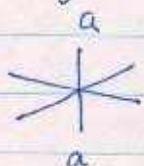
$M(AB)cd$ GI=2

$M(AB)_2$ GI x

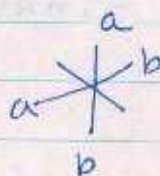
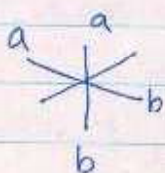
• C.N. = 6

PDS

1) same ligands present at trans

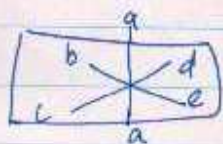


2) two trans are identical



Ma_2bcde

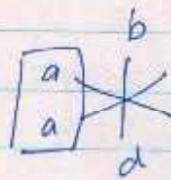
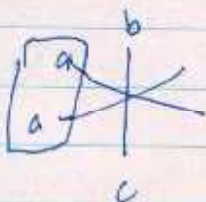
(a,a) trans



$$\Rightarrow GI = 3$$

$$OA = 0$$

(a,a) cis



...

$\Rightarrow$

$$GI = 6$$

$$OA = 12$$

$$GI = 9$$

$$OI = 15$$

$$SI = 15$$

$M_{(AA)} bcde$

$$GI = 6 ; OA = 12 ; OI = 12 ; SI = 12$$

consider only (a,a) is of $M_a, bcde$

$M_{(AF)} bcde$

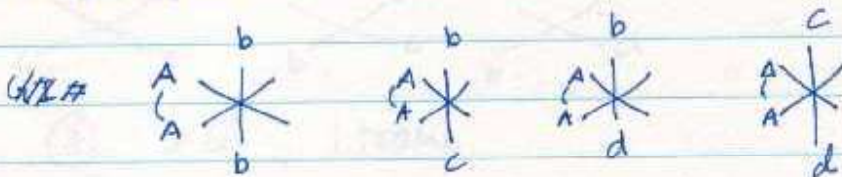
$$GI = 12 ; OI = 24 ; SI = 24$$

(24+0)

$M_{abc} def$

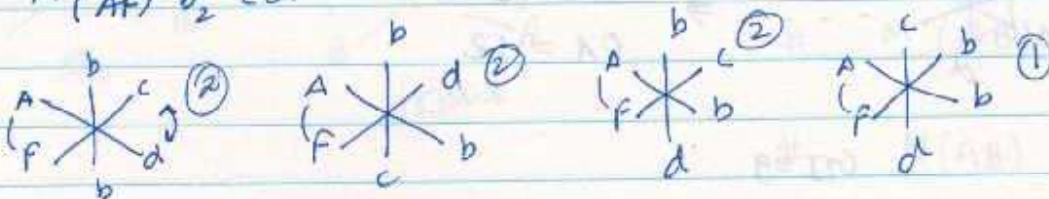
$$GI = 15 ; OI = 30 ; SI = 30$$

$M_{(AA)} b_2 cd$



$$GI = 4 \quad OA = 8$$

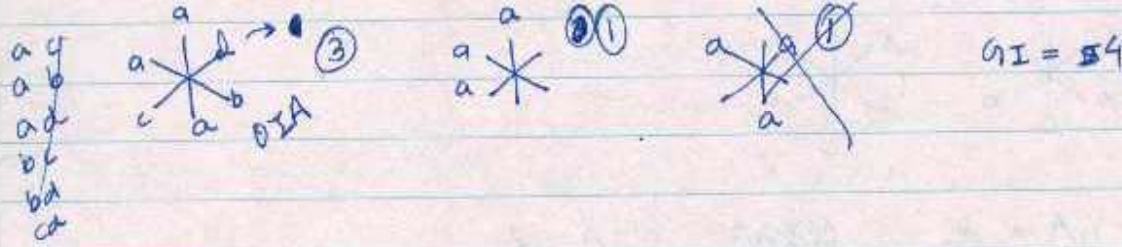
$M_{(AF)} b_2 cd$



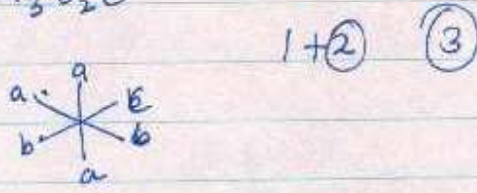
$$GI = 7$$

$$OI = 10$$

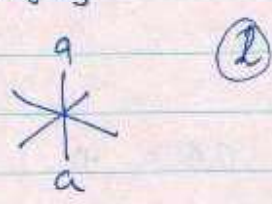
Ma_3bcd



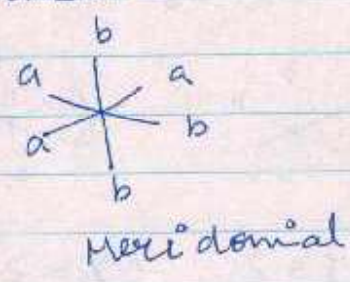
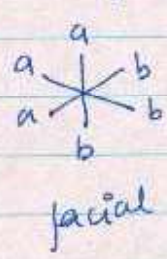
Ma_3b_2c



Ma_3b_3

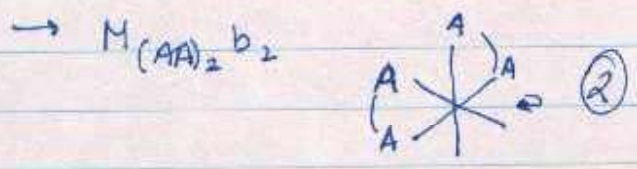


fac-mer isomerism

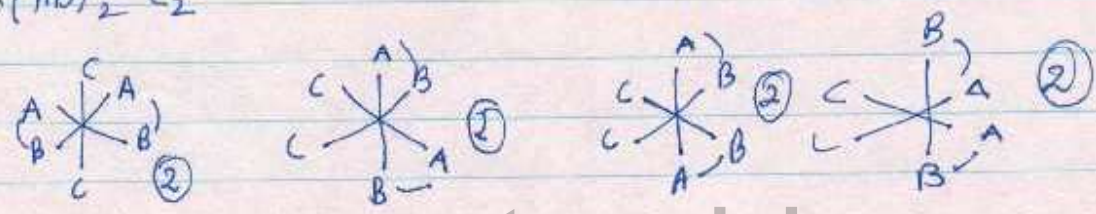


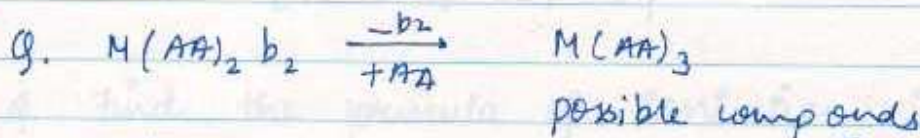
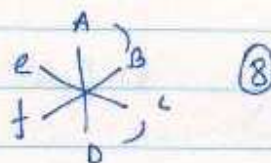
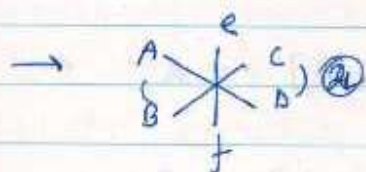
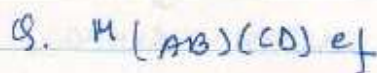
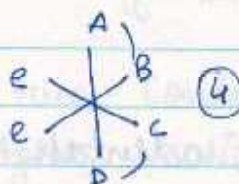
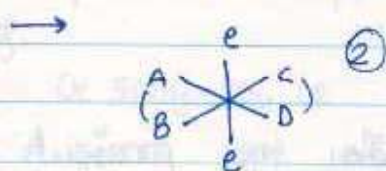
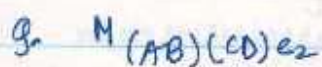
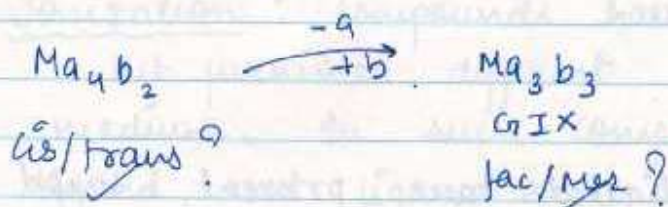
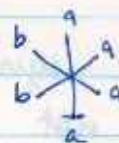
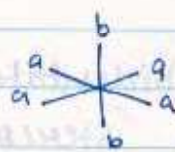
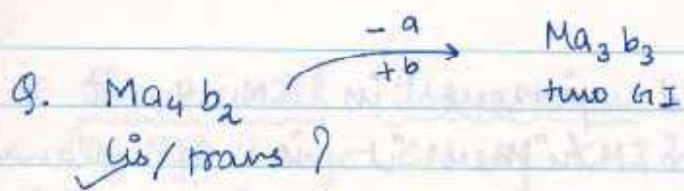
Q $M(AA)_4$ $\xrightarrow[+AA]{-b_2}$ compd Z

Find total no. of G.I. of Z



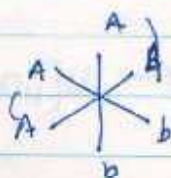
$M(AB)_2C_2$



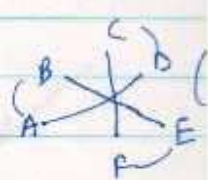
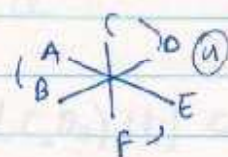
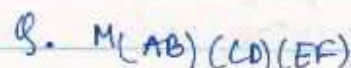
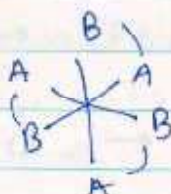
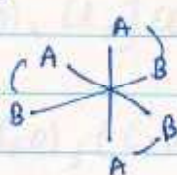
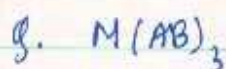


which
isomer

(1) (DA)



only cis



- when symm. bidentate ligand present in CN 4 it does not show stereoisomerism. either it is sq. planar or tetrahedral.
- In CN 6,
 - i) when two bidentate ligands are present, compd (cis) is always optically active.
 - ii) when three bidentate ligands are present compd is always OA
 - iii) when 3 identical monodentate ligands are present only 1 cis isomer exists.

* Structural Isomerism :-

[Do NOT change CN of CMI]

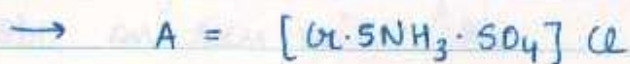
- 1) Ionization : compounds having same molecular formula but furnishes different diff. counter ions in aq. medium. In such complexes counter ion can act as ligand and ligand as counter ion.

They can be distinguished by conductivity measurement, qualitative analysis.



Q.

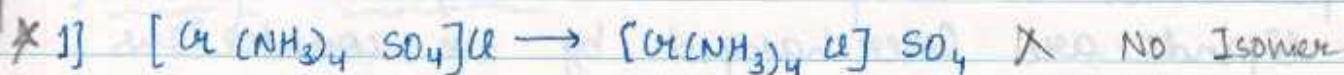
$Cr \cdot 5NH_3 \cdot SO_4 \cdot Cl$ has two ionisation isomers A & B. A gives ppt with $AgNO_3$ solⁿ while B gives ppt with $BaCl_2$ solⁿ. Identify A & B & Arrange them in correct order of conductivity.



$B > A$

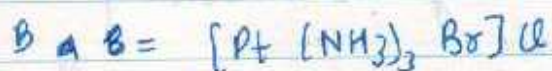
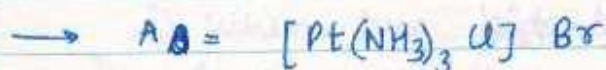


Q. Find the formula of ionisation isomers of given complexes.





8. Find the structural formula of A & B having same M.F. $\text{Pt} \cdot 3\text{NH}_3 \cdot \text{Cl} \cdot \text{Br}$. If $\Delta A > \Delta B$



Also write down the observation when A & B reacts with excess AgNO_3 solⁿ.

A = Pale yellow

B = white

2] Hydrate Isomer :

This type of isomerism arises due to presence of H_2O in complex compound.

When no. of water molecules inside & outside the coordination sphere are different then compounds are hydrate isomers of each other.

- Complex must be complex cation & simple anion.
- H_2O ligands are interchanged by -ve counterions

They can be distinguished by

- i) Conductivity measurement
- ii) Quantitative analysis
- iii) Weight loss measurement on rxn with conc. H_2SO_4 (removal of uncoordinated water)
- iv) Δ splitting / stability of complex

$\frac{2-4}{\sqrt{15}}$

- A) $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$
 - B) $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$
 - C) $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$
 - ~~$[\text{Cr}(\text{H}_2\text{O})_3\text{Cl}_3] \cdot 3\text{H}_2\text{O}$~~
- } Hydrate Isomers

Conductivity order : $A > B > C$

Moles of AgCl obt. when : 3, 2, 1
1 mol of each reacts with excess AgNO_3

Order of M. Wt. of product : $A > B > C$
obt. on rxn with conc. H_2SO_4

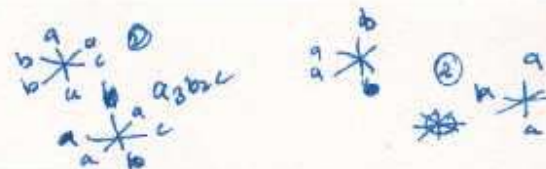
Stability order : $A > B > C$

3] linkage isomerism :

Complexes having same MF also having ligands having same MF but co-ordinated by diff. diff. donor site with CMI are linkage isomers of each other. This type of isomerism arises when complex contains ambidentate ligand.

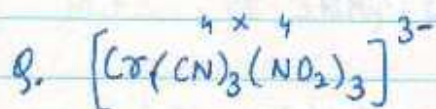
eg,



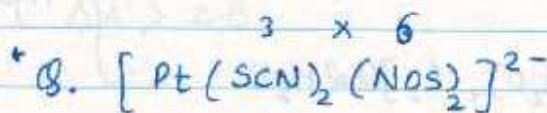


Q. Find total no. of linkage isomers of $[Pt(CN)_4]^{2-}$.

→ 4 (5)



→ 16

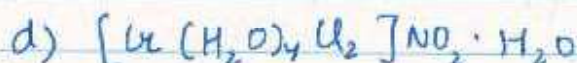
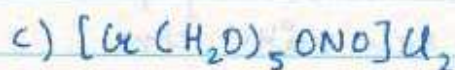
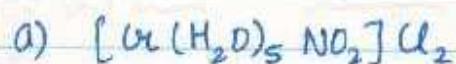


→ 18

Q. Find total no. of isomers of $[Cr(NH_3)_3(CN)_3]$

→ 10

Q. Find the isomerism present b/w given pair of complexes



→ ab → hydrate, ~~isomerism~~

ac → linkage

ad → hydrate, I

bc → linkage, ~~I~~, H

bd → I

cd → H, I

Note : I when counter "ions" get different

$$7 + 12 + 15 + 15 + 12 + 7$$

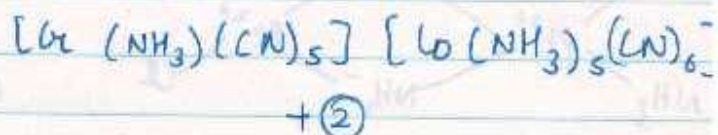
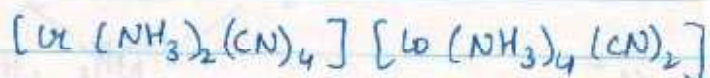
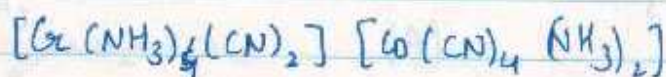
4) Co-ordination Isomerism :

This type of isomerism arises when compound contains complex cation as well as complex anion.

Compounds having same MF but ligand or grp of ligands are present in diff. diff. co-ordination sphere, are co-ordination isomers of each other.



⑥



linkage + CN
↳ ⑥

Q. Find total no. of C.N. isomers of this $[Co(NH_3)_6]^{+3} [Co(C_2O_4)_3]^{+3}$

→ ①

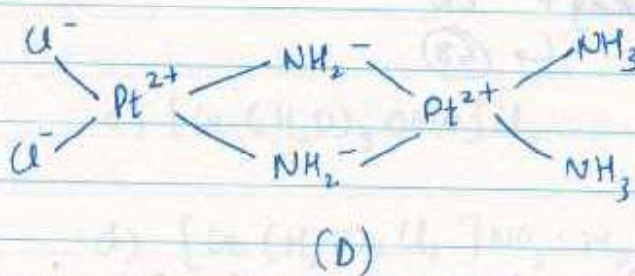
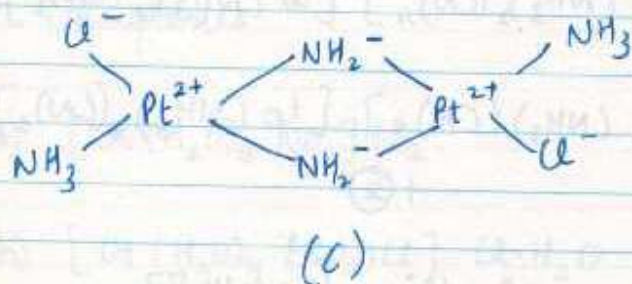
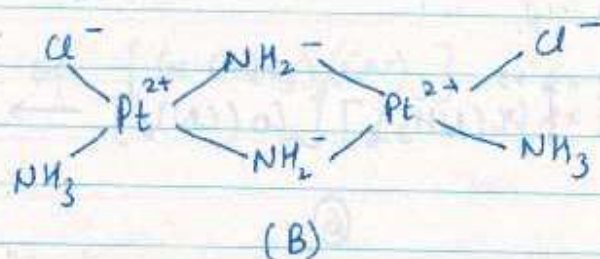
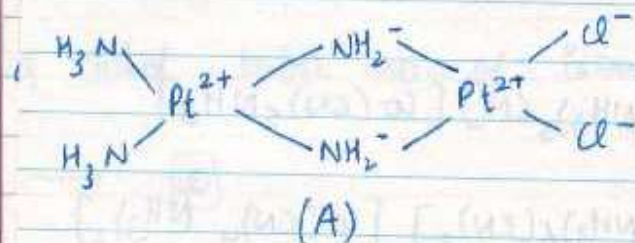
Q. $[Pt(NH_3)_4]^{+2} [PtCl_4(SCN)_2]^{+2}$ C.N. isomers.

→ 7

5) Co-ordination position :

This arises in binuclear complexes. Binuclear complexes having same MF but ligand or group of ligands are co-ordinated with diff. diff metal ion are co-ordination position isomers of each other.

eg,



AB) CPI

AC) CPI

AD) identical

BC) CI

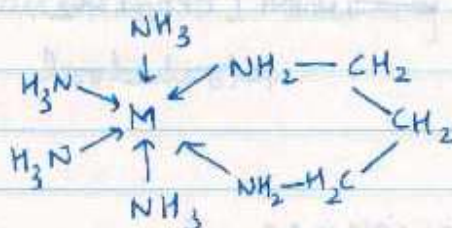
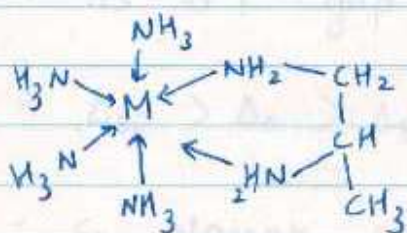
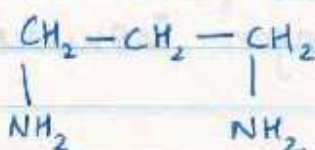
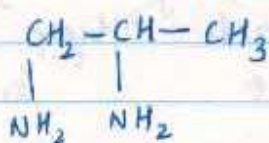
BD) CPI

DC) CPI

6) Ligand :

When complex contains ligand which can show position isomerism. then complex can show ligand isomerism.

eg,



Tetraammine - (1,2-diamine propane) M.

7) Polymerisation :

This is not true isomerism. complexes having diff. MF but same empirical formula are polymer isomers of each other.

	Pt	NH ₃	Cl
[Pt(NH ₃) ₂ Cl ₂]	1	2	2
[Pt(NH ₃) ₄][PtCl ₄]	2	4	4
[Pt(NH ₃) ₃ Cl] ₂ [PtCl ₄]	3	6	6

} PI

When a neutral complex contains, neutral ligand as well as anionic ligand it can show polymerisation Isomerism.

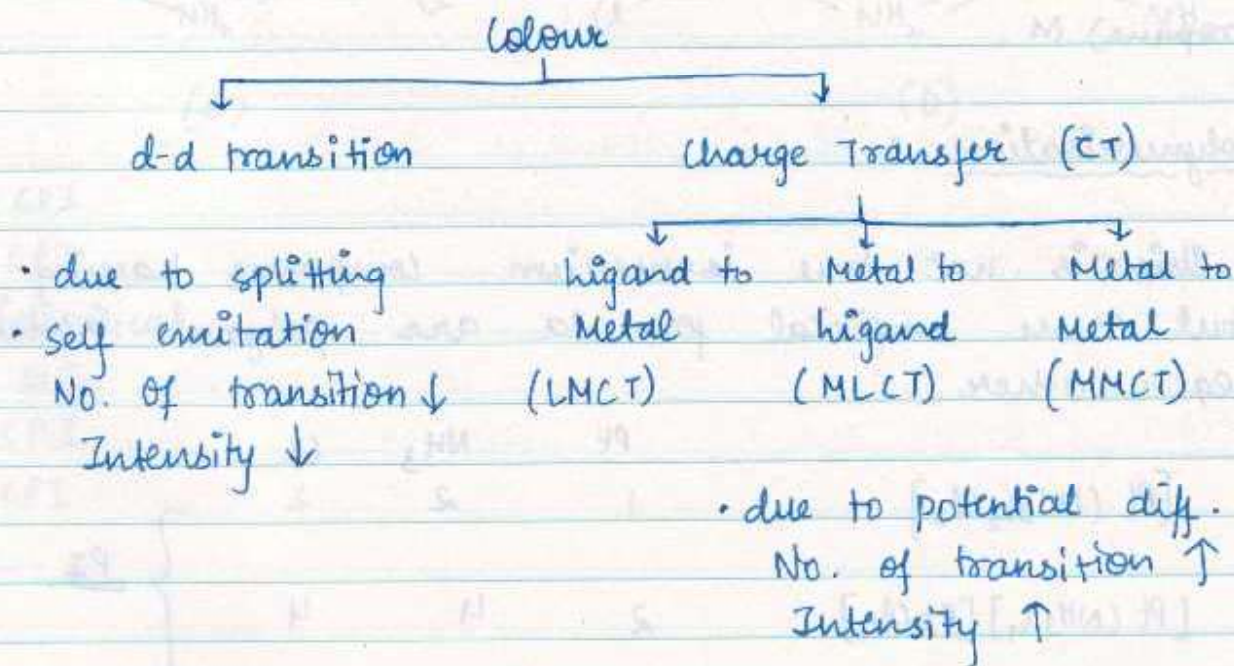
* Allophen Isomerism :

Complexes having same MF but different geometry around CMI are allophen isomers of each other.

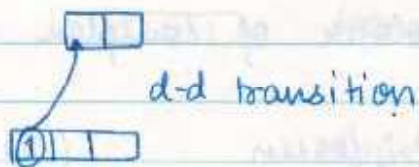
eg,



* COLOUR OF COMPLEXES :



* d-d transition



⇒ d^0, d^8, d^5 - WFL (d^5 -WFL symmetrically filled)

⇒ d^{1-9} ⇒ $\Delta = \frac{hc}{\lambda_{ab}}$

⇒ As $\Delta \uparrow$ gap $\uparrow$ transition $\downarrow$ intensity $\downarrow$

⇒ $\Delta_{sp} > \Delta_o > \Delta_t$

• Sq planar complexes are generally colourless or light yellow.

Tetrahedral complexes are more intense in comparison of octahedral

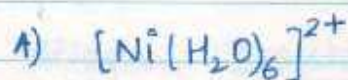
* $CoCl_4^{2-} \rightarrow$ intense blue
 $[Co(H_2O)_6]^{2+} \rightarrow$ Pink

• Octahedral cyano complexes are yellow in colour.

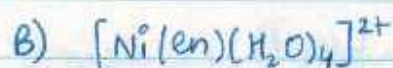
Q. Match them correctly

Complex

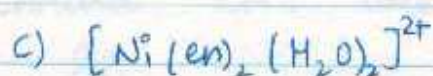
Colour of complex



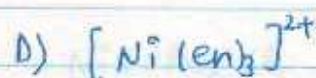
P) Green (R)



Q) Blue-green



R) Blue (B)



S) Violet (Y)

→ A - P, B - Q, C - R, D - S

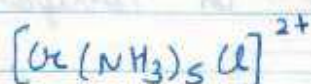
VIBGYOR

$\lambda_{\text{abs}} \Rightarrow R > Q > P > S$

Q. Complex

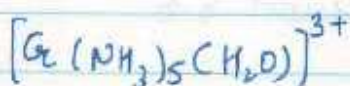
Colour

$\Delta T \quad \lambda_{\text{abs}} \downarrow$



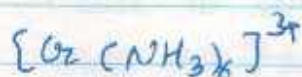
→

Yellow



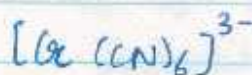
→

Green



→

Blue



→

Violet

→ For abs. colour,

A - S

B - R

C - Q

D - P

d^3 



$d-d$
transition

excited state





No change in spin
(extra pairing)

Spin allowed

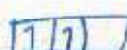
d^5 



~~$d-d$~~
transition

excited state





change in spin
(extra pairing ✓)

highly unstable

Spin forbidden

(Spin not allowed)

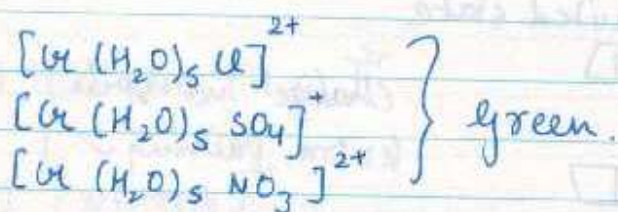
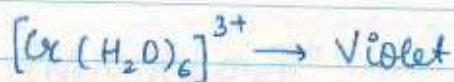
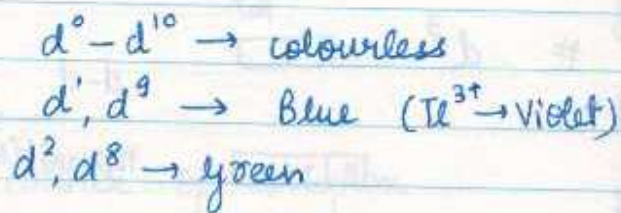
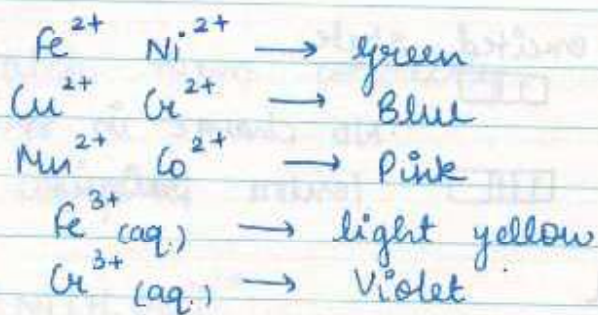
- Emerald green is a mineral beryl - $\text{Be}_3\text{Al}_2\text{Si}_6\text{O}_{18}$ cyclic silicate, is coloured due to presence of impurity Cr^{3+} in presence of Al^{3+} at octahedral site.

- Ruby (Red) is an impure Al_2O_3 . It contains impurity of Cr^{3+} which is the reason of colour.

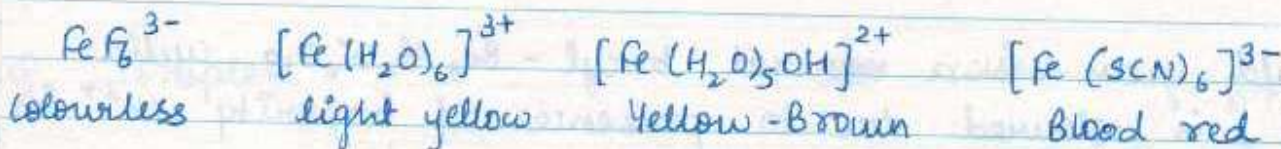
Colour of transition metal ions in aq. medium:

In presence of water, d -orbitals are splitted, so their solution is coloured due to $d-d$ transition except Fe^{3+} & Mn^{2+} .

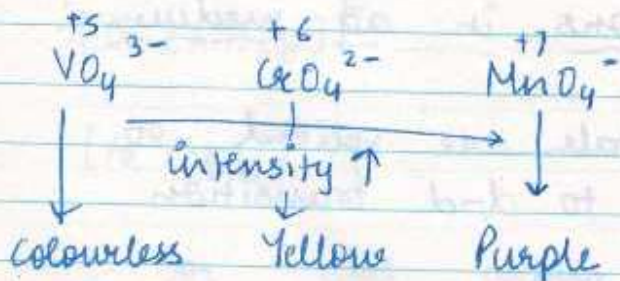
$\text{Fe}^{3+}(\text{aq.})$ & $\text{Mn}^{2+}(\text{aq.})$ are coloured due to absorbance, LMCT



• LMCT (Polarisation) $\rightarrow$
 $d^5 \text{ WFL}, d^{10}, d^0$



Polarizability $\text{F}^- < \text{H}_2\text{O} < \text{OH}^- < \text{SCN}^-$
 transition $\uparrow$ intensity $\uparrow$



* K_2MnO_4 d^1 d-d transition

• MLCT :-

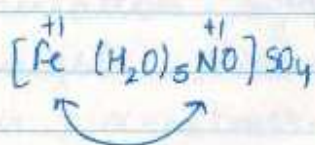
Charge on CMI $\leq$ ligand

low o.s. of CMI

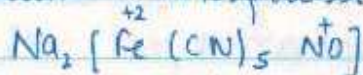
ligand : CO, NO⁺, π donor

eg,

Brown ring complex



Sodium Nitroprusside



~~MLCT~~

colourless

• MMCT :-

- At least 2 metal ions should be present, which belongs to d-block (can show variable o.s.)

- P.D. must be present b/w metal ions

$\text{M}^{+x} \leftarrow \text{N} \equiv \text{C} \rightarrow \text{Fe}^{2+}$ shows colour when e^- acceptance tendency of $\text{M}^{+x} > \text{Fe}^{2+}$

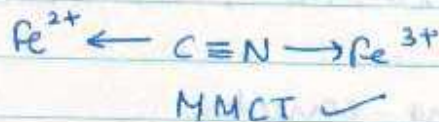
or $x > 2$ for $x = 2$, Zeff of $\text{M}^{+2} > \text{Fe}^{2+}$
 $\text{M}^{+x} \leftarrow \text{N} \equiv \text{C} \rightarrow \text{Fe}^{3+}$ is always coloured

- $\text{Fe}_2^{+2} [\text{Fe}^{+2}(\text{CN})_6]$ (white)
ferro ferro cyanide
Iron (II) hexacyanido ferrate (II)
2:2-complex



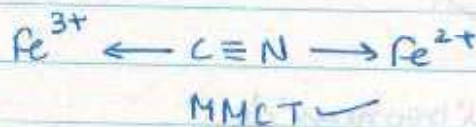
white ppt

- $\text{Fe}_4^{+3} [\text{Fe}^{+2}(\text{CN})_6]_3$
ferri ferro cyanide
3:2-complex
Prussian blue



Blue ppt

- $\text{Fe}_3^{+2} [\text{Fe}^{+3}(\text{CN})_6]_2$
ferro ferric cyanide
2:3-complex
turnbull's blue



Blue ppt

- $\text{Fe} [\text{Fe}(\text{CN})_6]$
3:3-complex
ferric ferricyanide



~~MMCT~~ LMCT ✓

Red-Brown colouration

Ag,

- $\text{Cu}_2 [\text{Fe}(\text{CN})_6]$ $\text{Cu}^{2+} \leftarrow \text{N} \equiv \text{C} \rightarrow \text{Fe}^{2+}$ coloured PP $\text{Cu}^{2+} > \text{Fe}^{2+}$

- $\text{Ag}_4 [\text{Fe}(\text{CN})_6]$ $\text{Ag}^+ \leftarrow \text{N} \equiv \text{C} \rightarrow \text{Fe}^{2+}$ white

- $\text{Zn}_2 [\text{Fe}(\text{CN})_6]$ $\text{Zn}^{2+} \leftarrow \text{N} \equiv \text{C} \rightarrow \text{Fe}^{2+}$ white

- $\text{Cd}_2 [\text{Fe}(\text{CN})_6] \rightarrow$ Bluish white

- $\text{Ag}_3 [\text{Fe}(\text{CN})_6] \rightarrow$ coloured

- $\text{Cu}_3 [\text{Fe}(\text{CN})_6]_2 \rightarrow$ coloured

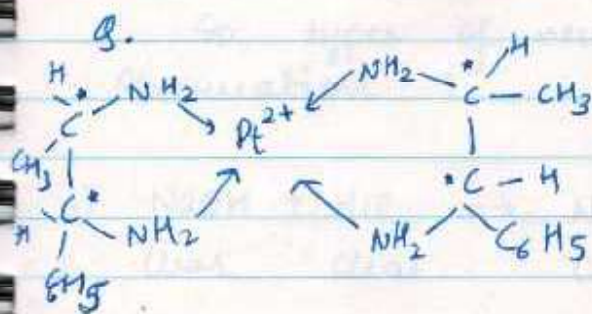
Applications :-

- Coordination compounds find use in many qualitative and quantitative chemical analysis. The familiar colour reactions given by metal ions with a number of ligands, as a result of formation of coordination entities, form the basis for their detection and estimation by classical & instrumental methods of analysis. Examples of such reagents include EDTA, DMG, α -nitroso- β -naphthol, cupron, etc.
- Hardness of water is estimated by simple titration with Na_2EDTA . The Ca^{2+} and Mg^{2+} ions form stable complexes with EDTA. The selective estimation of these ions can be done due to difference in the stability constants of calcium & magnesium complexes.
- Some important extraction processes of metals, like those of silver and gold, make use of complex formation. Gold, for example, combines with cyanide in the presence of oxygen and water to form the coordination entity $[\text{Au}(\text{CN})_2]^-$ in aqueous solⁿ. Gold can be separated in metallic form from this solution by the addition of zinc.
- Similarly, purification of metals can be achieved through formation and subsequent decomposition of their coordination compounds. For example, impure ~~nickel~~ nickel is converted to $[\text{Ni}(\text{CO})_4]$, which is decomposed to yield pure nickel.

- Coordination compounds are of great importance in biological systems. The pigment responsible for photosynthesis, chlorophyll, is a coordination compound of magnesium. Haemoglobin, the red pigment of blood which acts as oxygen carrier is a coordination compound of iron. Vitamin B₁₂, cyanocobalamin, the anti-pernicious anaemia factor, is a coordination compound of cobalt. Among the other compounds of biological importance with coordinated metal ions are the enzymes like, carboxypeptidase A and carbonic anhydrase (catalysts of biological systems).
- Coordination compounds are used as catalysts for many industrial processes. Examples include rhodium complex, $[(PPh_3)_3RhCl]$, a Wilkinson catalyst, is used for the hydrogenation of alkenes.
- Articles can be electroplated with silver and gold much more smoothly and evenly from solutions of the complexes, $[Ag(CN)_2]^-$ & $[Au(CN)_2]^-$ than from a solution of simple metal ions.
- In black and white photography, the developed film is fixed by washing with hypo solution which dissolves the undecomposed AgBr to form a complex ion, $[Ag(S_2O_3)_2]^{3-}$.
- There is growing interest in the use of chelate therapy in medicinal chemistry. An example is the treatment of problems caused by the presence of

Ex 3 A

metals in toxic proportions in plant/animal systems. Thus, excess of copper and iron are removed by the chelating ligands D-penicillamine and desferrioxime B via the formation of coordination compounds. EDTA is used in the treatment of lead poisoning. Some coordination compounds of platinum effectively inhibit the growth of tumours. Examples are: cis-platin and related compounds.



STEREO.

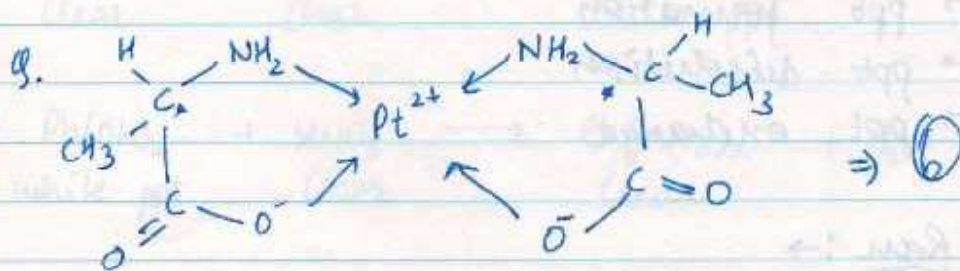
$⑥ \times ②$

$8 + 4 + 2 = 14$

cis $\rightarrow 10$ (6, 40A)

trans $\rightarrow 10$ (6, 40A)

20



$\Rightarrow ⑥$

Types of Reaction

- 1] Acid - Base rxn
- Neutralisation
 - Lewis acid - base
 - complex formation
 - Hydrolysis
(But not exactly Lewis acid - base)

2] Thermal decomposition

- 3] Redox rxn
- dispⁿ
 - compⁿ
 - displacement
 - decomp.
 - combination

- 4] Ion exchange
- ppt formation
 - ppt dissolution
 - ppt exchange

* Neutralisation Rxn :→

In this rxn, formation of H_2O takes place & energy is released so rxn moves in forward direction.

Acid : Compound of NM, generally they are molecules, exist in gaseous / liquid state. When dissolved in water, either ionisable (strong acid) or non-ionisable (weak acid).

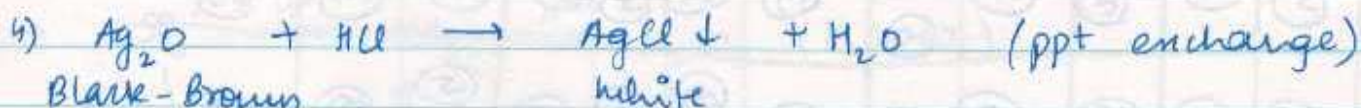
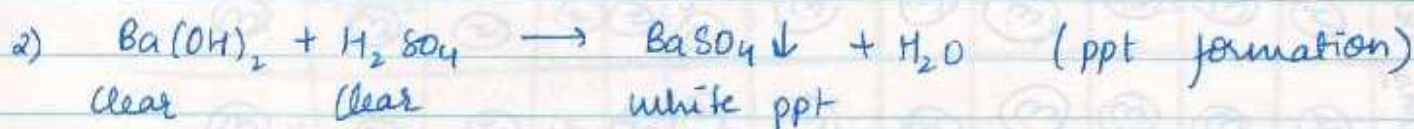
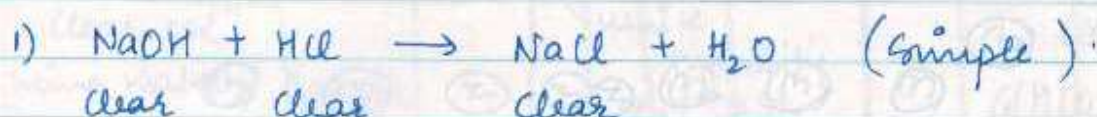
Weak acid are present in water in soluble form

due to dipole-dipole interaction / H-bond with H_2O .
No ppt observed in water.

Base : compound of metal, ionic in nature, exists in solid state, either soluble or (strong base) or insoluble (weak base). Insoluble gives ppt in water.

Salt : Ionic compd, solid state, either soluble or insoluble.

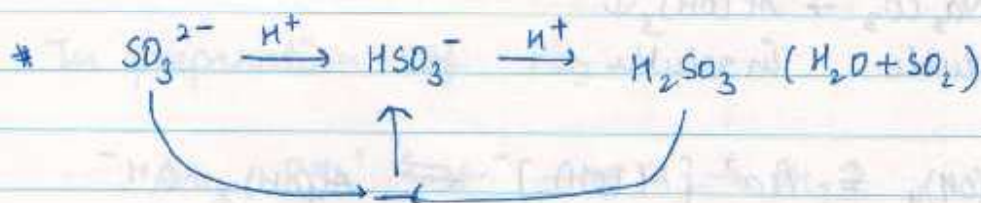
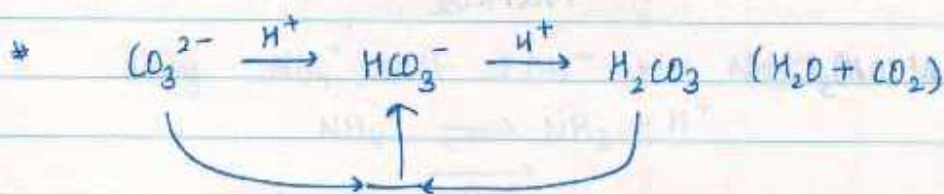
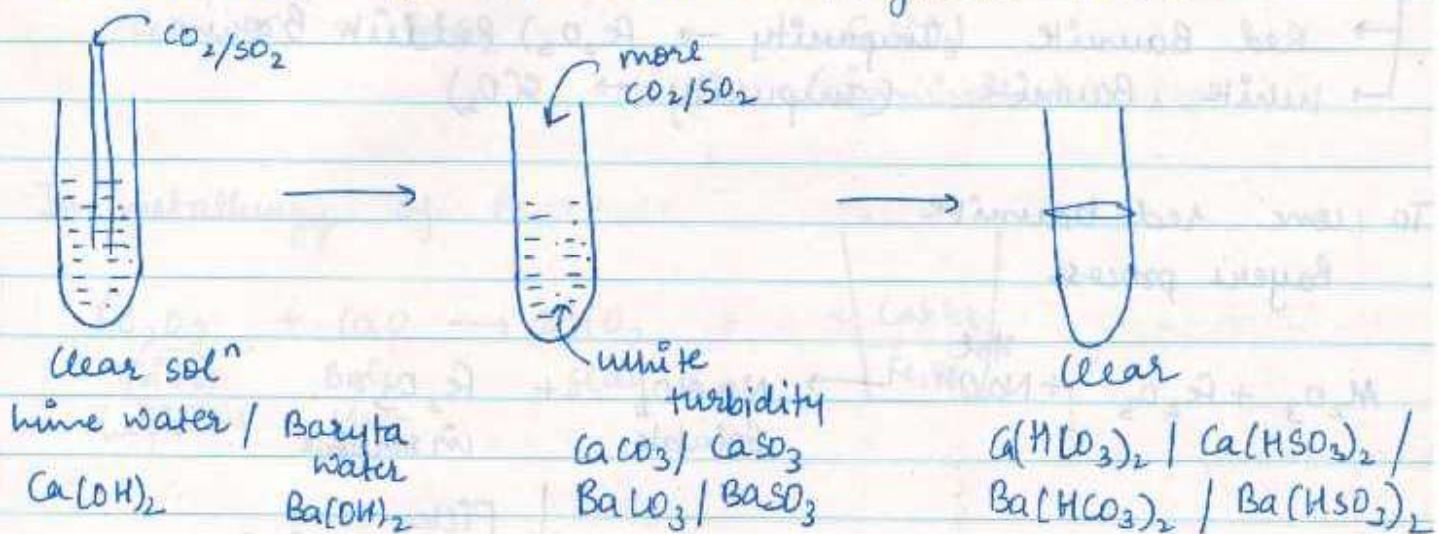
So, types of neutralisation rxn on the basis of observation :



Other Acid-Base Reactions:

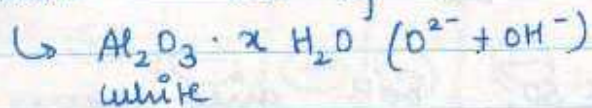
1) lime water / Baryta water test:

Used to identify CO_2 & SO_2 , both give same observation so we can't distinguish them.



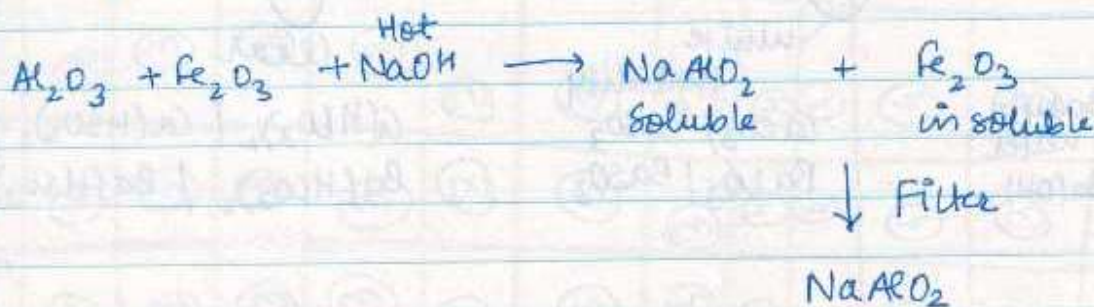
In metallurgy,

Bauxite $\rightarrow$ ore of Al

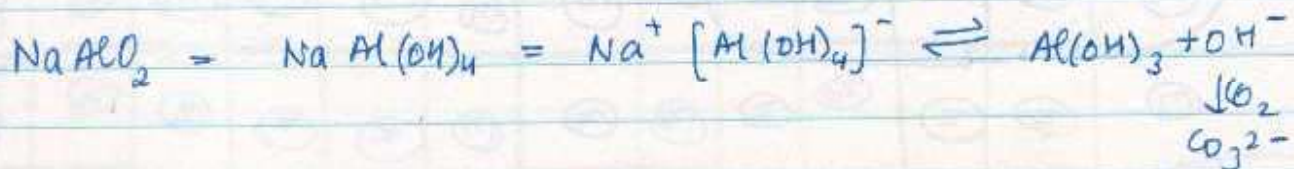
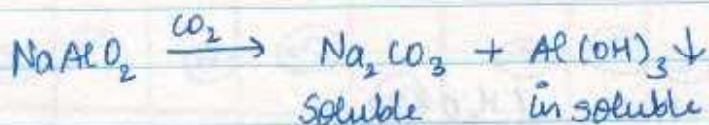


- $\rightarrow$ Red Bauxite (Impurity $\rightarrow \text{Fe}_2\text{O}_3$) Reddish Brown
- $\rightarrow$ White Bauxite (Impurity $\rightarrow \text{SiO}_2$)

To conc. red bauxite
Bayer's process



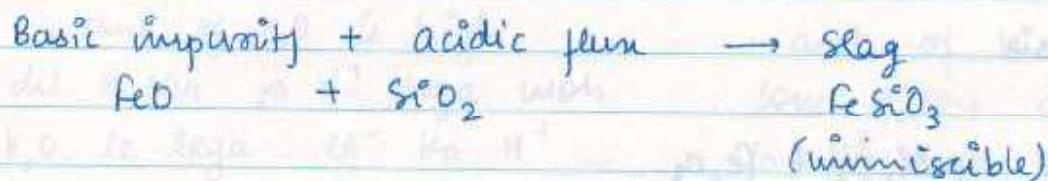
Recovery of $\text{Al}_2\text{O}_3 / \text{Al}(\text{OH})_3$
from filtrate



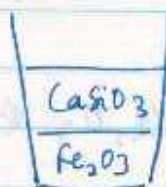
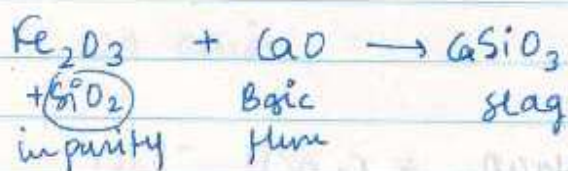
In furnaces

In metallurgy of Cu.

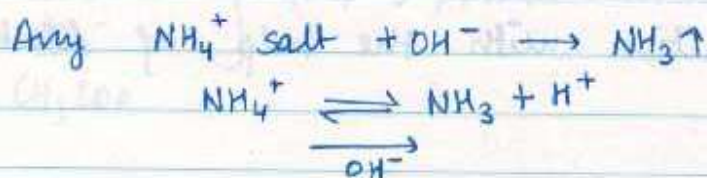
Cu_2S contains impurity of FeO



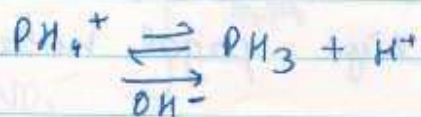
In metallurgy of Fe



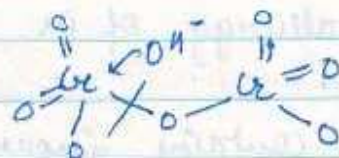
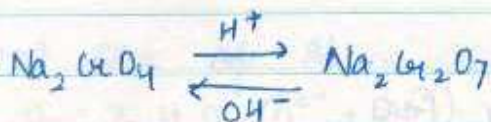
• In preparation of NH_3



In preparation of PH_3



- In preparation of dichromate

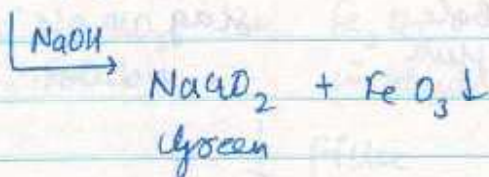


- In salt analysis

To separate Al_2O_3 & Fe_2O_3



To separate Cr_2O_3 & Fe_2O_3

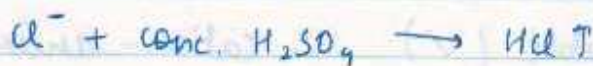
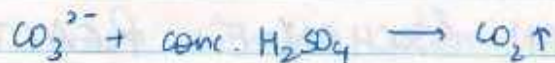
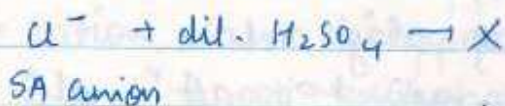
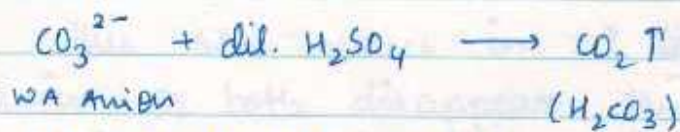


- To distinguish strong acid anion & weak acid anion from soluble salt solution with the help of dilute H_2SO_4 .

Stronger the acid weaker the base

Strong acid kaa anion weak base hota hai jiski H^+ lene ki tendency bahot kam hoti hai

Weaker the acid stronger the base weak acid kaa anion strong base hota hai jiski H^+ lene ki tendency bahot jyada hoti hai, H^+ ke liye pagal hota hai



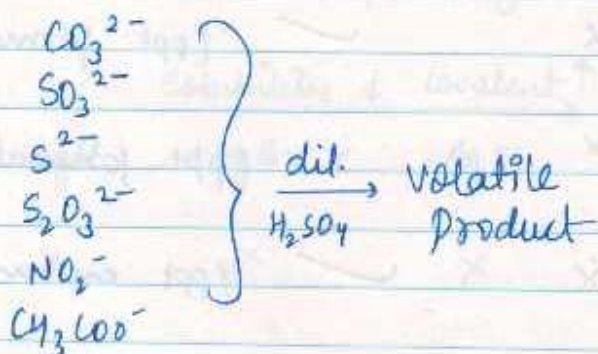
↑

amt. of H₂O is high
dil. H₂SO₄ jo H⁺ dega woh
H₂O le lega. Cl⁻ ko H⁺
lene ke liye koi pressure
nahi hai

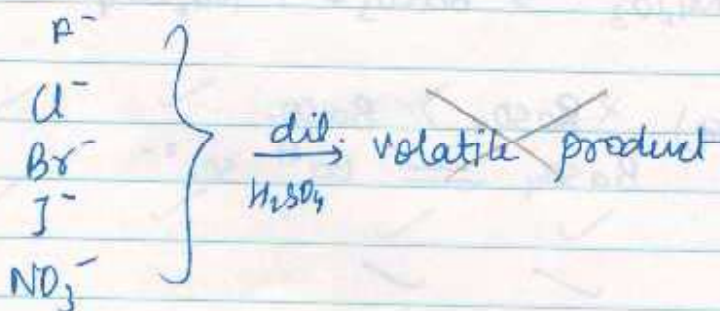
↑

amt. of H₂O is very low.
conc. H₂SO₄ strong acid hai
jo H⁺ dega hi dega, abh
Cl⁻ ko H⁺ lena hi padega

WA anion



SA anion



* ION EXCHANGE REACTION :

Soluble (✓) = cation-anion ka alag alag rehna.
cation-anion ke beech mai koi attⁿ nahi hai, dono ek dusre ko nahi chahte hai.

Insoluble (x) = cation anion ek saath rehna chahte hai, ek dusre ke liye pagal hai



①	✓	✓	✓	✓	(no rxn)
②	✓	✓	x	✓	(ppt. formation)
③	✓	✓	x	x	(ppt. formation)
④	x	✓	x	✓	(ppt. exchange)

Solubility $\text{Salt}_1 > \text{Salt}_3$



Solubility (K_{sp}) $\text{BaSO}_4 > \text{BaCO}_3$
 $\text{BaSO}_4 \rightleftharpoons \text{Ba}^{2+} \text{SO}_4^{2-}$

⑤ x ✓ ✓ ✓

ye rxn nahi hogi bina external force ke

This rxn moves in $\rightarrow$ dirⁿ when either cation or anion or both disappear from solⁿ.

• cation can disappear by,

i) ppt exchange rxn



ii) by complex formation rxn

coz during complex f. metal ion becomes CMI & loses its identity.

eg, * All Ag salts are soluble in excess KCN

* All Ag salts are soluble in excess hypophosphite except Ag_2S

* all Ag salts are soluble in excess NH_3 except AgI & Ag_2S

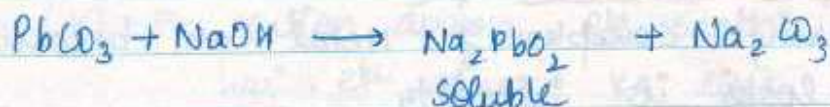
Solubility $\downarrow$ Covalent $\uparrow$

	AgCl	AgBr	AgI	Ag ₂ S	
H ₂ O	X	X	X	X	$[\text{Ag}(\text{H}_2\text{O})_2]^+$
Cold NH ₃	✓	X	X	X	} $[\text{Ag}(\text{NH}_3)_2]^+$
Hot NH ₃	✓	✓	X	X	
$\text{S}_2\text{O}_3^{2-}$	✓	✓	✓	X	$[\text{Ag}(\text{S}_2\text{O}_3)_2]^{3-}$ $\swarrow$ CN=4, chelation
CN ⁻	✓	✓	✓	✓	$[\text{Ag}(\text{CN})_2]^-$

Stability : $[\text{Ag}(\text{CN})_2]^- > [\text{Ag}(\text{S}_2\text{O}_3)_2]^{3-} > [\text{Ag}(\text{NH}_3)_2]^+ > [\text{Ag}(\text{H}_2\text{O})_2]^+$

As solubility of salt ↓
for dissolution, req. stability of complex ↑.

* All Pb(II) salts are soluble in excess NaOH except PbS due to formation of hydroxo complex

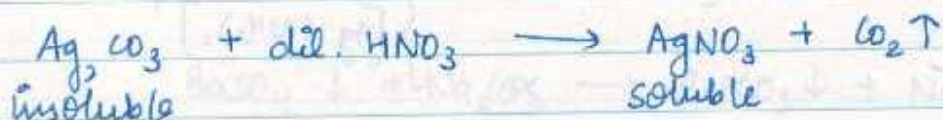


* All Zn(II) salts are soluble in excess NaOH except ZnS



• Anion can disappear by,
i) on reaction with strong acid

⇒ All carbonate salts are soluble in
dil. HCl, dil. H₂SO₄, dil. HNO₃, CH₃COOH



⇒ To Dissolve Acid which is not used for dissolution

Ag (I) salt
Pb (II) salt
Ba (II) salt
Ca (II) salt

HCl, CH₃COOH
dil. HCl, dil. H₂SO₄
dil. H₂SO₄
X

Acidic Nature $\text{H}_2\text{CrO}_4 < \text{HCl} / \text{HNO}_3 / \text{H}_2\text{SO}_4 / \text{CH}_3\text{COOH}$
SA WA

$$\text{H}_2\text{CrO}_4 < \text{HCl} / \text{HNO}_3 / \text{H}_2\text{SO}_4 / \text{CH}_3\text{COOH}$$

H_2CO_4 is non volatile while $H_2CO_3 (CO_2)$ is volatile.

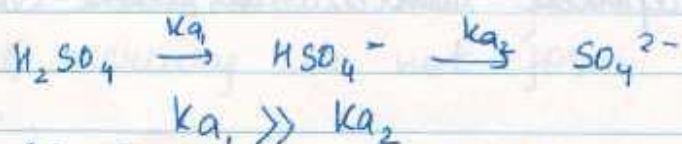
Weak acid CH_3COOH mai saare CO_3^{2-} soluble hai to CO_2 volatile hai, den ko $\rightarrow$ le jayega.

WA CH_3COOH mai ek hi CrO_4^{2-} soluble hai SrCrO_4 , Co_2 saare insoluble CrO_4^{2-} mai uski solubility sabse jyada hai, H_2CrO_4 non volatile hai, solⁿ mai pada rahega.

dil. H_2SO_4

Only $\text{BaCl}_2 \cdot 4\text{H}_2\text{O}$ is soluble in CH_3COOH among insoluble sulfate.

dil. H_2SO_4 , dil. HNO_3 , they are soluble in conc. a.c.



SO_4^{2-} is strong base in comparison of HSO_4^- so SO_4^{2-} can take H^+ from only from conc. acid from soluble HSO_4^-



H_2O jyada hai toh H^+ le lega



H_2O km hai majburi mai H^+ SO_4^{2-} ko lena padega.

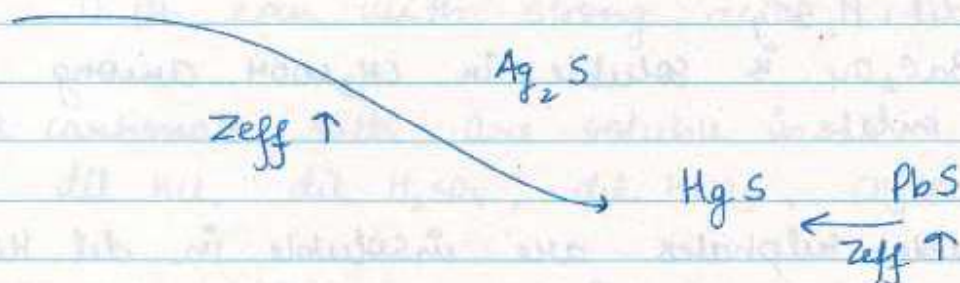
• Sulphide

dissolution tendency (coloured) > Black

(On the basis of covalent ch°)

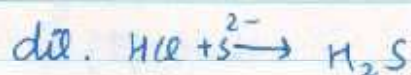
Black colour sulphide

FeS CoS NiS CuS

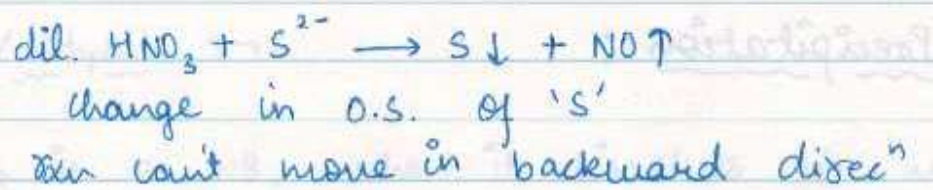


⇒ only black colour sulphide which insoluble in dil. HCl is FeS

⇒ Only sulphide which is insoluble in dil. HNO_3 is HgS



No change in o.s. of 'S'
 rxn can moves in backward dirⁿ



So HNO_3 is better reagent for dissolution of black colour sulphide in comparison of dil. HCl .

• By Redox change :

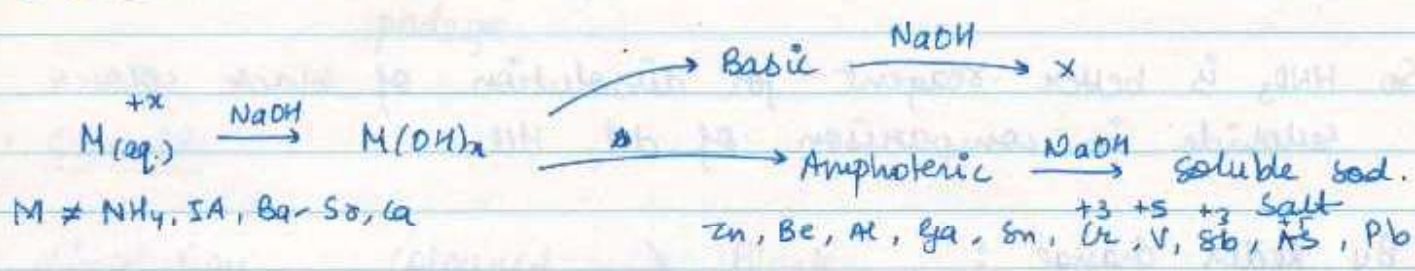


* COMPLEX FORMATION REACTION :-

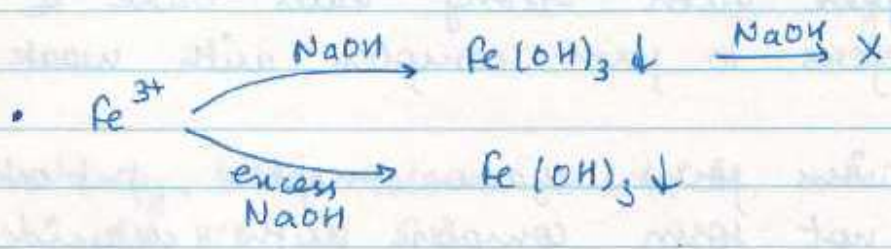
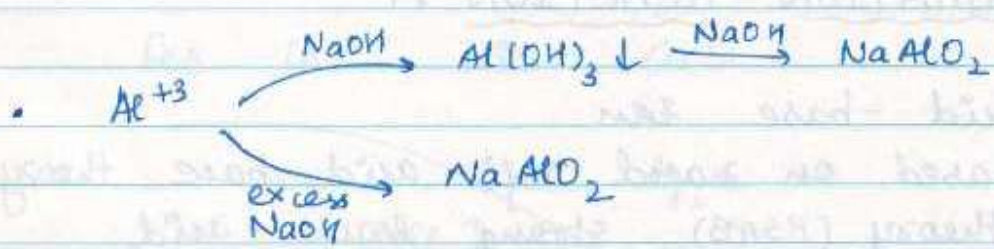
- It is lewis acid - base rxn.
- This rxn is based on hard soft acid base theory. Accⁿ to this theory (HSAB), strong lewis acid prefers to form complex with strong lewis base & weak lewis acid prefers to form complex with weak lewis base.
- eg, d-block metal ions form cyano complex, p-block & mercury do not form complex with cyanide ion.
- d-block cations except Fe, Mn, Hg form complex with NH_3 .
- p-block cation & Hg form iodo complex.
- d-block as well as p-block cations form complex with hypo ($\text{S}_2\text{O}_3^{2-}$), weak field ligand but forms stable complex due to chelation.

• Reagents of Precipitation:

1) NaOH $\rightarrow$



- S, p block $d^{10} \rightarrow$ white
- $Cr^{3+}, Fe^{2+}, Ni^{2+} \rightarrow$ green
- $Cu^{2+}, Co^{2+} \rightarrow$ Blue
- $Mn^{2+}, Co^{2+} \rightarrow$ pink
- $Fe(OH)_3 \rightarrow$ Reddish Brown.

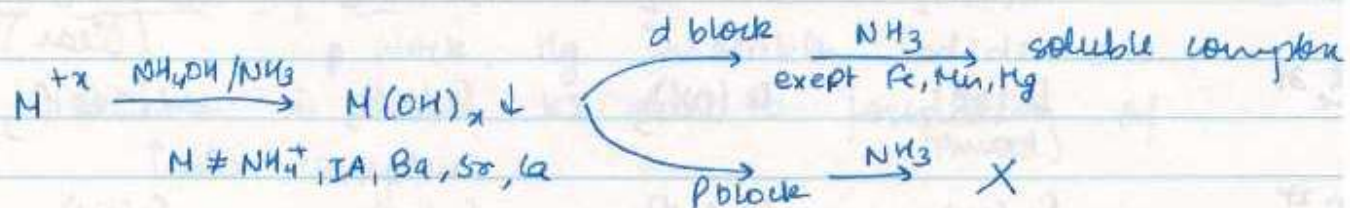


2) NH_3 / NH_4OH $\rightarrow$

- H.NH_3 is a WB, when it dissolves in water initially it gives OH^- so reacts like NaOH but as it is a weak base, it gives a limited amount of OH^- in water.

On adding excess NH_3 / NH_4OH , NH_3 present in water in molecular form, forms H-bond with H_2O .
So in excess NH_3 , main reagent is NH_3 & limited NH_3 main reagent is OH^- .

In excess NH_3 , NH_3 acts as ligand, shows complex formation rxn, it is a SPL so forms complex with d block cation.



$\Rightarrow$ Hg shows complex formation as like p block cation due to large size, absence of $(n-1)d$ orbitals

$\Rightarrow$ NH_3 acts as WFL for Mn^{2+} , Fe^{2+}

$\Rightarrow$ K_{sp} of $\text{Fe}(\text{OH})_3$ is very very low so it is perfect ppt in presence of NH_3 .

$d^{10} \rightarrow$ colourless
 $(\text{Cu}^{2+}(\text{amm.}), \text{Ni}^{2+}(\text{amm.})) \rightarrow \text{Blue}$

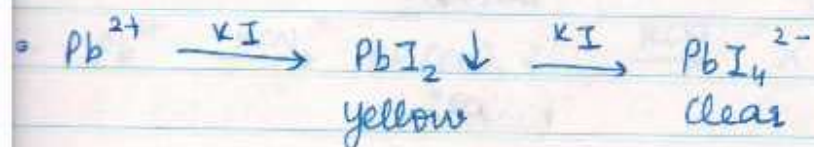
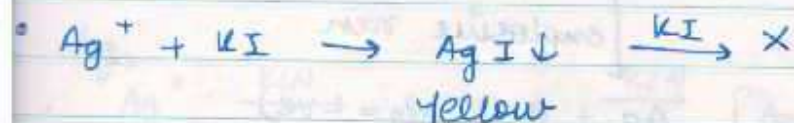
H.W.

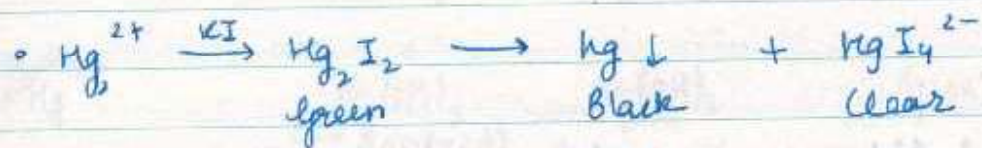
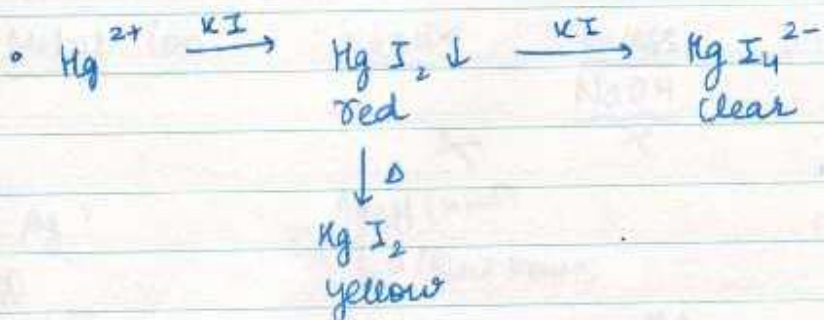
<u>Metal Ion</u>	<u>NaOH</u>	<u>excess NaOH</u>	<u>NH₃ / NH₄OH</u>	<u>excess NH₄OH</u>
Ag ⁺	AgOH (white) RT → Ag ₂ O (Black brown)		AgOH	[Ag(NH ₃) ₂]
Pb ²⁺	Pb(OH) ₂	Na ₂ PbO ₂	Pb(OH) ₂	Pb(OH) ₂
Hg ²⁺	Hg(OH) ₂ RT → HgO (red)	Hg(OH) ₂	Hg(OH) ₂	HgO · HgNH ₂ Cl
Cu ²⁺	Cu(OH) ₂ (Blue)	Cu(OH) ₂	Cu(OH) ₂	[Cu(NH ₃) ₄] ²⁺ (Blue)
Bi ³⁺	Bi(OH) ₃	Bi(OH) ₃	Bi(OH) ₃	Bi(OH) ₃
Cd ²⁺	Cd(OH) ₂	Cd(OH) ₂	Cd(OH) ₂	[Cd(NH ₃) ₄] ²⁺ (clear)
Fe ³⁺	Fe(OH) ₃ (brown)	Fe(OH) ₃	Fe(OH) ₃	Fe(OH) ₃
Fe ²⁺	Fe(OH) ₂ (green)	Fe(OH) ₂ → on standing, reddish brown	Fe(OH) ₂	Fe(OH) ₂ (Fe ²⁺ → Fe ³⁺)
Cr ³⁺	Cr(OH) ₃ (green)	NaCrO ₂	Cr(OH) ₃	[Cr(NH ₃) ₆] ³⁺ (Pink)
Al ³⁺	Al(OH) ₃	NaAlO ₂	Al(OH) ₃	Al(OH) ₃
Mn ²⁺	Mn(OH) ₂ (pink)	Mn(OH) ₂	Mn(OH) ₂	Mn(OH) ₂
Co ²⁺	Co(OH) ₂ (pink)	Co(OH) ₂	Co(OH) ₂	[Co(NH ₃) ₆] ²⁺ (Yellow)
Zn ²⁺	Zn(OH) ₂	Na ₂ ZnO ₂	Zn(OH) ₂	[Zn(NH ₃) ₄]
Ni ²⁺	Ni(OH) ₂ (green)	Ni(OH) ₂	Ni(OH) ₂	[Ni(NH ₃) ₆] (Blue)

$$\text{Ca}^{2+} \quad \times \quad \times \quad \times \quad \times$$

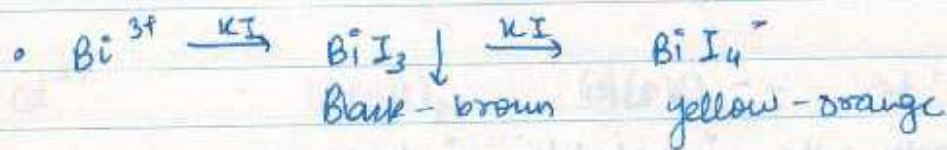
→ solubility :- All iodide are water soluble except Ag^+ , Bi^{3+} , Cu^+ , $\text{Hg}_2^{2+} / \text{Hg}^{2+}$, Pb^{2+}

→ In excess of KI :- $I^- \rightarrow WFL$
p block, Hg insoluble iodide are soluble in excess KI due to formation of iodo complex.

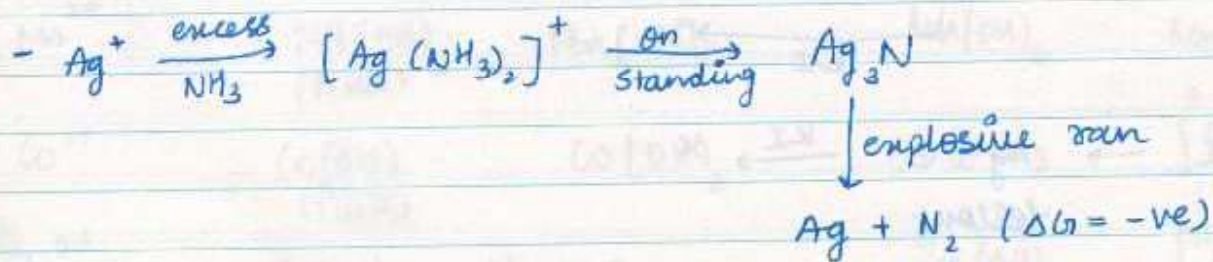
$$\bullet \text{ Cu}^{2+} + \text{KI} \rightarrow \underset{\text{white}}{\text{CuI} \downarrow} + \text{I}_2 / \text{KI}_3 \quad \text{Brown}$$


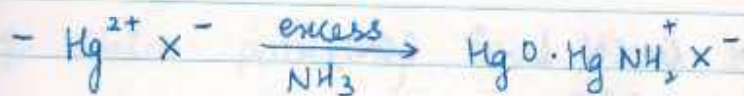


- Hg_2^{2+} dis^r in presence of ligand $\text{Hg} + \text{Hg}^{2+}$
Black ppt



* Note :





due to high p.p. of Ag^+ & Hg^{2+}

4) KCN $\Rightarrow$

- All cyanides are water insoluble except NH_4^+ , IA, IIA, $\text{Hg}(\text{CN})_2$

- CN^- polyatomic anion, p block, polarizability $\downarrow$

- s, p block, $d^{10} \rightarrow$ white

d^{1-9} (gen.) $\rightarrow$ yellow

- excess KCN: all insoluble d-block cyanide are soluble in excess KCN

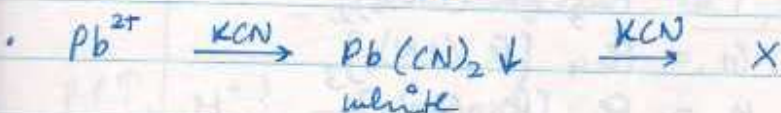
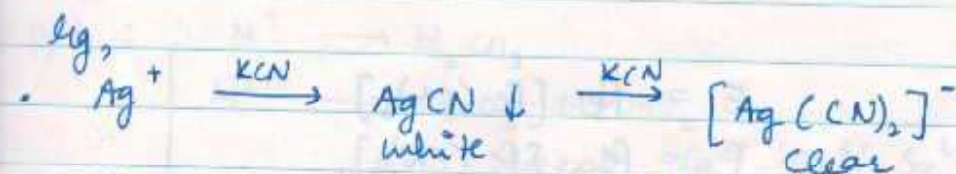
$\text{CN}^- \rightarrow$ strongest ligand (complex of cyano)

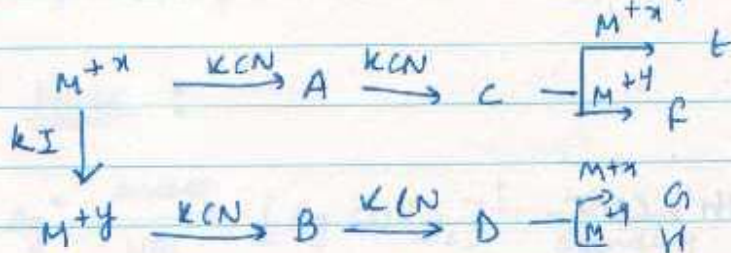
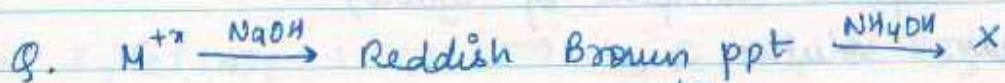
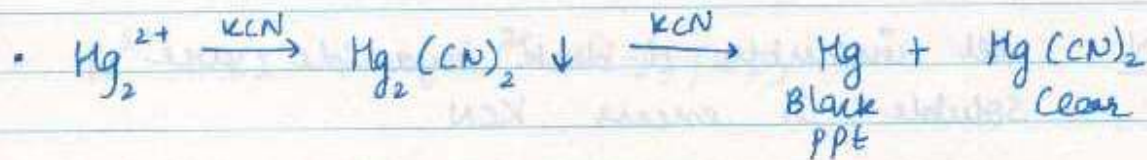
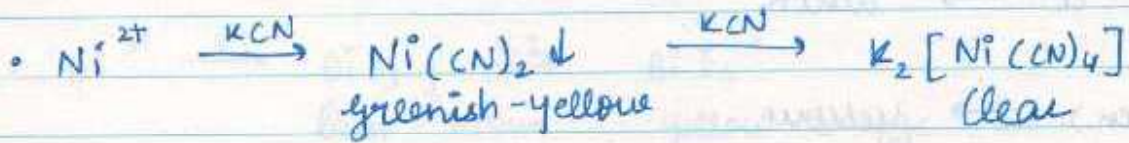
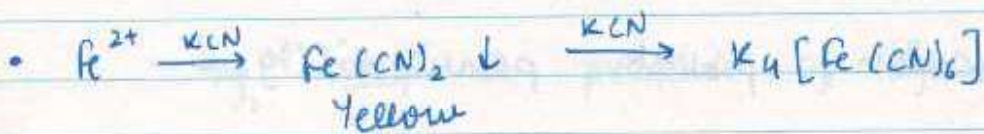
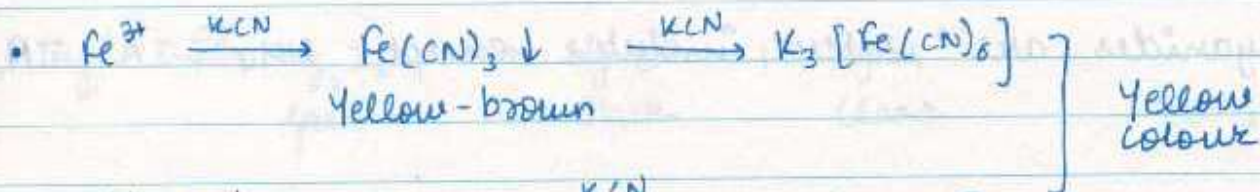
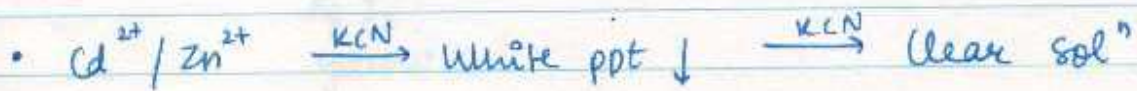
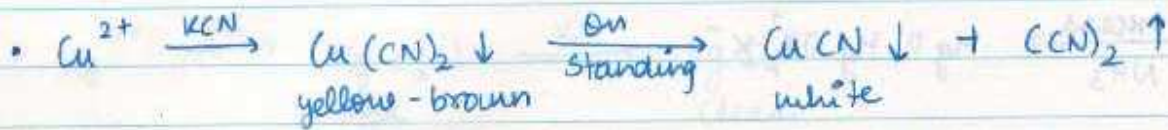
$d^{10} \rightarrow$ clear solution

d^{1-9} (gen.) $\rightarrow$ yellow

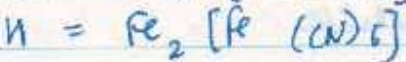
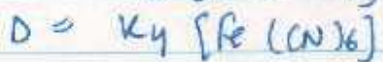
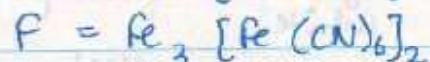
(Octahedral)

- Cu^{2+} shows redox with CN^-





Identify $A^{-1} \in M^n$.



Q. Which pair of ions can be separated by NaOH?

- A) Pb^{2+} ; Cu^{2+}
- B) Al^{3+} ; Ca^{2+}
- C) Ni^{2+} ; Zn^{2+}
- D) Cr^{3+} ; Fe^{3+}

Q. Which d-block metal ion is soluble in excess NH_3 as well as excess NaOH but when limited NaOH is used it gives white coloured ppt?

→ Zn^{2+}

Q. A metal ion gives ppt & coloured solⁿ when reacts with KI. Find the formula of compd when metal ion reacts with excess KCN.

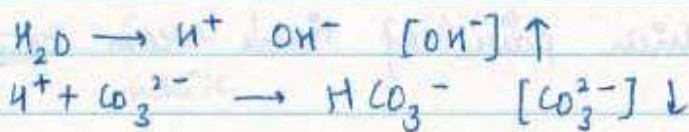
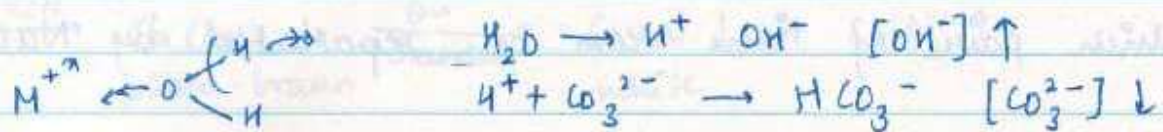
→ $K_3 [Cr(CN)_6]$

5] Na_2CO_3 :-

- All CO_3^{2-} water insoluble except NH_4^+ , IA

• Colour : s/p block, Zn^{2+} → white
d block → coloured (except Zn^{2+})

• ppt :
 M^+ → M_2CO_3
 M^{2+} → $MCO_3 \cdot M(OH)_2$
except $Ca^{2+}, Ba^{2+}, Sr^{2+}, Fe^{2+}$ → soluble OH^- (value of K_{sp} is high)
P.P.T ↓ M^{3+} → $M(OH)_3$



- 6) Na₂CrO₄ $\rightarrow$

- $$\text{Hg}_2^{2+} \xrightarrow{\text{CeO}_4^{2-}} \text{Brown ppt (Unknown comp.)} \xrightarrow{\Delta} \text{Hg}_2\text{CrO}_4 \text{ (Red)}$$

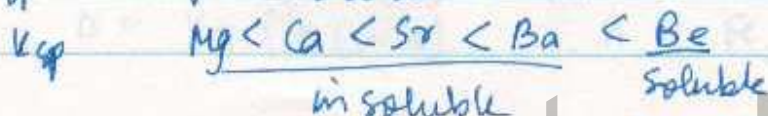
- CrO_4^{2-} is anion of WA H_2CrO_4 which is ~~so~~ non volatile. All insoluble CrO_4^{2-} are soluble in SA (dil. HCl , HNO_3 , H_2SO_4) but only one insoluble CrO_4^{2-} is soluble in CH_3COOH which is SrCrO_4 due to its high K_{sp} among insoluble CrO_4^{2-} .

7] Na₂C₂O₄ :-

- All $C_2O_4^{2-}$ are water insoluble except NH_4^+ , IA, FeC_2O_4 , BeC_2O_4
colour : white except $Fe_2(C_2O_4)_3$

(Blue) $\Rightarrow$ used in blueprints

- $\text{H}_2\text{C}_2\text{O}_4$ is a WA, non volatile in nature, $\text{C}_2\text{O}_4^{2-}$ is SB so soluble in SA but only one insoluble $\text{C}_2\text{O}_4^{2-}$ is soluble in CH_3COOH which is BaC_2O_4 due to its high K_{sp} among insoluble oxalate



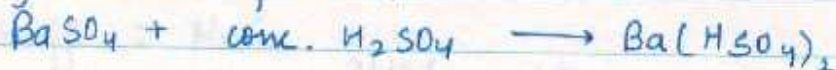
8] $\underline{\text{Na}_2\text{SO}_4}$:-

- All sulphates are water soluble except Hg^{2+} , Sr^{2+} , Ba^{2+} , Pb^{2+}
Colour : Anhydrous sulphate = white

- SO_4^{2-} is anion of SA so insoluble SO_4^{2-} are not soluble in any dil. acid



Insoluble SO_4^{2-} are soluble in conc. HCl / HNO_3 / H_2SO_4 due to formation of HSO_4^-

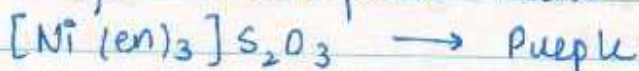


9] $\underline{\text{Na}_2\text{S}_2\text{O}_3}$:-

(aq. solⁿ of Hypo.)

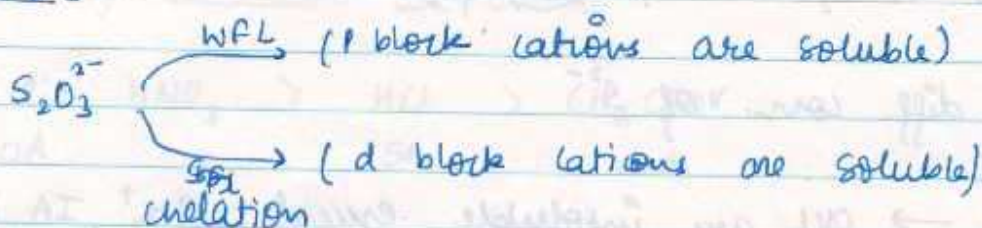
Solubility : All thio sulphates are water soluble except Ag^+ , Bi^{3+} , Cu^+ , $[\text{Ni(en)}_3]^{2+}$, $\text{Hg}_2^{2+}/\text{Hg}^{2+}$, Ba^{2+} , Pb^{2+}

Colour : Simple thiosulphate salts $\rightarrow$ white



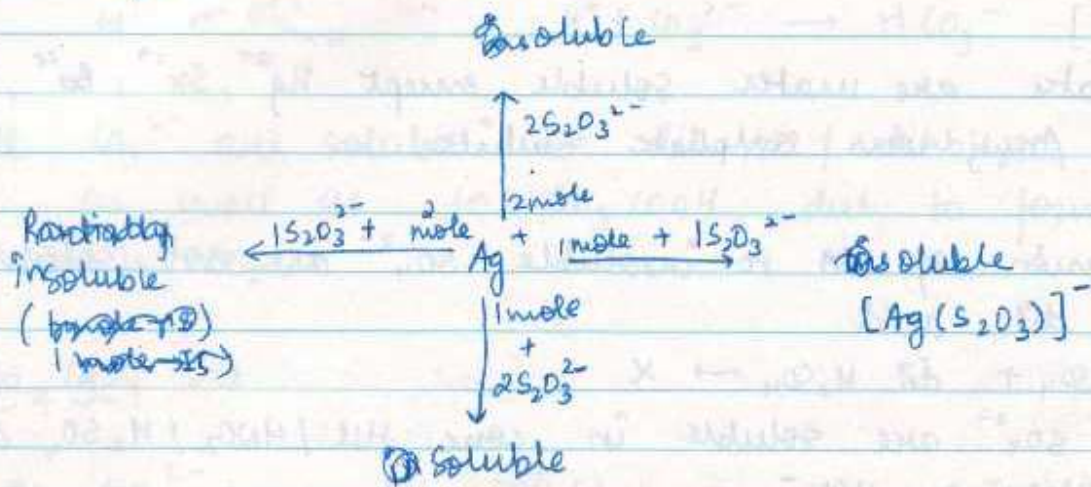
coloured due to d-d transition (complex cation)

Excess $\text{Na}_2\text{S}_2\text{O}_3$:

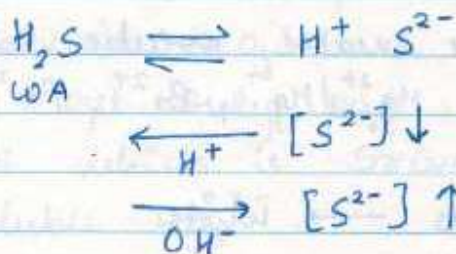


- All insoluble simple thio sulphates are soluble in excess thiosulphate except BaS_2O_3 .

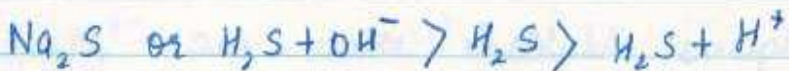
g. soluble / insoluble



10) $\underline{\underline{H_2S}} \rightarrow$
(Na_2S , $H_2S + H^+$)

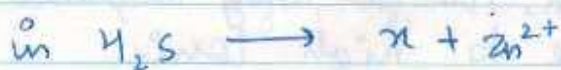
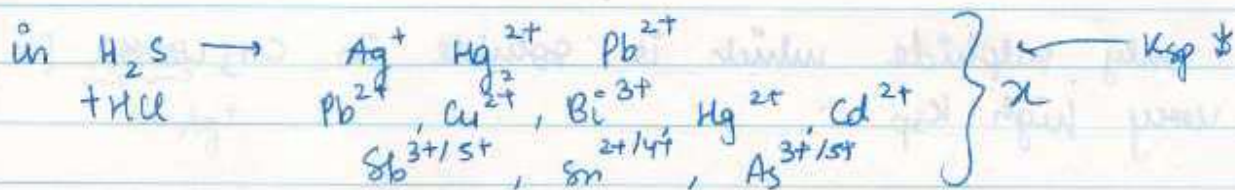


Conc. of S^{2-}



- ppt in diff. con. of S^{2-}

in $\text{Na}_2\text{S} \rightarrow$ all are insoluble except NH_4^+ , IA, IIA



• Colour of insoluble sulphide

Black : $PbS, HgS, Ag_2S, FeS, CuS, CoS, NiS$

White : ZnS

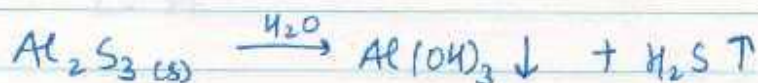
Pink / buff : MnS

Yellow : $Cd^{2+}, As^{3+/5+}, Sn^{4+}$

Orange : $Sb^{3+/5+}$

Black-Brown : Sn^{2+}, Bi^{3+}

- MgS, Al_2S_3, Ga_2S_3 exist only in solid state in aq. medium they get hydrolysed



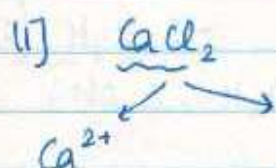
- Fe^{3+} shows redox with S^{2-}

• Dissolution of insoluble sulphides :



- HgS is only sulphide which is insoluble in hot & dil. HNO_3 .
- due to its lowest K_{sp} among sulphides.

- MnS is only sulphide which is soluble in CH_3COOH due to very high K_{sp} .
- Among black coloured sulphides, FeS is only sulphide which is soluble in dil. HCl due to high ionic ch^+ (low P.P. of Fe^{2+}).
- Those who precipitate in presence of HCl are insoluble in dil. HCl .
- Only ZnS , MnS & FeS are soluble in dil. HCl .

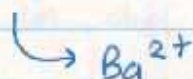
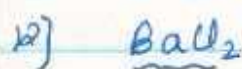


All Cl^- are water soluble
except Ag^+ , Cu^+ , Hg_2^{2+} , Pb^{2+}

Insoluble

$(CaF_2, CaCO_3, CaSO_3, Ca_3(PO_4)_2, CaC_2O_4)$

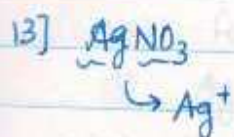
are
all soluble in dil. HCl .



$\swarrow$ soluble in CH_3COOH

Insoluble : $BaCO_3, BaC_2O_4, BaSO_3, Ba_3(PO_4)_2,$
 $BaS_2O_3, BaSO_4, BaCrO_4$ — coloured

$\swarrow$ Insoluble in dil. HCl



Soluble : AgF , Ag_2SO_4 , AgNO_3 , all bi salts

Solubility : - All Ag(I) salts are soluble in hot & dil. HNO_3
 except AgCl , AgBr , AgI
 acidic nature $\text{HCl/HBr/HI} > \text{HNO}_3$

- All Ag(II) salts are soluble in NH_4OH except
 AgI & Ag_2S

- All Ag(I) salts are soluble in hypo except Ag_2S

- All Ag(I) salts are soluble in KCN .



Soluble : $(\text{CH}_3\text{COO})_2\text{Pb}$, $\text{Pb}(\text{NO}_2)_2$, $\text{Pb}(\text{NO}_3)_2$, all bi salts

Solubility : - All Pb(II) salts are soluble in NaOH
 except PbS

- All Pb(II) salts are soluble in hot & dil.
 HNO_3 except PbSO_4 , PbCl_2 , PbBr_2 , PbI_2

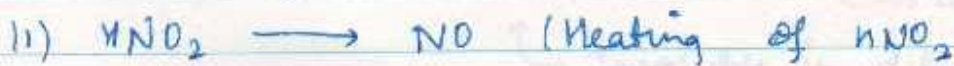
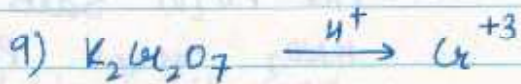
- Only white ppt which is soluble in
 hot water is PbCl_2

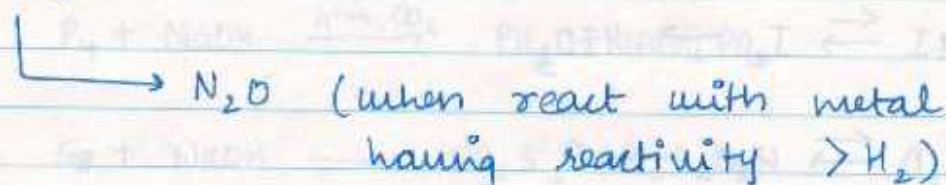
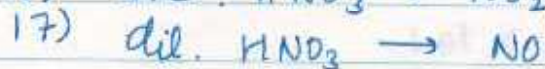
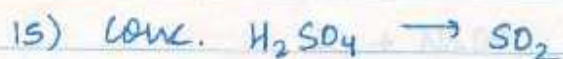
	$PbCl_2$	$PbBr_2$	PbI_2	
$H_2O(25^\circ C)$	X	X	X	✓ Soluble
Hot Water	✓	X	X	X Insoluble
Boiling Water	✓	✓	✓	

* REDOX $\therefore \rightarrow$

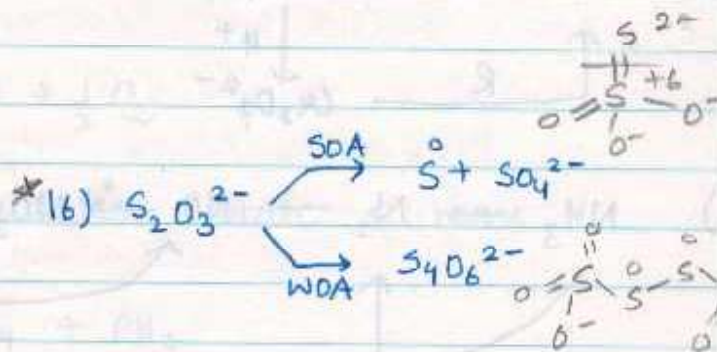
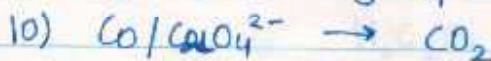
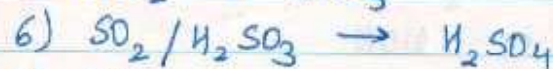
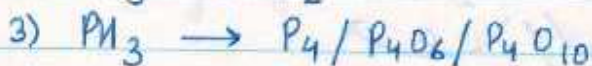
- Oxidising agent & reducing agent :

- Oxidising agent : 1) Halogens





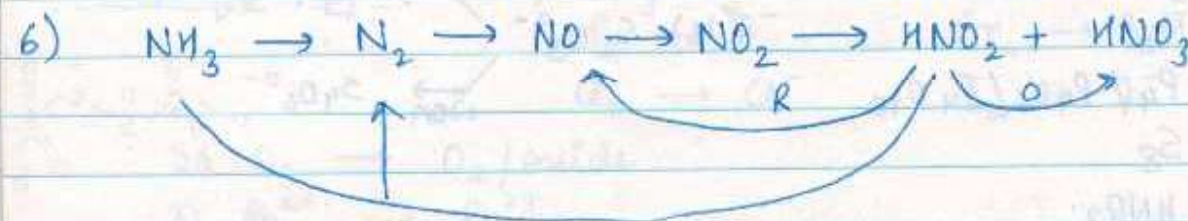
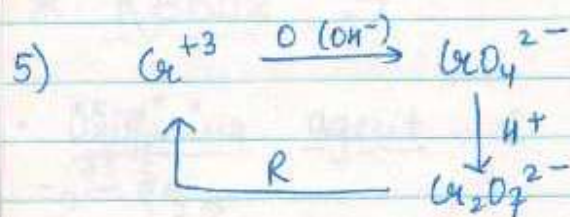
- Reducing agents :



* Imp. redox series :



Stable in
basic med.



* Types of Redox Rxn :

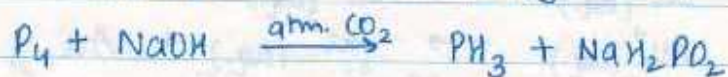
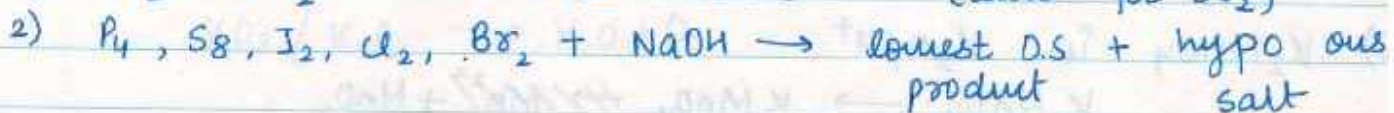
1) Intramolecular

- Disproportionation
- Thermal decomposition

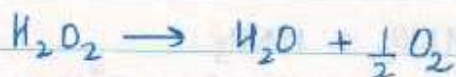
2) Intermolecular

- Combination
- Displacement

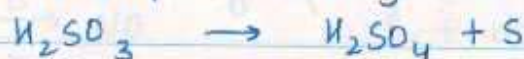
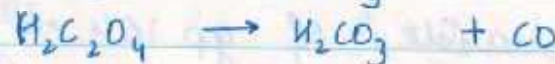
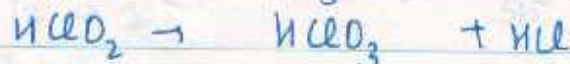
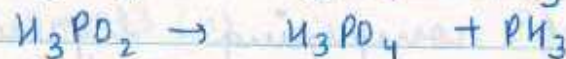
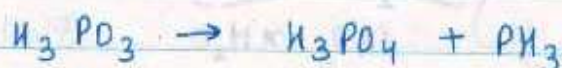
• Disproportionation :



3) Decomposition of H_2O_2



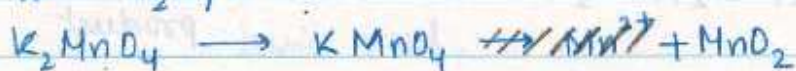
4) Heating of -ous acid / oxidation state < max.



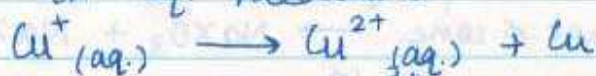
5) Hydrolysis of KO_2



6) K_2MnO_4 in $\text{H}_2\text{O}/\text{H}^+$

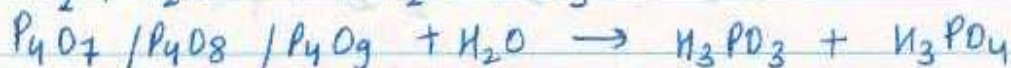


7) Cu^+ & Au^+ in aq. medium

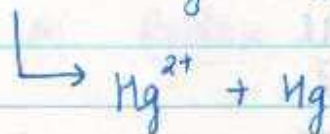


Reactivity $\text{H}_2 > \text{Cu/Au}$

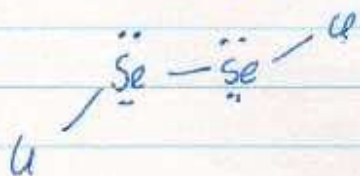
8) Mixed anhydride of p-block + water
gp. 15, 16, 17



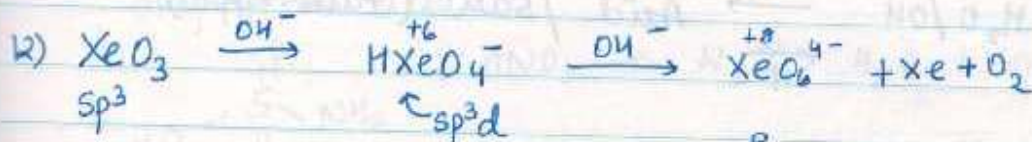
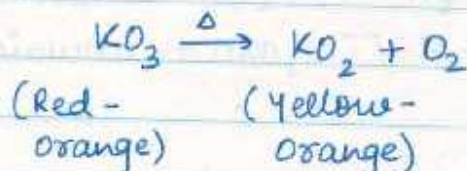
9) Hg^{2+} in presence of complexing ligand
(eg: NH_3 , I^- , CN^- , S^{2-})



10) Heating of mono halide of gp. 16 $\rightarrow$

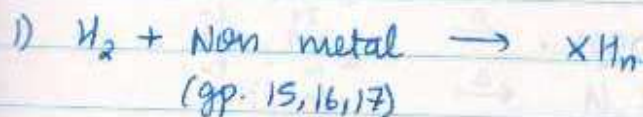


1) Heating of Ozonide / peroxide / superoxide :
 (O_3^-, O_2^{2-}, O_2^-)

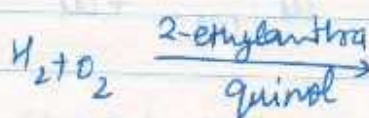
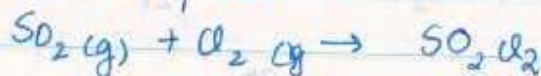
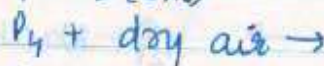
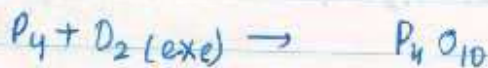


perxenate ion

• Combination Reaction $\Rightarrow$

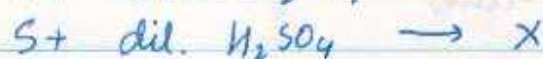
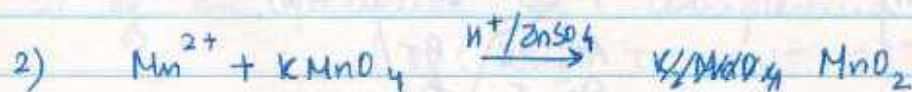
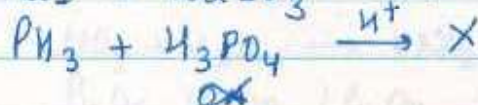
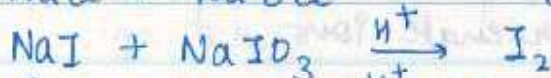
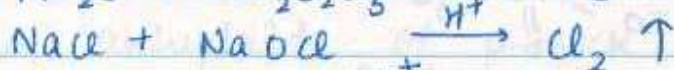
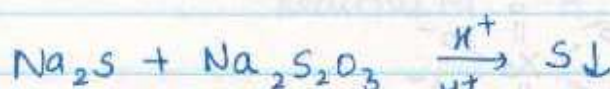
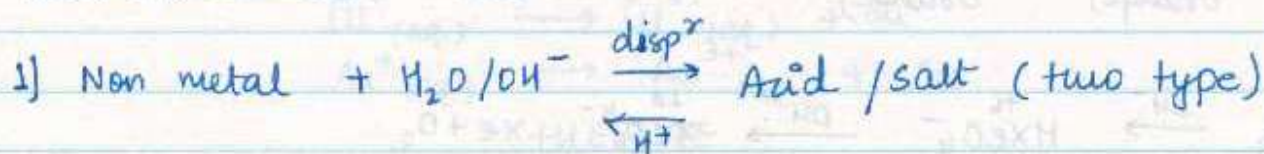


EXO.

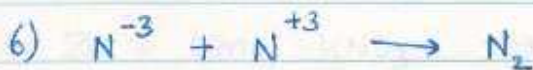




• Comproportionation Reaction :-

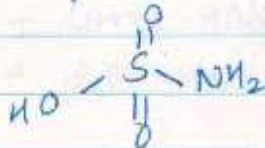


←
when
insoluble compd of Cu^+ forms

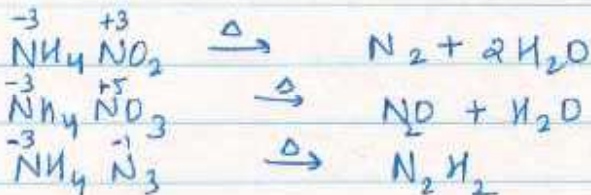


↓ Fe(OH)₃
Blood red

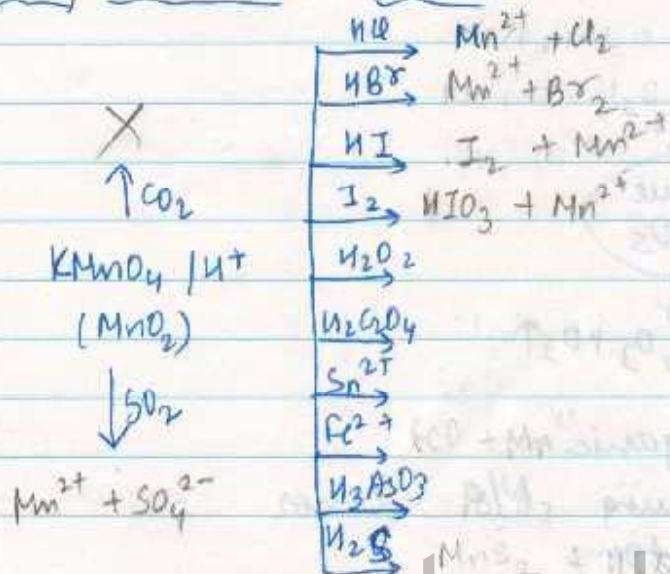
Sulphamic acid

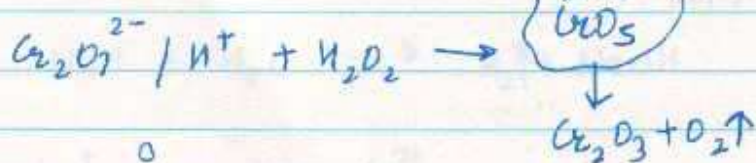
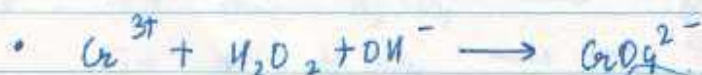
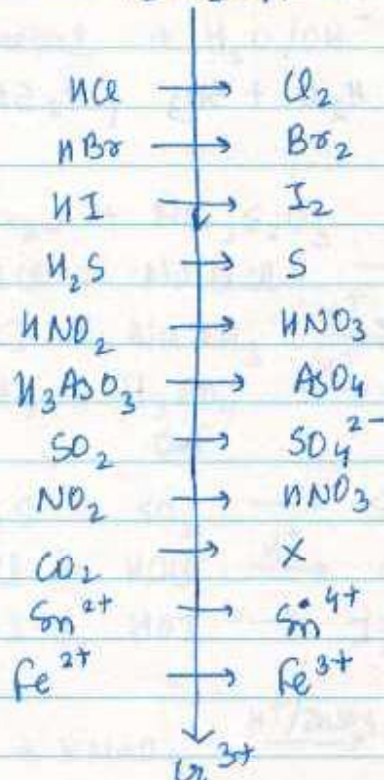
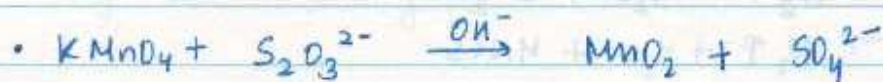
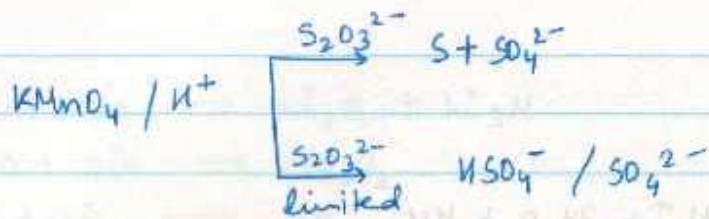


7) Heating of Ammonium Salts :
(containing anion 'N' atom)



* Displacement Reactions :-





Stable in organic solvent having N/O donor atom



Cu^+ is formed only
when insoluble
product is formed.

